CHAPTER 02

Isomers

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: 0c10d482a6512ea86eb518d8bd0784cce6a7e1723ecc19b96ece785efd9381e9

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

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ch02:molecule:ethanol | ethanol

ch02:molecule:ether | ether

ch02:molecule:isobutane | isobutane

ch02:molecule:butane | butane

ch02:molecule:cyclohexane | cyclohexane

ch02:molecule:butanol2 | butanol2

ch02:molecule:butanediol | butanediol

ch02:molecule:e-butene | e-butene

ch02:molecule:z-butene | z-butene

ch02:curriculum | scope-1, scope-2, scope-3

Curriculum scope

scope-1: 2.1 Structural Isomers

scope-2: 2.2 Stereoisomers

scope-3: 2.3 Relative and Absolute Configurations

Objective: Classify isomer relationships

Objective: Assign CIP and E/Z descriptors

Objective: Analyze conformations, symmetry, and optical mixtures

Objective: Calculate specific and observed rotation with explicit units

Objective: Distinguish optical sign, enantiomeric excess, and chiral resolution

Supplied Kaplan curriculum: Chapter 2; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter02b.xhtml

IUPAC optical rotatory power definition | https://old.goldbook.iupac.org/html/O/O04313.html

IUPAC stereochemical terminology: Fischer projection and conventions | https://iupac.qmul.ac.uk/stereo/FM.html

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

connectivity | Connectivity vs. spatial arrangement

Same formula is only the first test

connectivity:rule | Core Rules

Constitutional isomers share a molecular formula but differ in connectivity. Stereoisomers have the same connectivity and differ in spatial arrangement. Conformers interconvert by rotations about single bonds.

Visual card: Isomer Classification & Index of Hydrogen Deficiency (IHD / Degrees of Unsaturation)

Chemistry diagram: Same elemental composition, different atom connectivity: alcohol versus ether.

IHD = [2C + 2 + N − H − X] / 2 (O and divalent S have 0 effect on IHD)

IHD = (2C+2+N-H-X)/2

Ring OR π Bond = 1 IHD

Constitutional = Diff Connectivity

Stereoisomer = Diff 3D

Same Connectivity + 3D | Identical Molecule | Conformer rotation / flipped drawing

Diff Atom-to-Atom Bonds | Constitutional Isomers | Same formula, different IUPAC name

Same Bonds, Diff 3D Space | Stereoisomers | Enantiomers or Diastereomers

1 IHD Unit — — Equals 1 ring OR 1 π bond (alkene C=C or carbonyl C=O).

2 IHD Units — — Equals 1 alkyne (C≡C), 2 double bonds, 2 rings, or 1 ring + 1 double bond (benzene = 4 IHD).

connectivity:cue | Memory Cues

Formula → connections → space. Compare in that order.

Visual card: 3-Question Flowchart: Identical vs. Constitutional vs. Stereoisomer

Compare two structures: how do they relate?

Different Molecular Formula? — NOT Isomers — Check atom counts first! Different formulas (e.g., C_6H_{12} vs C_6H_{14}) are completely different compounds, not isomers.

Same Formula, Different IUPAC Connectivity? — Constitutional (Structural) Isomers — Atoms are wired in a different order (e.g., propan-1-ol vs propan-2-ol vs methoxyethane) → different physical & chemical properties.

Same Connectivity, Non-Superimposable 3D? — Stereoisomers (Enantiomers / Diastereomers) — Identical bond-by-bond graph, but groups point differently in 3D space and cannot interconvert by free single-bond rotation.

connectivity:trap | Common Mistakes / MCAT Traps

A different drawing orientation does not prove isomerism. Rotating the whole molecule changes neither connectivity nor configuration.

Visual card: Mistaking a Rotated Conformer for a Constitutional Isomer

Common Trap — — Looking at a kinked zig-zag alkane drawing and calling it a branched constitutional isomer without naming it.

Correct Reasoning — — Generate the IUPAC name for both drawings: if the parent chain and locants are identical, it is the SAME molecule.

connectivity:app | MCAT Applications

An alcohol and an ether can share a formula but differ in hydrogen-bond donation, boiling point, and reactivity.

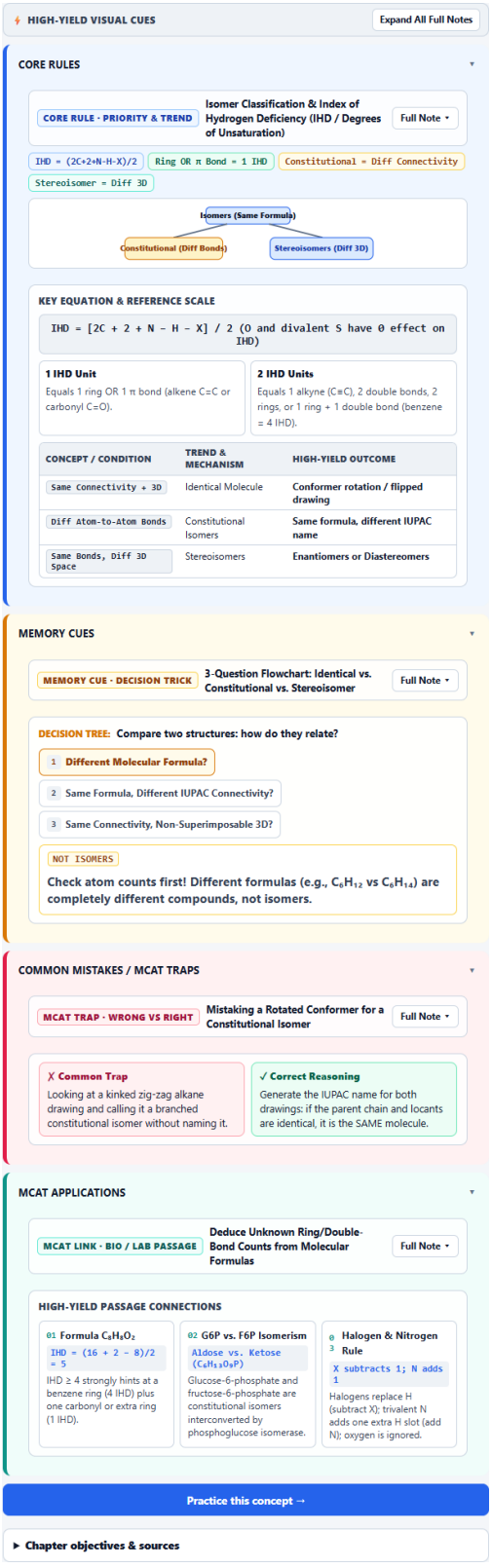
Visual card: Deduce Unknown Ring/Double-Bond Counts from Molecular Formulas

Formula $C_8H_8O_2$ — IHD = (16 + 2 − 8)/2 = 5 — IHD ≥ 4 strongly hints at a benzene ring (4 IHD) plus one carbonyl or extra ring (1 IHD).

G6P vs. F6P Isomerism — Aldose vs. Ketose ($C_6H_{13}O_9P$) — Glucose-6-phosphate and fructose-6-phosphate are constitutional isomers interconverted by phosphoglucose isomerase.

Halogen & Nitrogen Rule — X subtracts 1; N adds 1 — Halogens replace H (subtract X); trivalent N adds one extra H slot (add N); oxygen is ignored.

Screenshot: connectivity-visuals



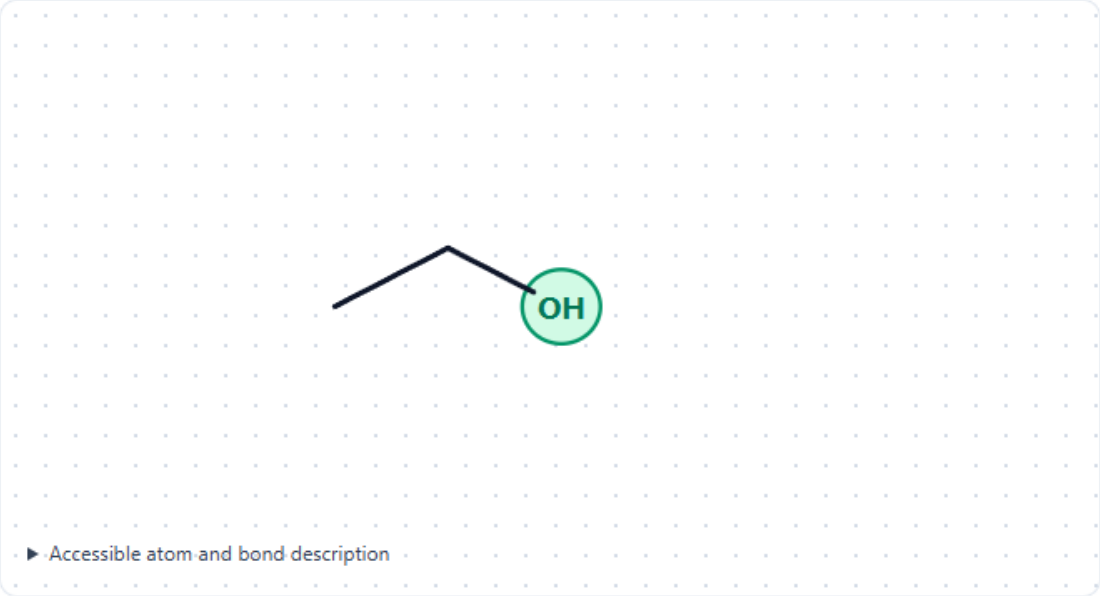
connectivity:step-1 | 1. Alcohol connectivity

Count two carbons, six hydrogens, and one oxygen. The oxygen bears H and can donate a hydrogen bond.

Displayed molecule ID: ethanol

{"highlights": {"prin": [2]}, "revealName": true}

Screenshot: connectivity-step-1



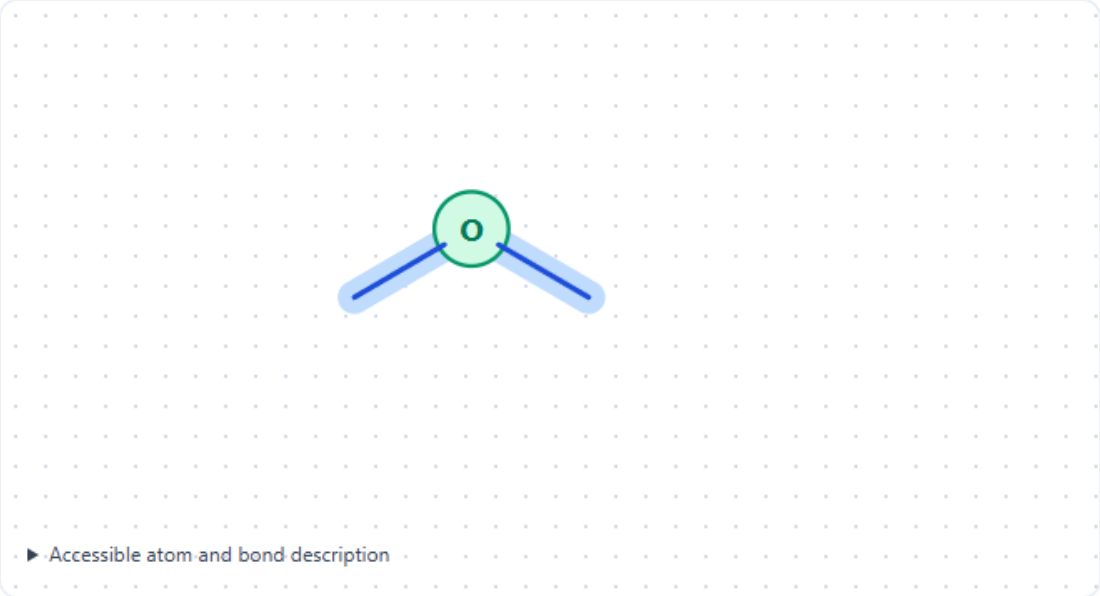
connectivity:step-2 | 2. Ether connectivity

The same formula now has oxygen between two carbons. No O–H bond is present. This is a constitutional isomer, not a different conformation.

Displayed molecule ID: ether

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Screenshot: connectivity-step-2



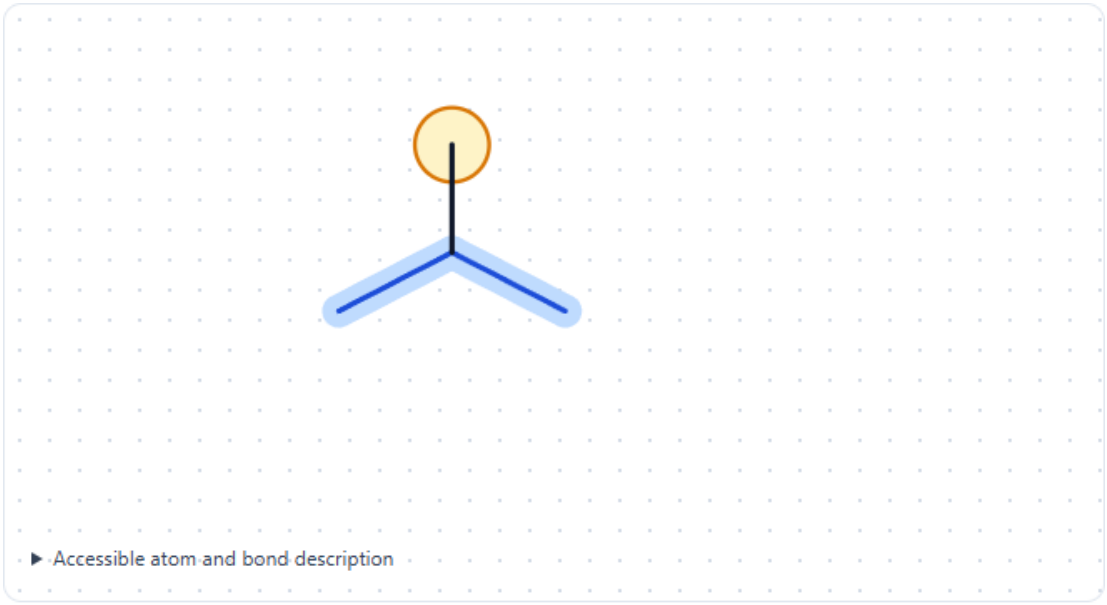
connectivity:step-3 | 3. Branched framework

Butane and 2-methylpropane are another constitutional-isomer pair: identical formula, different carbon skeletons.

Displayed molecule ID: isobutane

{"highlights": {"path": [0, 1, 2], "sub": [3]}, "revealName": true}

Screenshot: connectivity-step-3



conformations | Conformations and ring strain

Rotation changes energy, not identity

conformations:rule | Core Rules

Angle strain is deviation from preferred bond angles; torsional strain comes from eclipsing bonds; nonbonded (steric) strain comes from crowding of atoms not bonded to each other. Butane anti and gauche conformations are staggered, but gauche has greater methyl–methyl crowding. A cyclohexane chair has nearly ideal angles and staggered bonds; bulky substituents usually prefer equatorial sites to reduce 1,3-diaxial crowding.

Visual card: Newman Dihedral Energy & Cyclohexane Chair Ring-Flip Mechanics

Chemistry diagram: Newman projection of the lesson graph, shown at the anti teaching angle.

Anti (180°) < Gauche (60°) < Eclipsed < Syn (0°)

Chair Flip: Axial ⇄ Equatorial

Up Stays Up!

Newman Anti (180°) | Lowest energy staggered | Bulky groups maximally separated

Newman Syn (0°) | Highest energy eclipsed | Severe torsional + steric clash

Chair Equatorial | Bulky groups point outward | Avoids 1,3-diaxial steric clash

Newman Staggered Conformers — — Anti (180° between bulky groups) is lowest energy; Gauche (60°) has a small steric penalty (~3.8 kJ/mol for butane methyls).

Newman Eclipsed Transitions — — Eclipsed (120°) has torsional strain; Syn-periplanar (0° bulky-bulky) is the highest-energy barrier (~19 kJ/mol in butane).

Cyclohexane Chair Ring Flip — — Eliminates angle and torsional strain (0° eclipsing); ring flip converts ALL Axial ⇄ Equatorial while UP stays UP and DOWN stays DOWN.

conformations:cue | Memory Cues

For the butane methyl–methyl dihedral shown: anti = 180°, gauche = 60°, ordinary eclipsed = 120°, and totally (fully) eclipsed = 0°. At 0° the methyl groups eclipse one another, giving the highest energy of these conformations.

Visual card: 1,4- vs 1,3- vs 1,2-Disubstituted Cyclohexane Cis/Trans Cheat Sheet

What changes and what stays fixed during a cyclohexane chair flip?

Axial vs. Equatorial Status — ALWAYS Flips (Ax ⇄ Eq) — Every axial bond becomes equatorial and every equatorial bond becomes axial to relieve 1,3-diaxial steric clashes.

Up / Down Wedge-Dash Face — NEVER Changes (Up = Up) — A ring flip is a conformational rotation, not bond breaking: cis remains cis and R/S stereocenters never invert.

tert-Butyl Conformational Lock — Locks in Equatorial (>99.9%) — A bulky tert-butyl group experiences massive 1,3-diaxial strain if axial, locking the chair so t-Bu stays equatorial.

conformations:trap | Common Mistakes / MCAT Traps

Axial is not synonymous with up: axial directions alternate. A chair flip swaps axial/equatorial positions while preserving up/down and cis/trans. Boat flagpole contacts are steric interactions; do not call every unfavorable contact angle strain.

Visual card: Equating "Cis" with "Both Equatorial" or Thinking Ring Flips Invert R/S

Common Trap — — Assuming cis-1,2-dimethylcyclohexane is diequatorial or that a chair flip turns cis into trans.

Correct Reasoning — — Alternate Up/Down around the ring: 1,2-cis is (ax,eq), 1,3-cis is (eq,eq) or (ax,ax), and 1,4-trans is (eq,eq) or (ax,ax).

conformations:app | MCAT Applications

Conformational preferences affect binding. Cyclobutane puckers and cyclopentane can adopt envelope-like shapes to reduce eclipsing; cyclohexane has higher-energy boat and twist-boat forms, and its chair-flip pathway passes through half-chair transition states.

Visual card: β-D-Glucose All-Equatorial Stability & E2 Anti-Periplanar Requirement

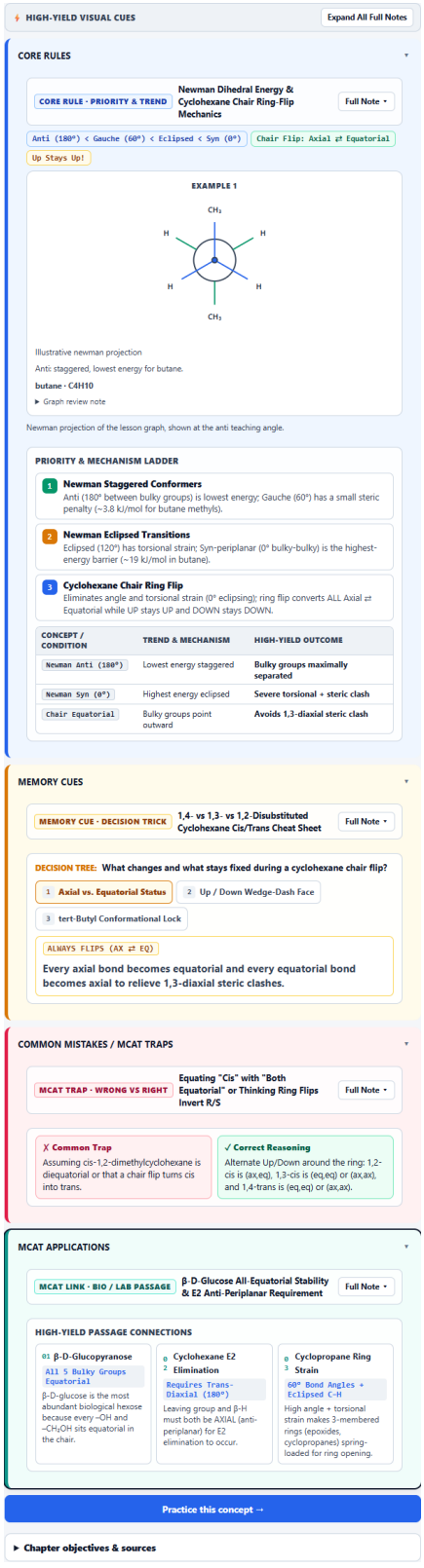
Chemistry diagram: Illustrative graph focus from the corresponding walkthrough explanation; no transformation is implied.

β-D-Glucopyranose — All 5 Bulky Groups Equatorial — β-D-glucose is the most abundant biological hexose because every –OH and $–CH_2OH$ sits equatorial in the chair.

Cyclohexane E2 Elimination — Requires Trans-Diaxial (180°) — Leaving group and β-H must both be AXIAL (anti-periplanar) for E2 elimination to occur.

Cyclopropane Ring Strain — 60° Bond Angles + Eclipsed C–H — High angle + torsional strain makes 3-membered rings (epoxides, cyclopropanes) spring-loaded for ring opening.

Screenshot: conformations-visuals



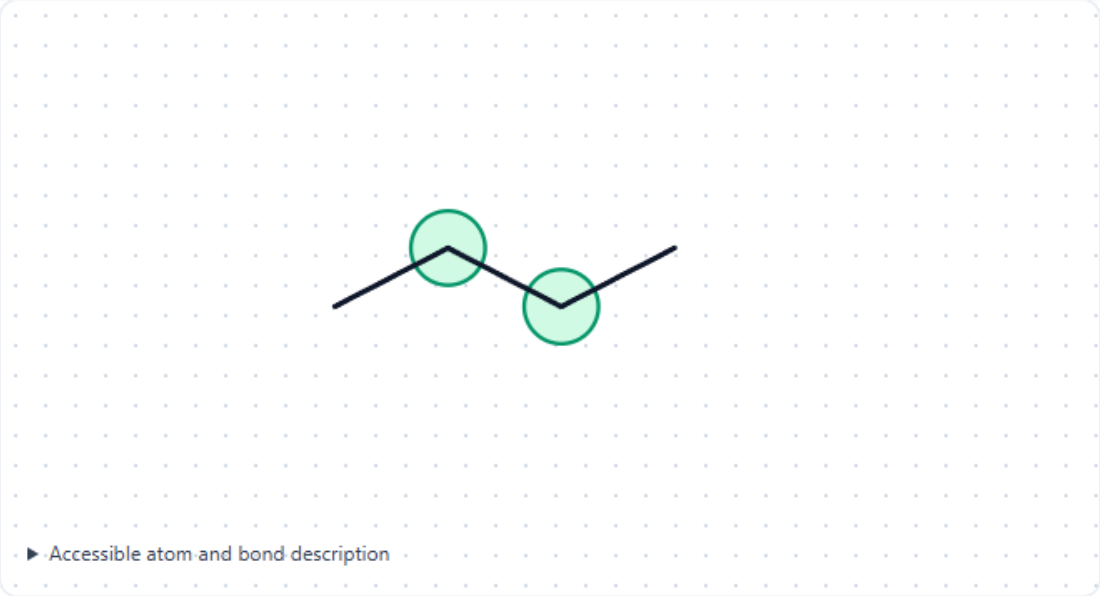
conformations:step-1 | 1. Rotate a single bond

Look down the middle C–C bond of butane. Adjust the angle to compare torsional and methyl–methyl interactions.

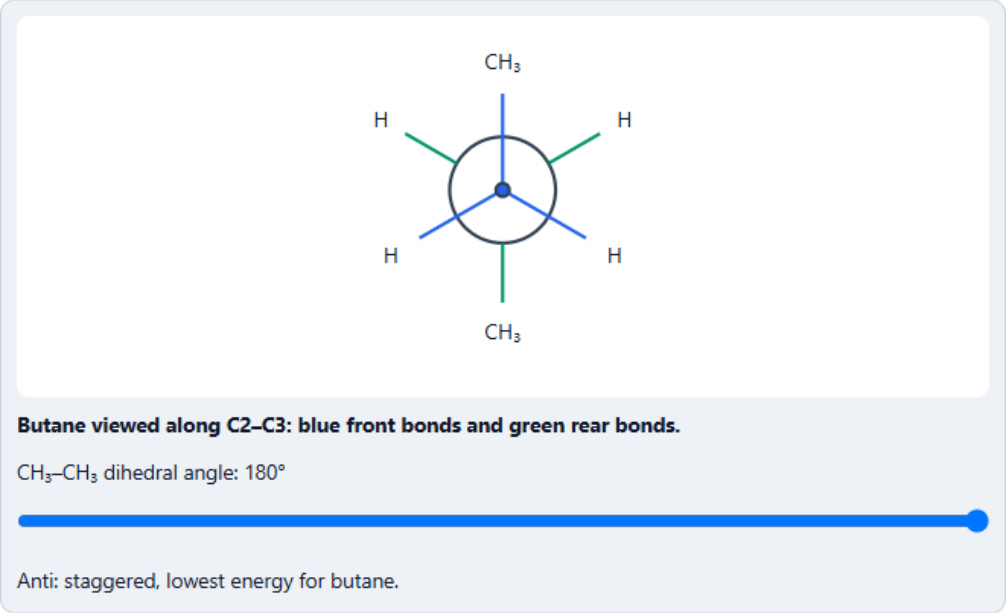
Displayed molecule ID: butane

{"highlights": {"prin": [1, 2]}, "revealName": true}

Screenshot: conformations-step-1



Screenshot: conformations-step-1-diagram



{"diagramSpecification": {"type": "newman", "caption": "Butane viewed along C2–C3: blue front bonds and green rear bonds."}}

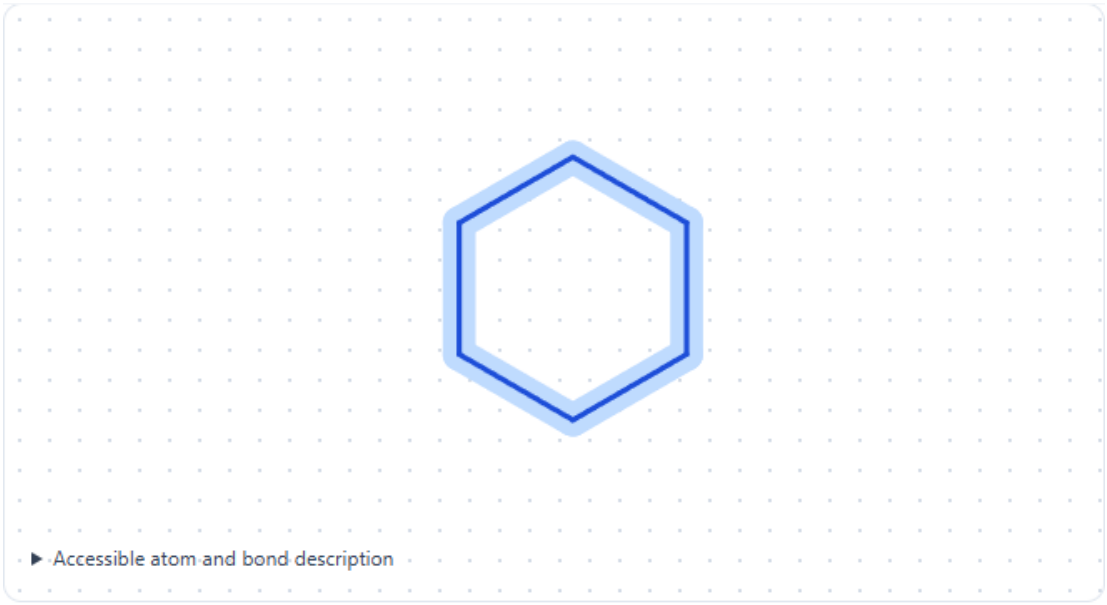
conformations:step-2 | 2. Ring constraints

This hexagon shows connectivity only, not a flat equilibrium geometry. Cyclohexane normally adopts a chair. An axial substituent interacts with axial groups at C3 and C5; an equatorial position avoids much of this crowding.

Displayed molecule ID: cyclohexane

{"highlights": {"path": [0, 1, 2, 3, 4, 5]}, "revealName": true}

Screenshot: conformations-step-2



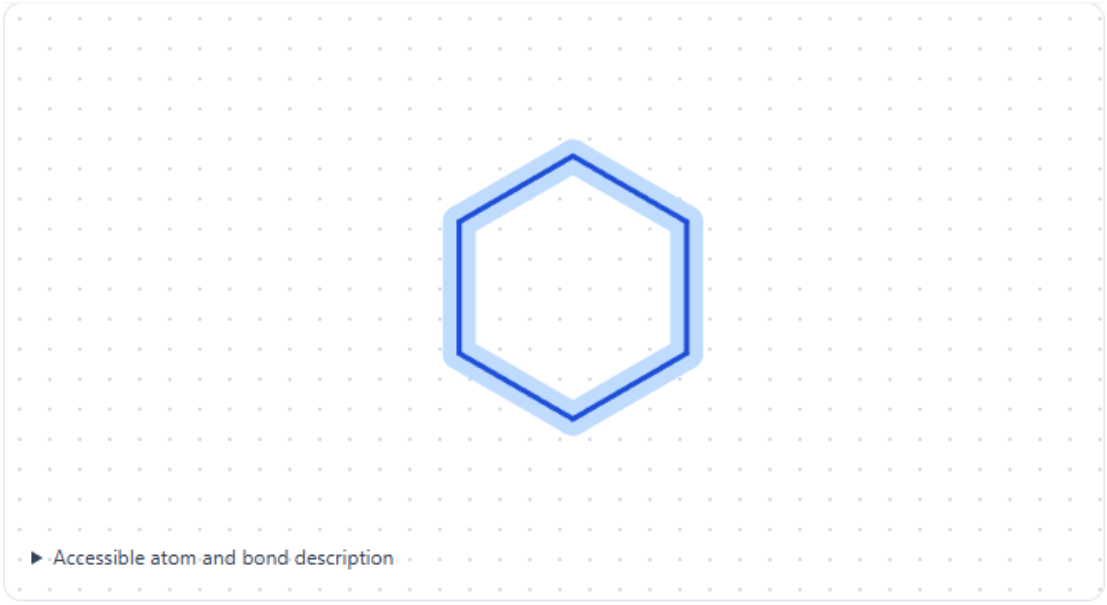
conformations:step-3 | 3. Ring-flip pathway

This hexagon is a connectivity map, not a chair drawing. A chair flip passes through higher-energy shapes: half-chair transition states and twist-boat regions; the boat has flagpole crowding and eclipsing interactions. The final chair exchanges axial/equatorial sites without changing up/down.

Displayed molecule ID: cyclohexane

{"highlights": {"path": [0, 1, 2, 3, 4, 5]}, "revealName": true}

Screenshot: conformations-step-3



chirality | Chirality, enantiomers, and meso compounds

Mirror image ≠ always a different molecule

chirality:rule | Core Rules

A tetrahedral carbon with four different substituents is a common stereogenic center. Enantiomers are nonsuperimposable mirror images. Diastereomers are stereoisomers that are not mirror images; E/Z pairs qualify too.

Visual card: Chirality, Enantiomers, Diastereomers & Meso Internal Symmetry Planes

Chemistry diagram: Inspect the four substituents at the OH-bearing carbon. Flat connectivity alone does not assign R/S.

Max Stereoisomers = 2^n (Reduced when an internal symmetry plane σ creates a Meso compound)

Chiral = No σ / i Symmetry

Max Stereoisomers = 2^n

Meso = Chiral Centers + Plane σ

Enantiomers = Opposite R/S

Enantiomers | Invert ALL chiral centers | Identical BP/MP; opposite [α]D

Diastereomers | Invert SOME chiral centers | Different BP/MP/polarity (separable!)

Meso Compound | Chiral centers + internal σ plane | Achiral & optically inactive ([α]=0)

Chiral Molecule — — Non-superimposable on its mirror image; lacks an internal mirror plane (σ) or inversion center (i).

Meso Compound — — Contains ≥2 stereocenters with identical substituent sets and opposite configurations (e.g., (2R,3S)-tartaric acid) → achiral!

chirality:cue | Memory Cues

One inverted center in a one-center molecule gives its enantiomer. With several centers, compare the entire molecule.

Visual card: Instant Stereochemical Relationship Classifier

Compare the R/S configurations at the stereocenters of two molecules:

Every Center Inverted: (2R,3R) vs (2S,3S) — Enantiomers (Unless Meso!) — Exact non-superimposable mirror images — check first that neither molecule has an internal mirror plane (meso).

One Inverted, One Same: (2R,3R) vs (2R,3S) — Diastereomers — Stereoisomers that are NOT mirror images; they have different melting points, boiling points, and TLC Rf values.

Mirror Plane Cuts Molecule in Half: (2R,3S) — Meso (Same Achiral Molecule) — Flipping (2R,3S)-tartaric acid 180° in the page superimposes it on (2S,3R)-tartaric acid — they are identical!

chirality:trap | Common Mistakes / MCAT Traps

2^n is a maximum number of stereoisomers, not a guarantee. Internal symmetry can make a molecule with stereocenters achiral (meso). R/S does not predict the sign of optical rotation.

Visual card: Assuming Every Molecule with Stereocenters Is Chiral (The Meso Trap)

Common Trap — — Counting n = 2 stereocenters in 2,3-dibromobutane and immediately answering "4 optically active stereoisomers".

Correct Reasoning — — Always check for a plane of symmetry: symmetric 2,3-disubstituted chains have ONLY 3 stereoisomers (1 enantiomeric pair + 1 achiral meso).

chirality:app | MCAT Applications

Enantiomers have the same ordinary properties in achiral environments but opposite optical rotations under matched conditions; chiral receptors can distinguish them. A suitable single-enantiomer resolving agent can convert a racemate into diastereomeric salts or derivatives. Their different properties permit separation, followed by regeneration of the original enantiomers. Chiral chromatography is another approach.

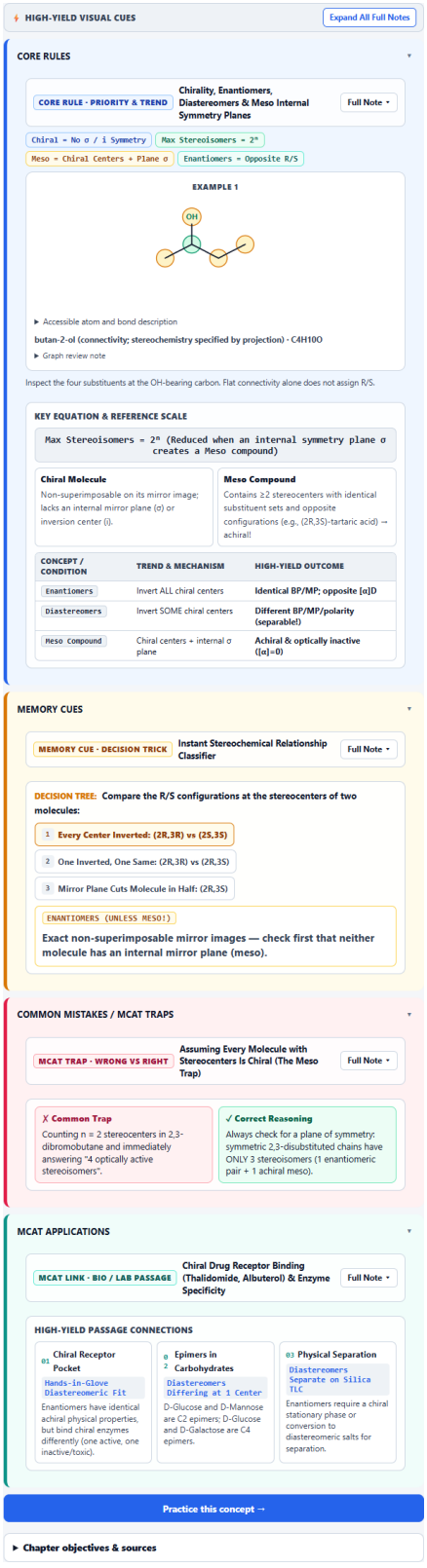
Visual card: Chiral Drug Receptor Binding (Thalidomide, Albuterol) & Enzyme Specificity

Chiral Receptor Pocket — Hands-in-Glove Diastereomeric Fit — Enantiomers have identical achiral physical properties, but bind chiral enzymes differently (one active, one inactive/toxic).

Epimers in Carbohydrates — Diastereomers Differing at 1 Center — D-Glucose and D-Mannose are C2 epimers; D-Glucose and D-Galactose are C4 epimers.

Physical Separation — Diastereomers Separate on Silica TLC — Enantiomers require a chiral stationary phase or conversion to diastereomeric salts for separation.

Screenshot: chirality-visuals



chirality:step-1 | 1. Identify four groups

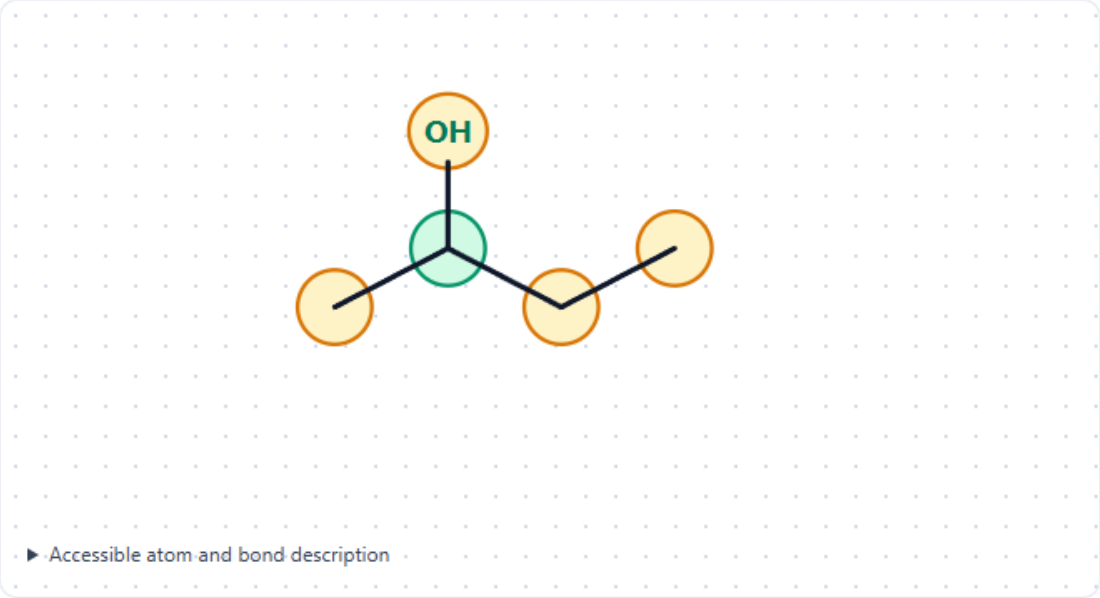
The alcohol-bearing carbon has OH, H, methyl, and ethyl as four distinct groups. Both carbons of the ethyl group are highlighted. The flat graph does not assign R or S. Try It Yourself removes these walkthrough highlights before asking for an

atom selection.

Displayed molecule ID: butanol2

{"highlights": {"prin": [1], "sub": [0, 2, 3, 4]}, "revealName": false}

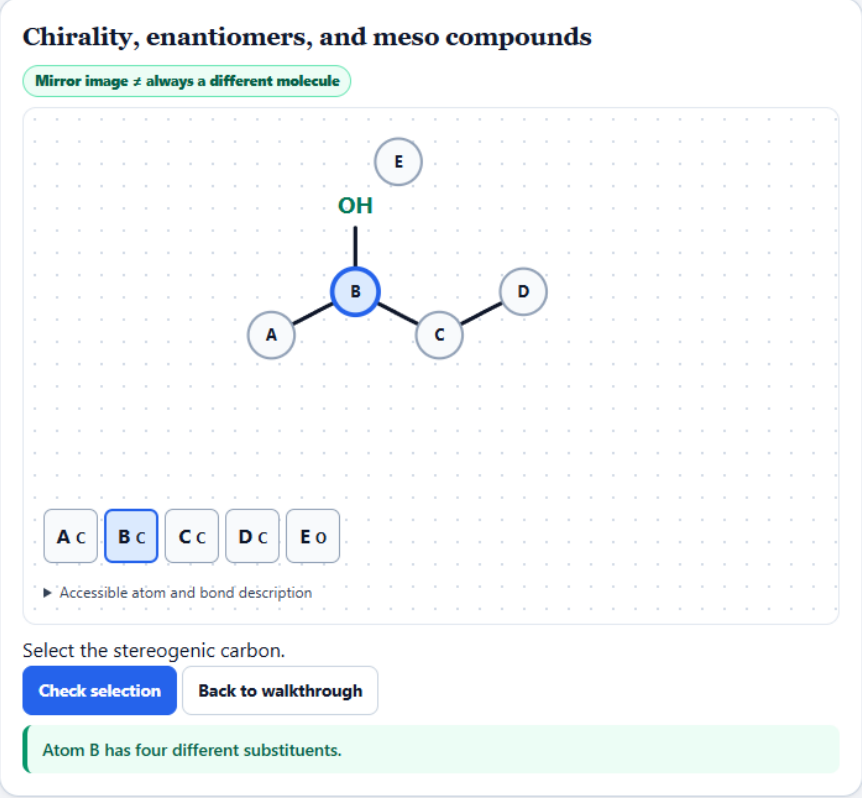
Screenshot: chirality-step-1



Embedded challenge: Select the stereogenic carbon.

{"prompt": "Select the stereogenic carbon.", "answer": [1], "explanation": "Atom B has four different substituents."}

Screenshot: chirality-step-1-challenge



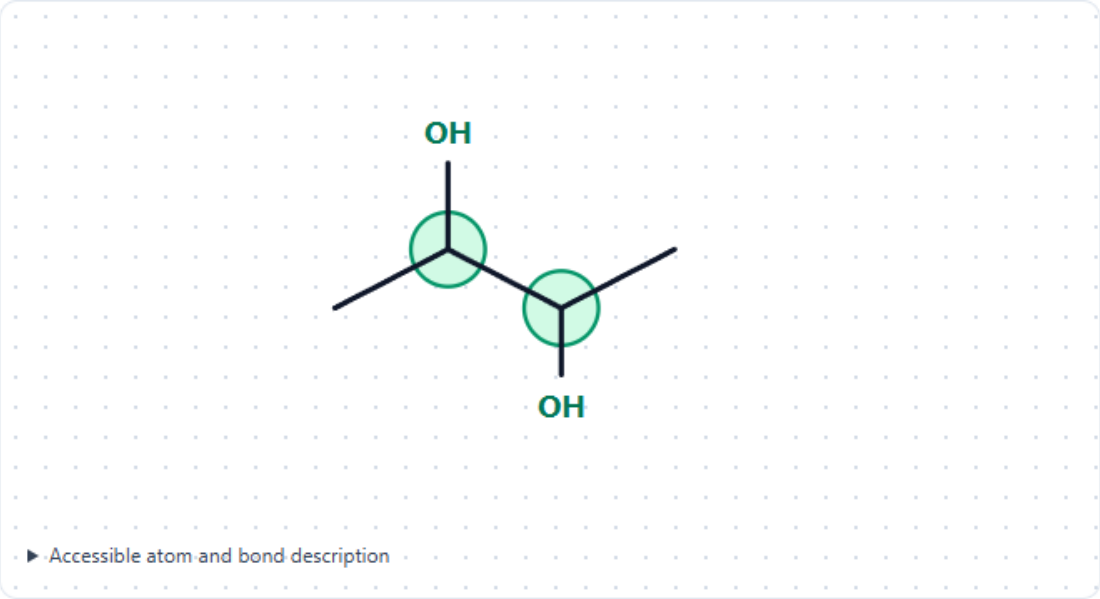
chirality:step-2 | 2. Count with symmetry

Butane-2,3-diol has two potential stereocenters. Its (2R,3S) and (2S,3R) descriptions refer to the same meso compound because of symmetry; the RR and SS forms are an enantiomeric pair. Total: three stereoisomers.

Displayed molecule ID: butanediol

{"highlights": {"prin": [1, 2]}, "revealName": true}

Screenshot: chirality-step-2



configuration | CIP priorities and Fischer projections

Assign priority before tracing direction

configuration:rule | Core Rules

Rank directly attached atoms by atomic number; isotopes by mass. Resolve ties one sphere at a time using sorted atom lists. Treat multiple bonds with duplicate-atom bookkeeping. With priority 4 away, clockwise 1→2→3 is R and counterclockwise is S. Absolute configuration assigns a stereochemical arrangement independently (for example R/S). Relative configuration describes a relationship to another stereocenter or reference compound. A remote chemical change can retain the local spatial arrangement while changing CIP priorities and therefore its R/S label; reassess priorities instead of equating a letter change with physical inversion.

Visual card: Cahn-Ingold-Prelog (CIP) Absolute Configuration (R / S) & Fischer Projections

Chemistry diagram: Fischer projections: horizontal bonds point forward; vertical bonds point back.

CIP: Atomic Number Z

Multiple Bond = Ghost Atoms

4th Group Dash → CW=R, CCW=S

Fischer H Horizontal = Flip!

4th Group on DASH | Read 1→2→3 directly | Clockwise = R, Counterclockwise = S

4th Group on WEDGE | Reverse your 1→2→3 read | Clockwise → S, Counterclockwise → R

Fischer 4th Horizontal | 4th group points AT you (wedge) | Reverse 1→2→3 result!

Rank 4 Groups by Atomic Number (Z) — — Compare directly attached atoms first (I > Br > Cl > S > P > F > O > N > C > H); heavier isotopes win ties (^{3}H > ^{2}H > ^{1}H).

Move Outward to First Point of Difference — — List attached atoms in decreasing Z order—the single highest atomic number wins (e.g., $–CH_2Br$ beats $–C(CH_3)_3$ because Br > C).

Orient Priority #4 & Trace 1 → 2 → 3 — — With group #4 on a DASH: Clockwise = R (Rectus), Counterclockwise = S (Sinister). If #4 is on a WEDGE, reverse the result.

configuration:cue | Memory Cues

Fischer: horizontal bonds come forward; vertical bonds go back. If group 4 is horizontal, reverse the apparent result. We draw C1 at the top here. A 180° rotation of the entire Fischer drawing preserves its configuration.

Visual card: Fischer Projection Crossbow Rule: Horizontal = Hugs You (Wedge)

Where is the lowest-priority group (#4, usually H) located?

Wedge-Dash: #4 on a Dash — Keep 1→2→3 Result (CW = R, CCW = S) — Standard steering-wheel view looking down the C–#4 bond.

Wedge-Dash: #4 on a Wedge — Invert Result (CW = S, CCW = R) — Because #4 points toward you, flip your clockwise/counterclockwise read.

Fischer: #4 on Horizontal Arm — Invert Result (CW = S, CCW = R) — Horizontal Fischer bonds are wedges hugging you! Vertical Fischer bonds are dashes pointing away.

configuration:trap | Common Mistakes / MCAT Traps

A 180° in-plane Fischer rotation preserves configuration; a 90° rotation does not. D/L is a relative-configuration convention, not a synonym for R/S or +/−.

Visual card: Adding Atomic Numbers Instead of Picking the Single Highest Atom

Common Trap — — Ranking $–C(CH_3)_3$ (three carbons: 6+6+6=18) higher than $–CH_2OH$ (O,H,H: 8+1+1=10).

Correct Reasoning — — Never add atomic numbers! Compare the single highest-Z atom at the tiebreaker shell: O (Z=8) beats C (Z=6), so $–CH_2OH$ wins.

configuration:app | MCAT Applications

Most proteinogenic L amino acids are S; L cysteine is R because sulfur changes CIP ranking. Glycine has no stereogenic alpha carbon.

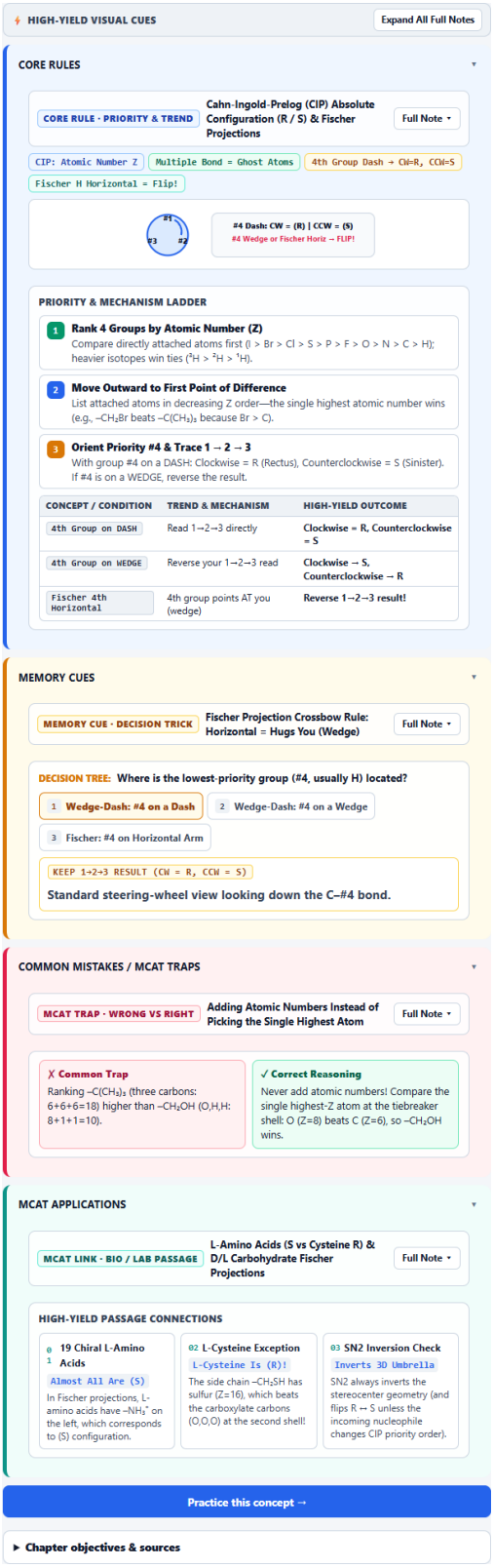
Visual card: L-Amino Acids (S vs Cysteine R) & D/L Carbohydrate Fischer Projections

19 Chiral L-Amino Acids — Almost All Are (S) — In Fischer projections, L-amino acids have $–NH_3^+$ on the left, which corresponds to (S) configuration.

L-Cysteine Exception — L-Cysteine Is (R)! — The side chain $–CH_2SH$ has sulfur (Z=16), which beats the carboxylate carbons (O,O,O) at the second shell!

SN2 Inversion Check — Inverts 3D Umbrella — SN2 always inverts the stereocenter geometry (and flips R ↔ S unless the incoming nucleophile changes CIP priority order).

Screenshot: configuration-visuals



configuration:step-1 | 1. Rank the substituents

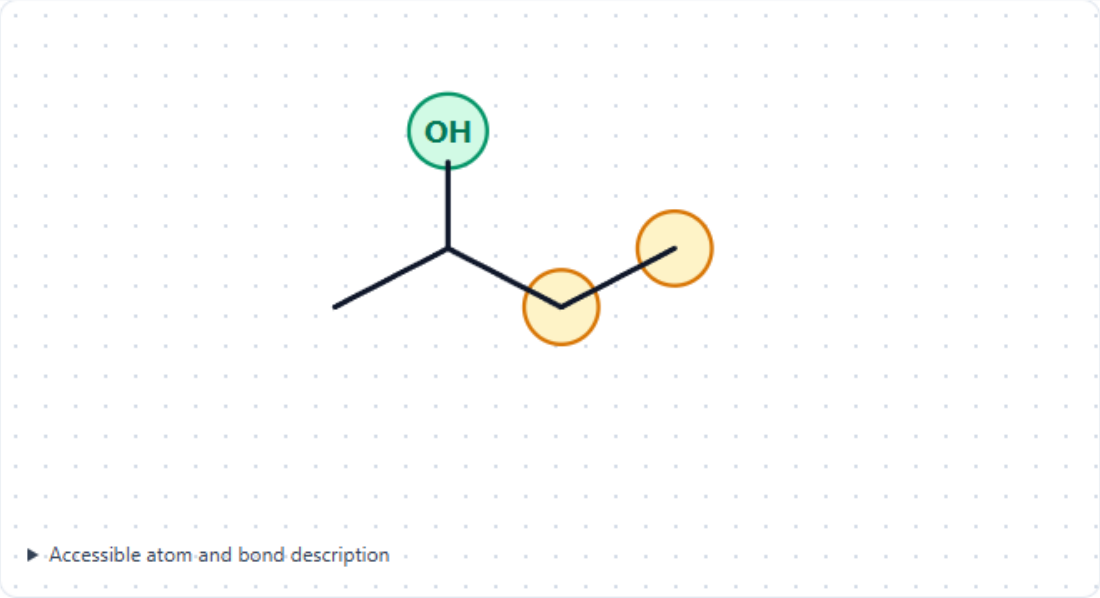
OH ranks first. Ethyl beats methyl at the next sphere: [C,H,H] beats [H,H,H]. The terminal CH_3 carbon of the ethyl group supplies the C that breaks the tie; both carbons of ethyl are highlighted in orange, while the competing methyl carbon is

unhighlighted. H ranks fourth.

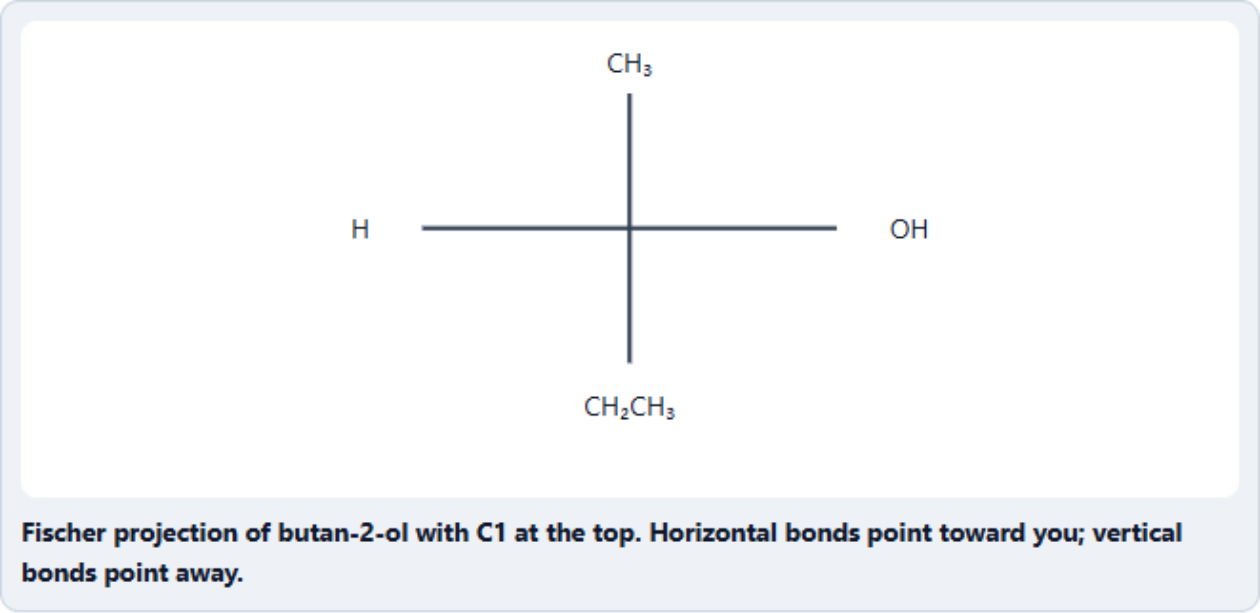
Displayed molecule ID: butanol2

{"highlights": {"prin": [4], "sub": [2, 3]}, "revealName": false}

Screenshot: configuration-step-1



Screenshot: configuration-step-1-diagram



Fischer projection of butan-2-ol with C1 at the top. Horizontal bonds point toward you; vertical bonds point away.

{"diagramSpecification": {"type": "fischer", "top": "CH_3", "bottom": "CH_2CH_3", "left": "H", "right": "OH", "caption": "Fischer projection of butan-2-ol with C1 at the top. Horizontal bonds point toward you; vertical bonds point away."}}

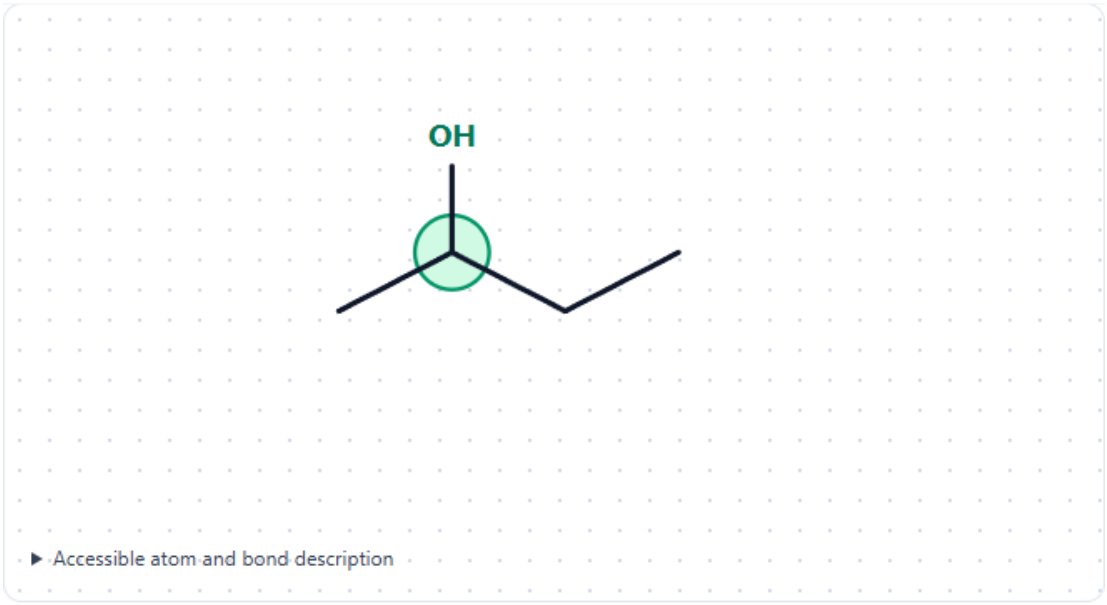
configuration:step-2 | 2. Orient then assign

Use the same Fischer projection with C1 at the top. Priorities are OH (right) > ethyl (bottom) > methyl (top) > H (left). The apparent 1→2→3 order is clockwise; H is horizontal and toward you, so reverse it: S.

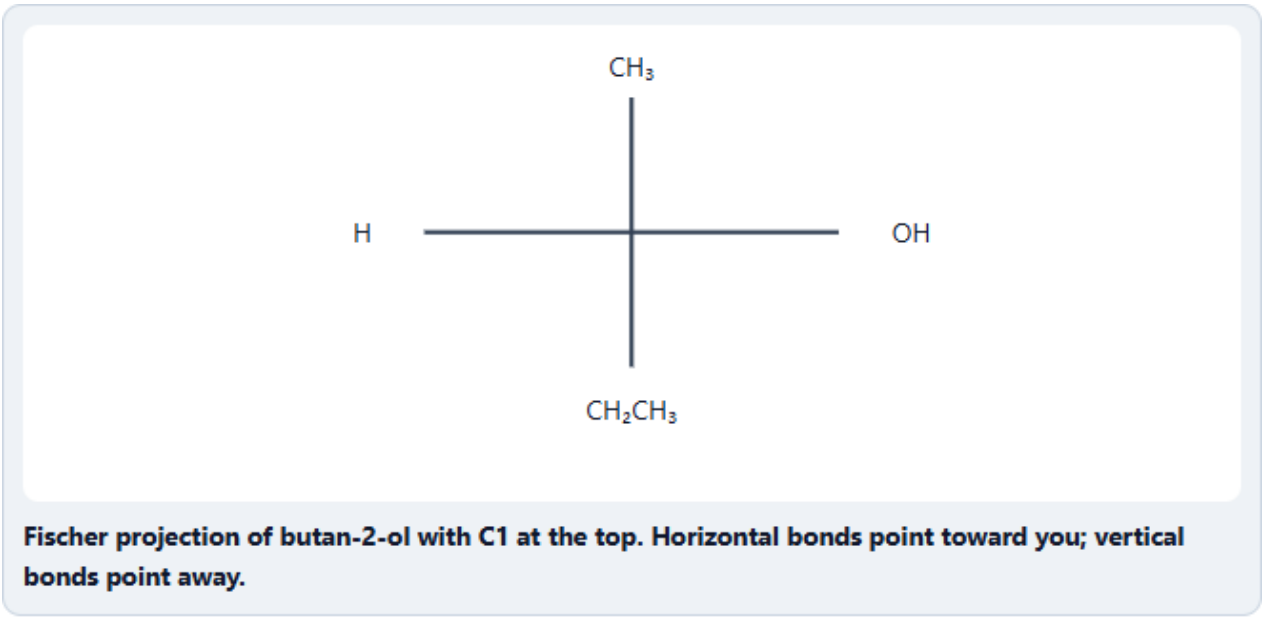
Displayed molecule ID: butanol2

{"highlights": {"prin": [1]}, "revealName": false}

Screenshot: configuration-step-2



Screenshot: configuration-step-2-diagram



{"diagramSpecification": {"type": "fischer", "top": "CH₃", "bottom": "CH₂CH₃", "left": "H", "right": "OH", "caption": "Fischer projection of butan-2-ol with C1 at the top. Horizontal bonds point toward you; vertical bonds point away."}}

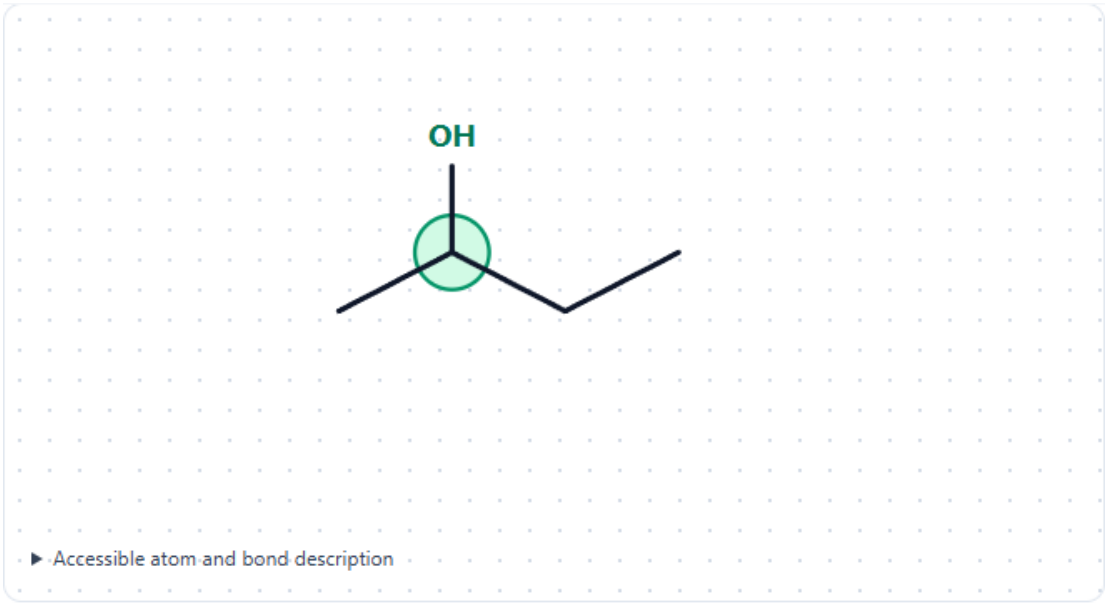
configuration:step-3 | 3. Mirror the projection

Interchanging H and OH inverts this single center. The mirrored projection is R and represents the enantiomer.

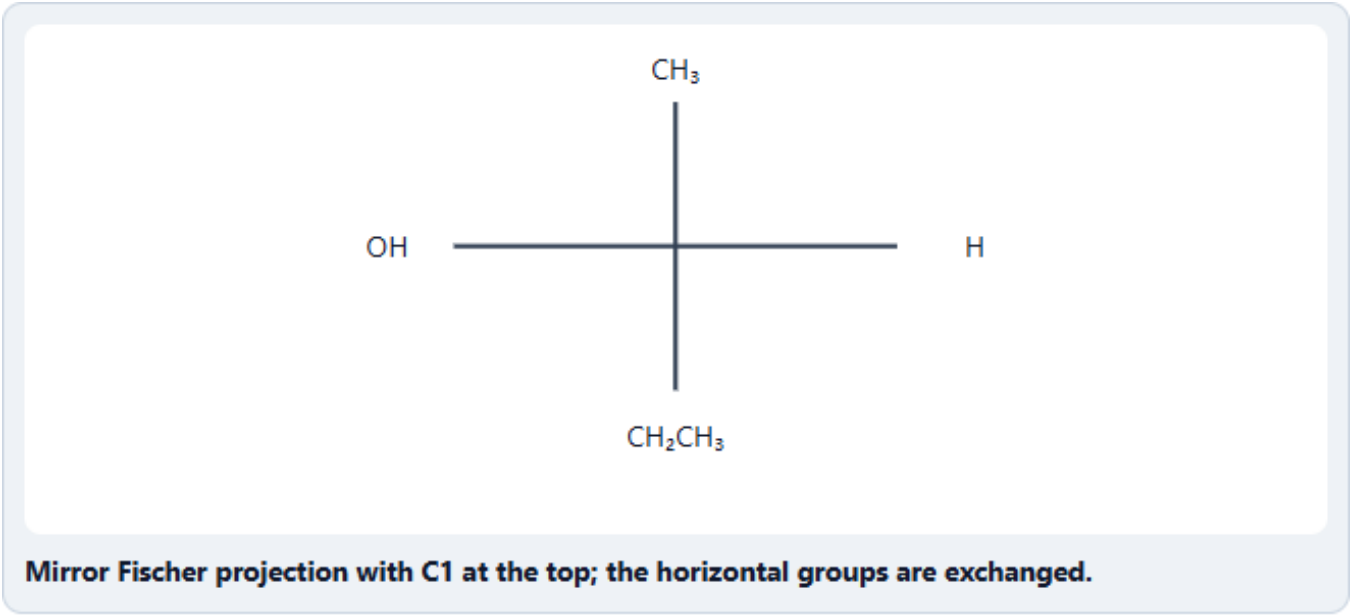
Displayed molecule ID: butanol2

{"highlights": {"prin": [1]}, "revealName": false}

Screenshot: configuration-step-3



Screenshot: configuration-step-3-diagram



{"diagramSpecification": {"type": "fischer", "top": "CH₃", "bottom": "CH₂CH₃", "left": "OH", "right": "H", "caption": "Mirror Fischer projection with C1 at the top; the horizontal groups are exchanged."}}

alkene | E/Z configuration and achirality

Rank each end separately

alkene:rule | Core Rules

E/Z is defined only if each double-bond carbon has two different substituents. Rank each end by CIP: Z puts the higher-priority groups on the same side; E puts them on opposite sides. E/Z geometry alone does not determine whether a whole molecule is chiral.

Visual card: Alkene E / Z Stereochemistry: Rigid π-Bond Barrier & CIP Priority Per Carbon

Chemistry diagram: Compare the substituents across C=C. Rank each alkene carbon separately.

Split C=C Vertically

Z = Zusammen (Zame Zide)

E = Entgegen (Opposite)

Cis/Trans ≠ E/Z Always

Z (Zusammen) | High-priority #1/#1 on SAME side | Mnemonic: "Zame Zide"

E (Entgegen) | High-priority #1/#1 on OPPOSITE sides | Mnemonic: "Epposite"

Terminal $=CH_2$ | Two identical groups on one alkene C | NO E/Z stereoisomerism possible

Cut Perpendicular to C=C First — — Treat Left C1 and Right C2 independently: rank the two groups on C1 by CIP atomic number, then rank the two groups on C2.

Compare the Two #1 Winners Across the C=C Axis — — Now slice along the C=C bond axis to see whether the two #1 priority winners sit on the same side or opposite sides.

Assign Z (Same Side) or E (Opposite Sides) — — Z (zusammen = "zame zide") when both #1 groups are on the same side; E (entgegen = opposite) when across the double bond.

alkene:cue | Memory Cues

Z: higher-priority groups together. E: higher-priority groups opposite. Identify E/Z first, then assess the symmetry of the whole molecule separately.

Visual card: Can This Alkene Show E/Z Isomerism?

Inspect the two substituents on each sp^2 alkene carbon:

Either Carbon Has 2 Identical Groups ($R_2C=CR'_2$) — Neither E nor Z (No Geometric Isomers) — Propene, 2-methylbut-2-ene, and terminal $=CH_2$ alkenes cannot have E/Z stereoisomers.

Both #1 CIP Winners on Same Face — Z Configuration ("Zame Zide") — Higher dipole moment in polar alkenes; often less stable than E due to steric clash between the two larger groups.

#1 CIP Winners on Opposite Faces — E Configuration (Entgegen) — Bulky groups sit diagonally across the rigid π bond, minimizing steric strain (higher heat of stability).

alkene:trap | Common Mistakes / MCAT Traps

Cis/trans cannot replace E/Z in every alkene. Both E- and Z-but-2-ene are achiral. Zero optical rotation can arise from an achiral compound or an equal mixture of enantiomers; it does not identify which situation applies.

Visual card: Assuming Cis Always Equals Z and Trans Always Equals E

Common Trap — — Looking at two methyl groups trans to each other in 2-bromobut-2-ene and automatically calling it (E).

Correct Reasoning — — Assign CIP atomic numbers independently on each carbon: –Br (Z=35) outranks –CH_3 (Z=6), which can make a "trans-methyl" alkene (Z)!

alkene:app | MCAT Applications

The pure Z isomer of but-2-ene is achiral and shows no ordinary optical activity. It is not a racemate. A racemate requires two distinct enantiomers in equal proportions.

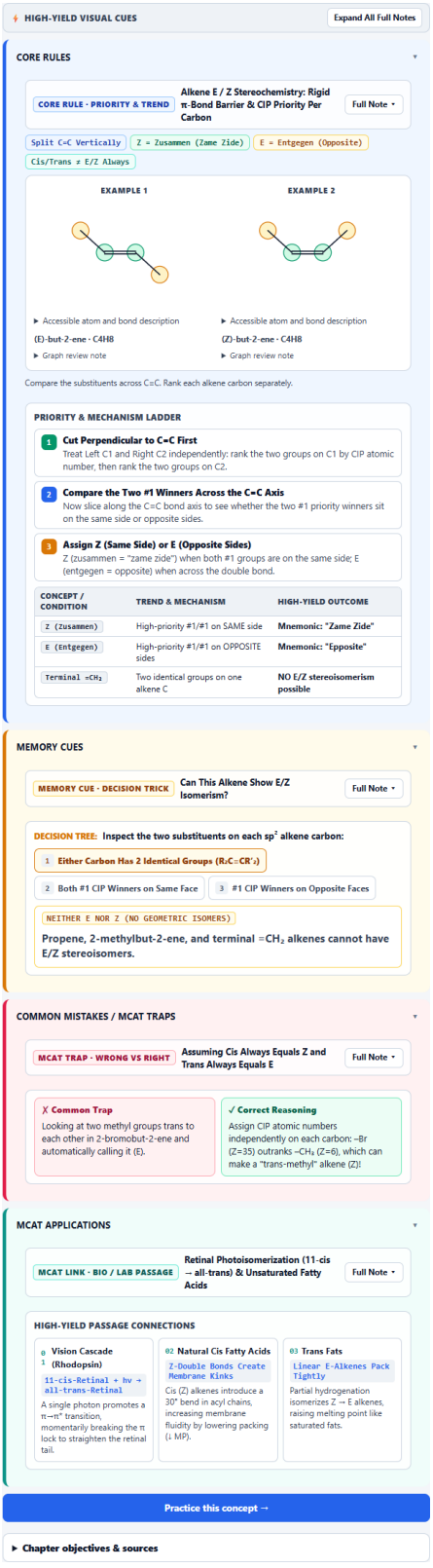
Visual card: Retinal Photoisomerization (11-cis → all-trans) & Unsaturated Fatty Acids

Vision Cascade (Rhodopsin) — 11-cis-Retinal + hν → all-trans-Retinal — A single photon promotes a π→π* transition, momentarily breaking the π lock to straighten the retinal tail.

Natural Cis Fatty Acids — Z-Double Bonds Create Membrane Kinks — Cis (Z) alkenes introduce a 30° bend in acyl chains, increasing membrane fluidity by lowering packing (↓ MP).

Trans Fats — Linear E-Alkenes Pack Tightly — Partial hydrogenation isomerizes Z → E alkenes, raising melting point like saturated fats.

Screenshot: alkene-visuals



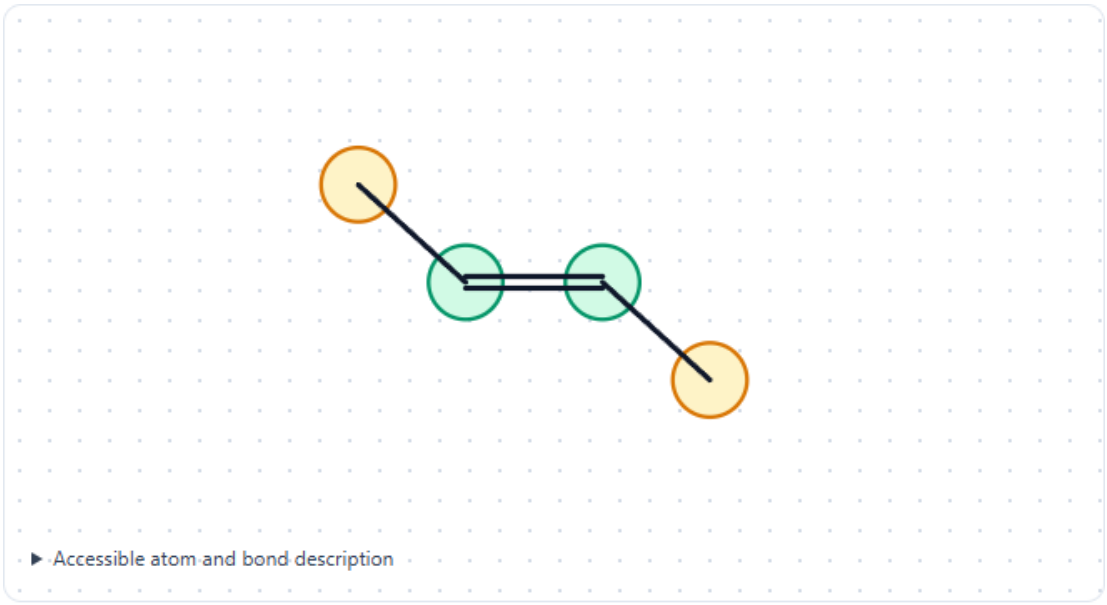
alkene:step-1 | 1. Compare the two ends

Each alkene carbon bears methyl and H. Methyl outranks H, and the methyl groups lie on opposite sides: E.

Displayed molecule ID: e-butene

{"highlights": {"prin": [1, 2], "sub": [0, 3]}, "revealName": true}

Screenshot: alkene-step-1



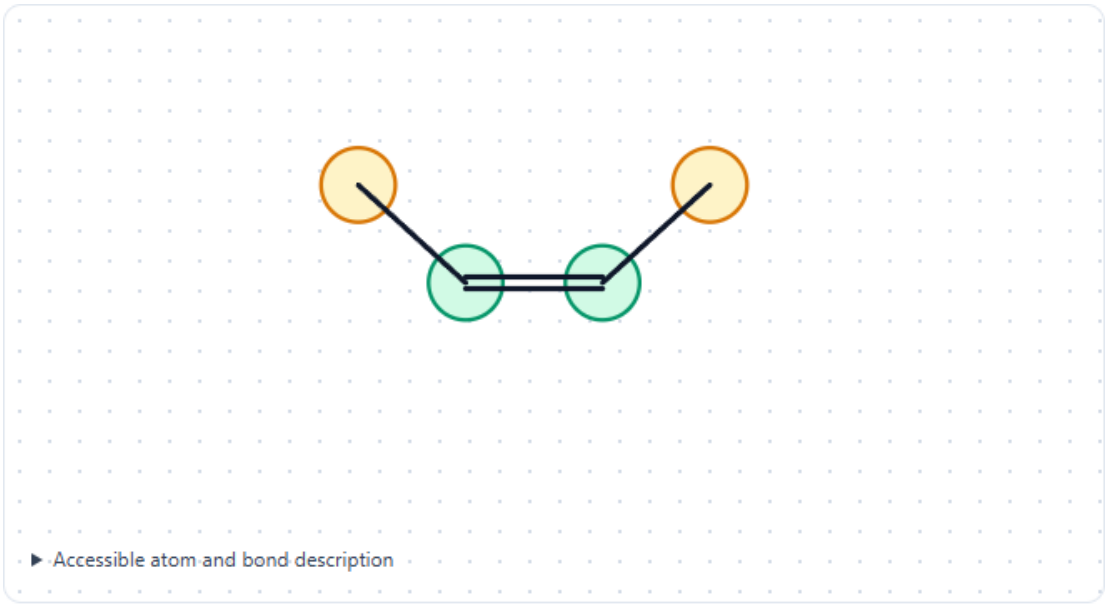
alkene:step-2 | 2. Same side

Both higher-priority methyl groups now lie on the same side: Z. Rotation about C=C would disrupt pi overlap, so these are configurational isomers.

Displayed molecule ID: z-butene

{"highlights": {"prin": [1, 2], "sub": [0, 3]}, "revealName": true}

Screenshot: alkene-step-2



optical | Optical activity, specific rotation, and resolution

Measure rotation; normalize units; keep configuration labels separate

optical:rule | Core Rules

Observed optical rotation αobs depends on concentration, path length, wavelength, temperature and solvent. In the common solution convention, [α] = αobs/(l × c), with l in dm and c in g/mL. If αobs is in degrees, the corresponding specific-rotation units are °·mL·g^{-1}·dm^{-1}. State wavelength, temperature and solvent when comparing measurements.

Visual card: Polarimetry, Specific Rotation [α]D, Enantiomeric Excess (ee) & Chiral Resolution

Chemistry diagram: Instrument schematic: rotation is measured after light passes through the sample; no measured angle is supplied.

[α]_D = α_obs / (c × l) | % ee = ([α]_mixture / [α]_pure) × 100 = |% R − % S|

[α] = α_obs / (c × l)

ee = %Major − %Minor

Racemate (50:50) → α = 0°

R/S ≠ (+)/(−) Sign!

Pure Enantiomer | 100% ee | Full specific rotation ±[α]D

Racemic Mixture | 50% R + 50% S (0% ee) | Optically inactive (α_obs = 0°)

Chiral Resolution | React racemate + pure chiral agent | Forms separable diastereomers!

Pathlength l (dm) & Conc c (g/mL) — — Doubling concentration c or tube length l doubles α_obs, but intrinsic specific rotation [α]_D stays constant!

Enantiomeric Excess (ee) — — Example: 75% (R) + 25% (S) gives 50% racemate (cancels out) + 50% net (R) excess → 50% optical purity.

optical:cue | Memory Cues

1 dm = 10 cm. Positive (+), dextrorotatory, means clockwise/rightward rotation as viewed by an observer looking toward the light source through the instrument; negative (−), levorotatory, is counterclockwise/leftward. Older texts use lowercase d/l for these signs. Uppercase D/L and R/S describe configuration and do not predict the rotation sign.

Visual card: Quick Polarimetry & Optical Purity Calculations

Chemistry diagram: Illustrative graph focus from the corresponding walkthrough explanation; no transformation is implied.

What polarimetry question is the MCAT asking?

Concentration or Tube Length Doubled? — α_obs Doubles; [α]_D Stays Constant! — [α]_D is an intrinsic physical constant at fixed T and λ (like density or boiling point).

Given % ee (e.g., 60% ee of + isomer) — % Major = 50 + (ee/2) = 80%; % Minor = 20% — Always split the remaining (100 − ee)% equally between the two enantiomers (40% = 20% + 20%).

How to Separate a 50:50 Racemate? — Convert to Diastereomeric Salts or Use Chiral HPLC — Standard silica TLC or distillation CANNOT separate enantiomers; react with pure (R)-base/acid to make diastereomers.

optical:trap | Common Mistakes / MCAT Traps

Do not put a length in cm directly into a formula requiring dm. Concentration in g per 100 mL needs conversion to g/mL in this convention. A racemate gives zero net rotation under matched ordinary conditions, but zero rotation alone does not prove a racemate. Specific rotation is not just an unnormalized angle.

Visual card: Assuming (R) Means Dextrorotatory (+) or That [α]D Doubles with Concentration

Common Trap — — Assuming an (R)-enantiomer always rotates light clockwise (+) or doubling [α]_D when c is doubled.

Correct Reasoning — — R/S is a paper geometry rule while (+)/(−) is an experimental measurement; they are completely independent!

optical:app | MCAT Applications

For an ideal mixture of only two enantiomers under matched conditions, ee = |%major − %minor| and observed rotation is the signed enantiomer excess times the pure-enantiomer observed rotation at the same total concentration and path length. A 75:25 mixture has 50% ee; if the major pure enantiomer gives +20°, the mixture gives +10°. These are illustrative practice data.

Visual card: Sugar Mutarotation, Racemization at α-Carbons & Enantiopure Drug Synthesis

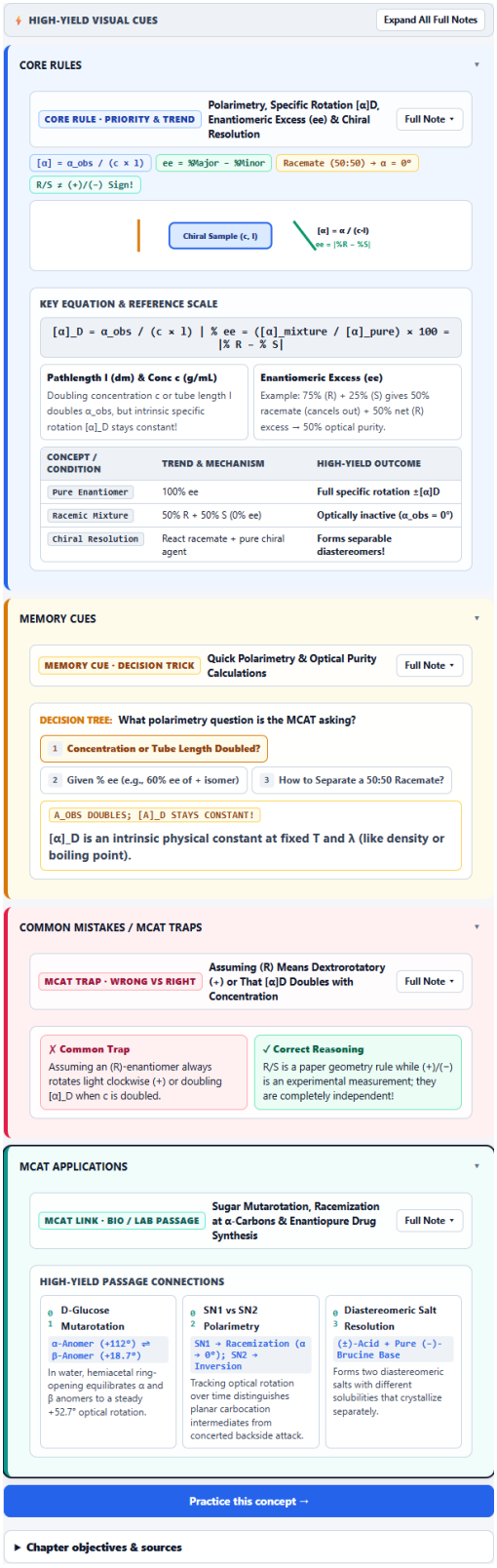
Chemistry diagram: Illustrative graph focus from the corresponding walkthrough explanation; no transformation is implied.

D-Glucose Mutarotation — α-Anomer (+112°) ⇌ β-Anomer (+18.7°) — In water, hemiacetal ring-opening equilibrates α and β anomers to a steady +52.7° optical rotation.

SN1 vs SN2 Polarimetry — SN1 → Racemization (α → 0°); SN2 → Inversion — Tracking optical rotation over time distinguishes planar carbocation intermediates from concerted backside attack.

Diastereomeric Salt Resolution — (±)-Acid + Pure (−)-Brucine Base — Forms two diastereomeric salts with different solubilities that crystallize separately.

Screenshot: optical-visuals



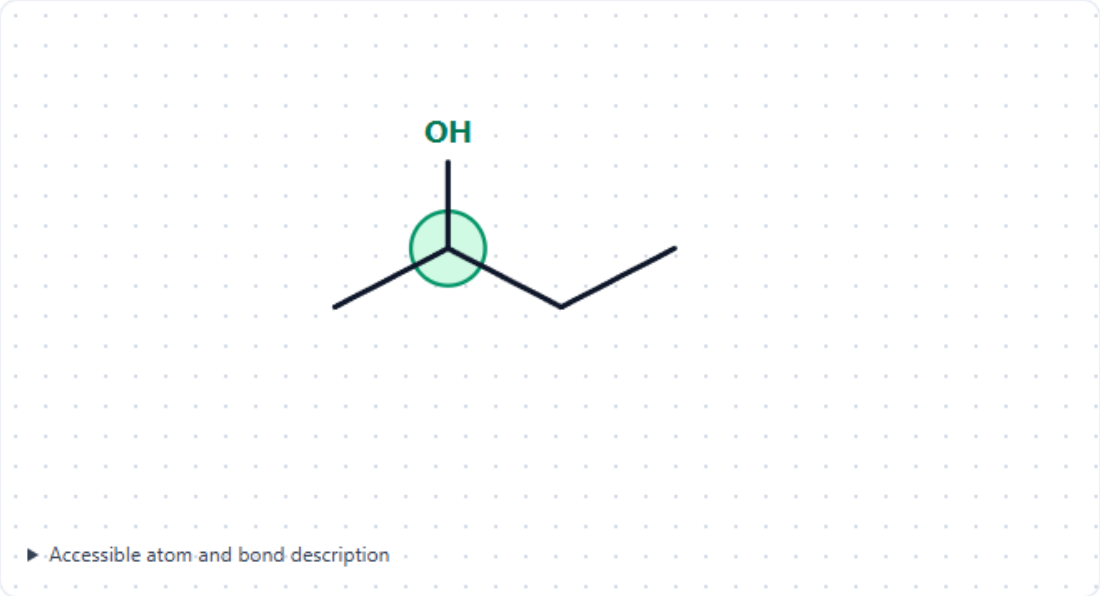
optical:step-1 | 1. Geometry is not optical sign

Butan-2-ol can exist as enantiomers. Its connectivity alone does not specify which enantiomer is present, and an R/S descriptor does not predict (+)/(−). Measure the rotation experimentally.

Displayed molecule ID: butanol2

{"highlights": {"prin": [1]}, "revealName": true}

Screenshot: optical-step-1



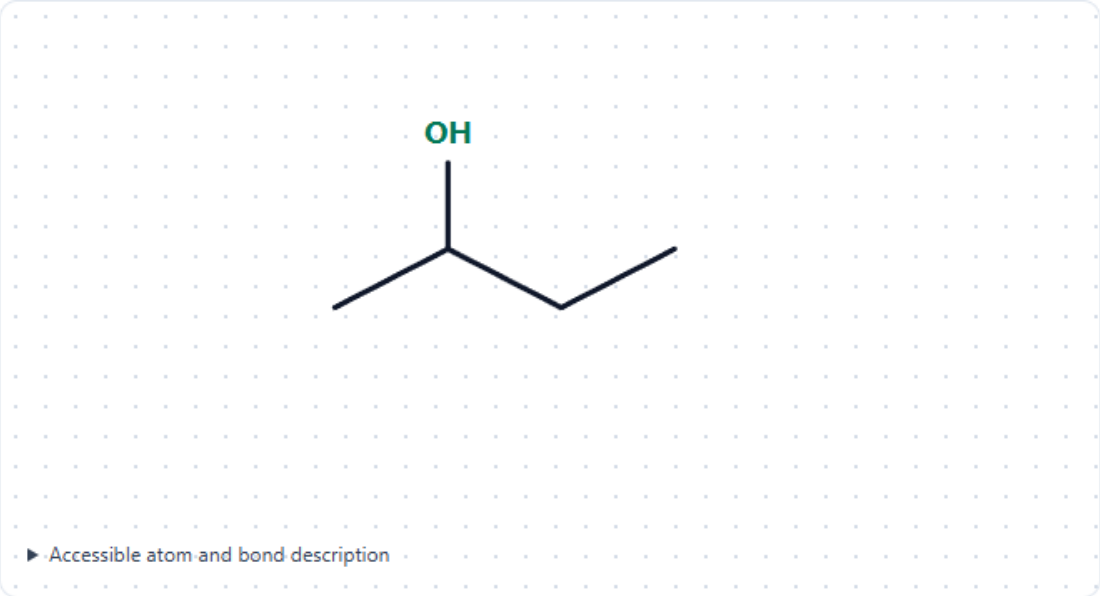
optical:step-2 | 2. Normalize a measurement

Illustrative data: αobs = +1.2°, c = 0.040 g/mL and l = 10 cm = 1.0 dm. Then [α] = +1.2/(0.040 × 1.0) = +30 °·mL·g^{-1}·dm^{-1}. These are practice numbers, not a tabulated value for the displayed connectivity.

Displayed molecule ID: butanol2

{"highlights": {}, "revealName": false}

Screenshot: optical-step-2



optical:step-3 | 3. Resolve a racemate

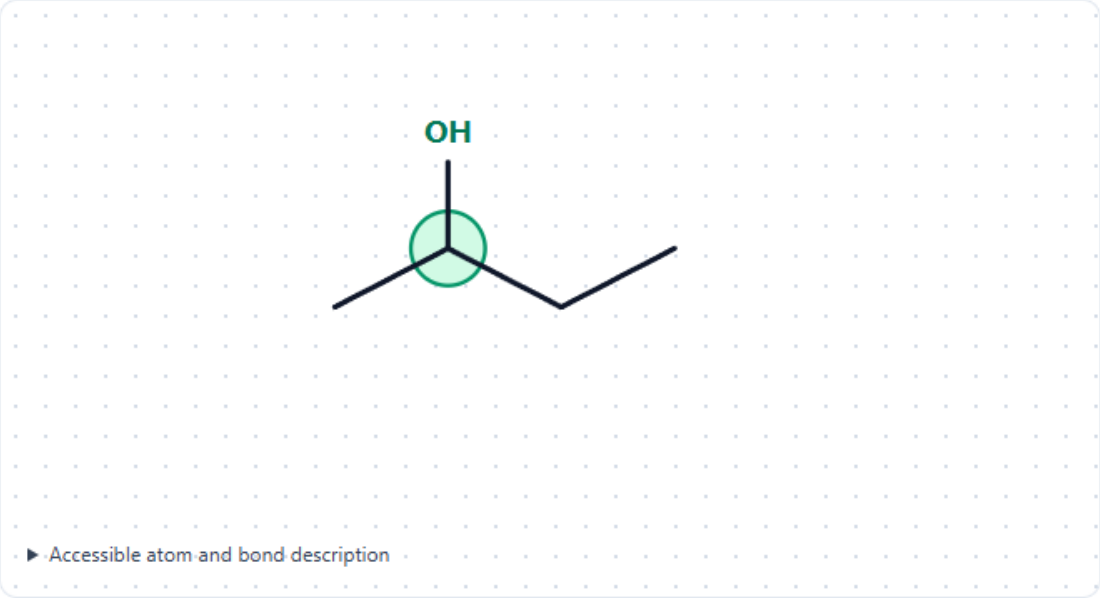
For a suitable substrate, use a single-enantiomer resolving agent to make diastereomeric salts or derivatives. Separate them by a suitable property difference, then regenerate the starting enantiomers. Ordinary achiral distillation does not generally

separate enantiomers, which share the same boiling point under achiral conditions.

Displayed molecule ID: butanol2

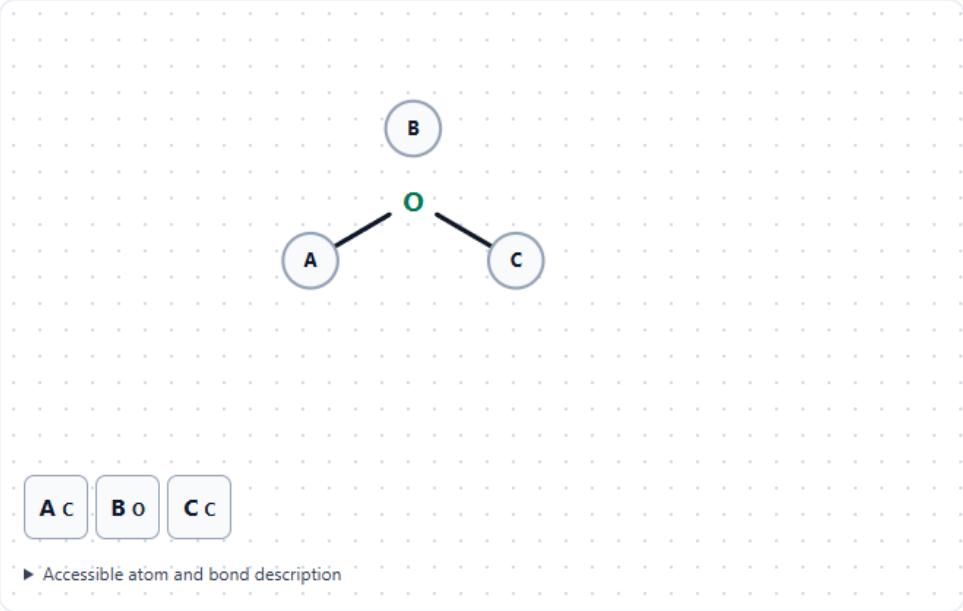
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Screenshot: optical-step-3



isomer-pair | Classify an isomer pair

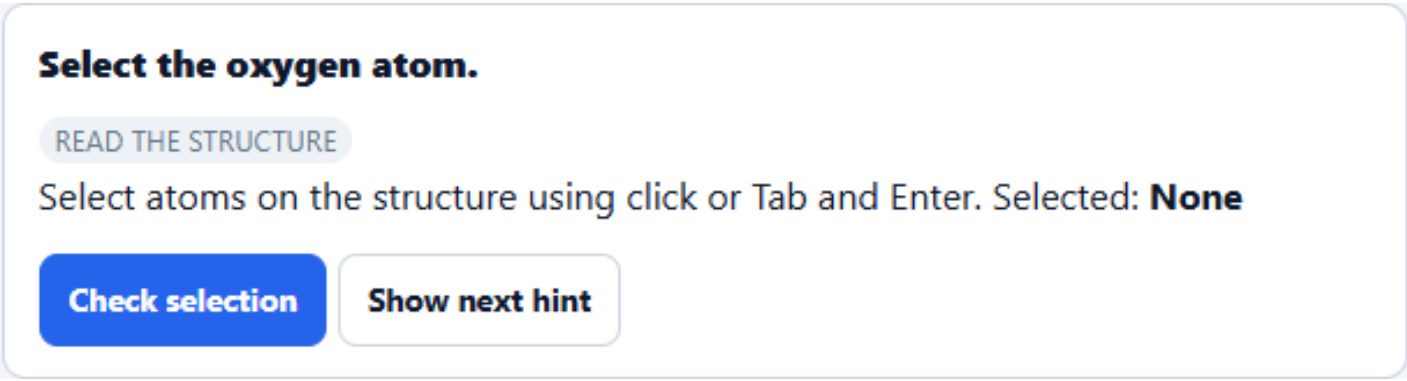
Screenshot: isomer-pair-stage-1-molecule



isomer-pair:stage-1 | Select the oxygen atom.

Answer type: atoms

Screenshot: isomer-pair-stage-1-question



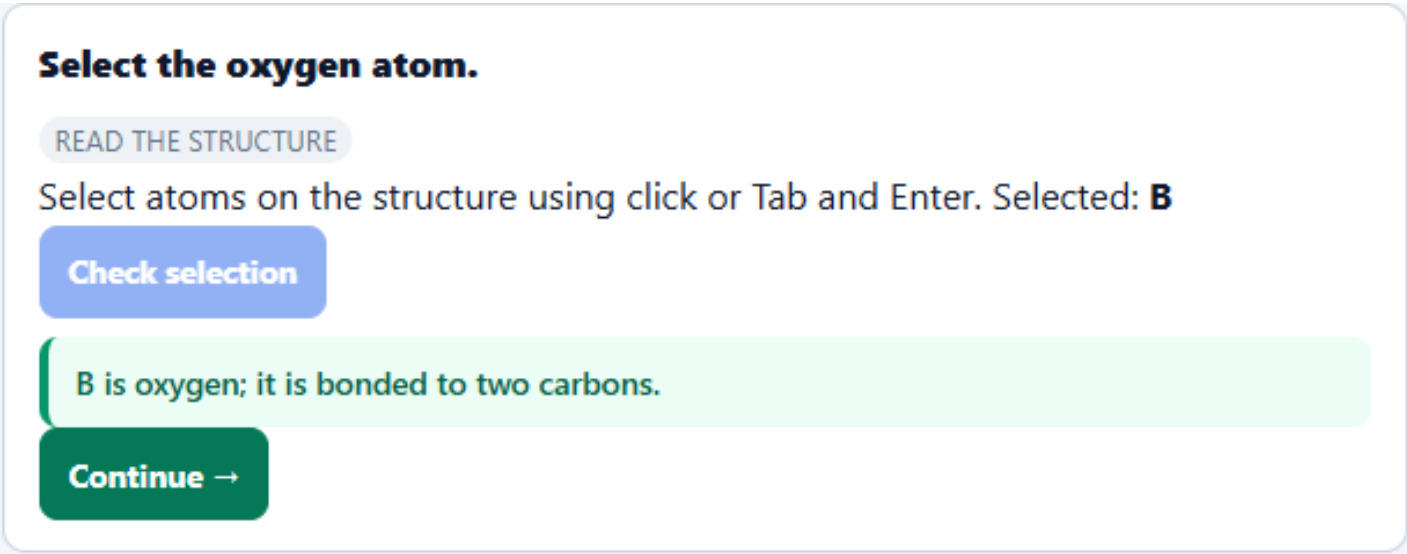
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1]}}

Hint 1: Read the element label.

Hint 2: The oxygen is between the carbon endpoints.

Explanation: B is oxygen; it is bonded to two carbons.

Screenshot: isomer-pair-stage-1-feedback



isomer-pair:stage-2 | Compared with ethanol (C2H6O), this compound is a...

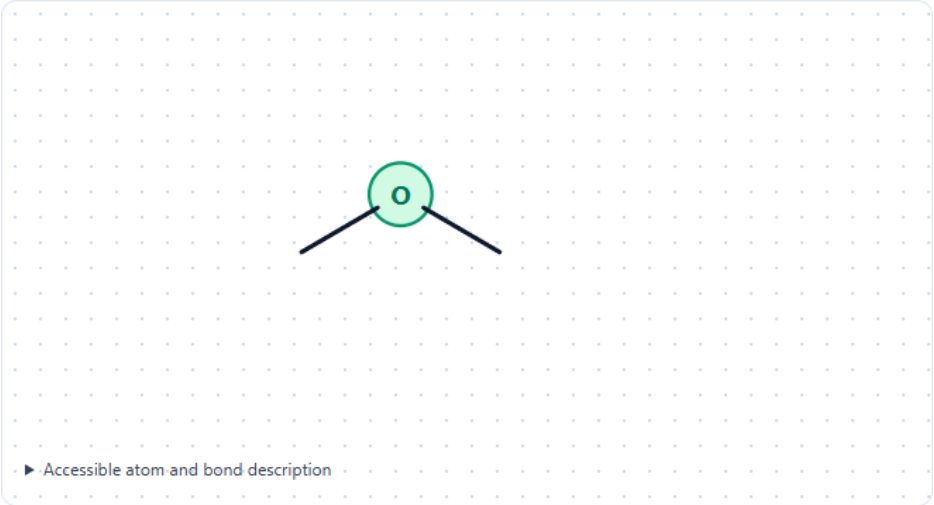
Answer type: choice

Option 0 (A): Constitutional isomer

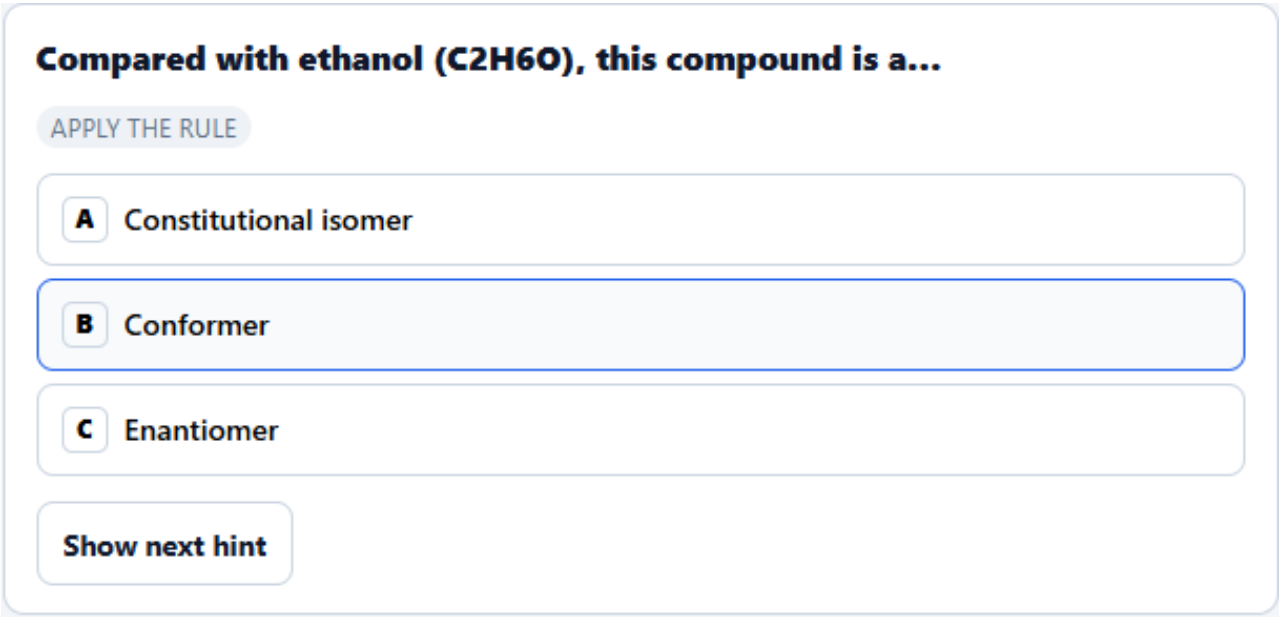
Option 1 (B): Conformer

Option 2 (C): Enantiomer

Screenshot: isomer-pair-stage-2-molecule



Screenshot: isomer-pair-stage-2-question



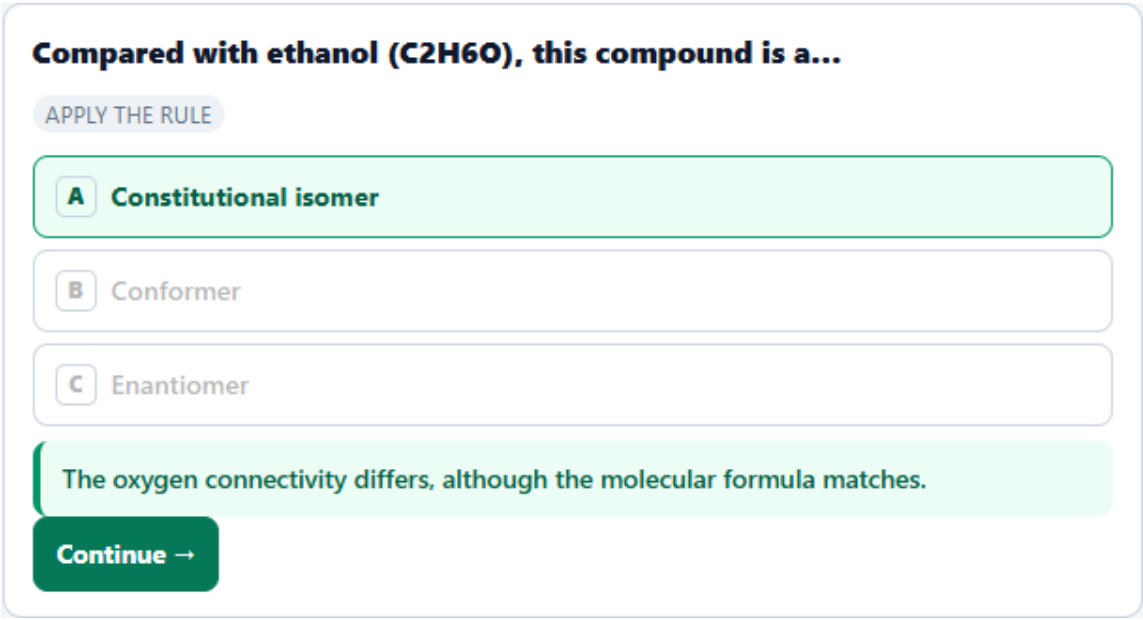
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Compare bonds before spatial arrangements.

Hint 2: Ethanol contains an O–H bond.

Explanation: The oxygen connectivity differs, although the molecular formula matches.

Screenshot: isomer-pair-stage-2-feedback



isomer-pair:stage-3 | Which can donate an O–H hydrogen bond?

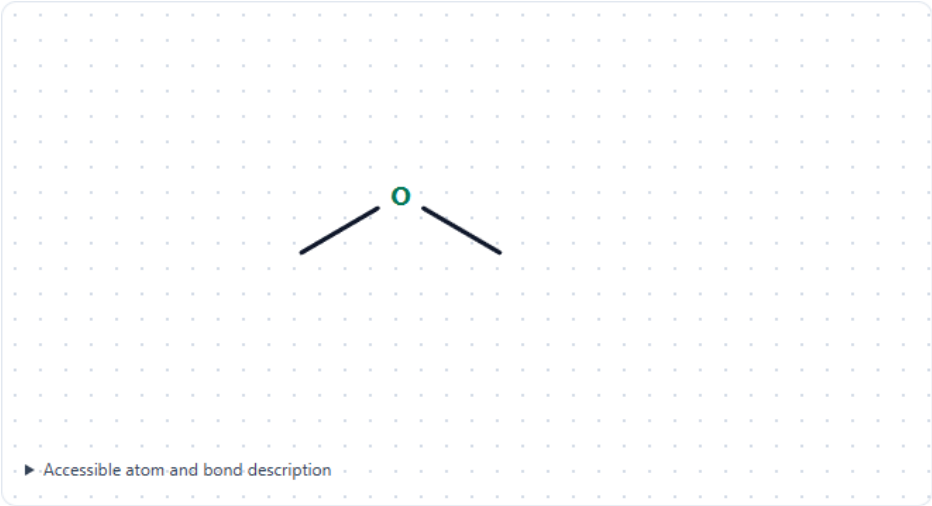
Answer type: choice

Option 0 (A): Both

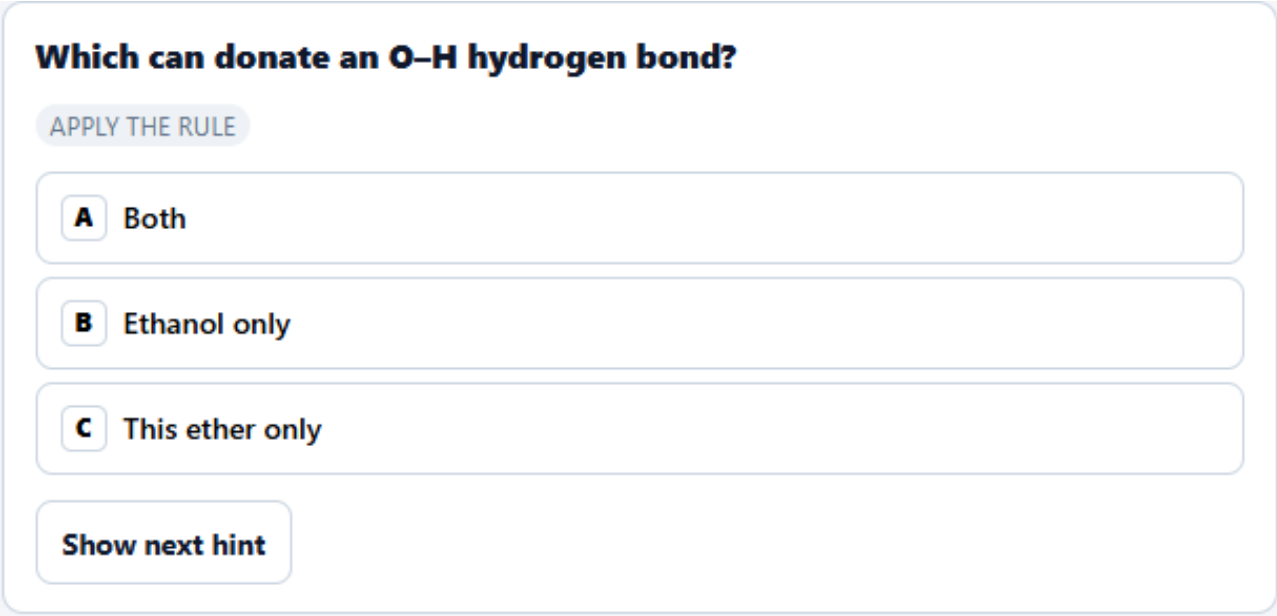
Option 1 (B): Ethanol only

Option 2 (C): This ether only

Screenshot: isomer-pair-stage-3-molecule



Screenshot: isomer-pair-stage-3-question



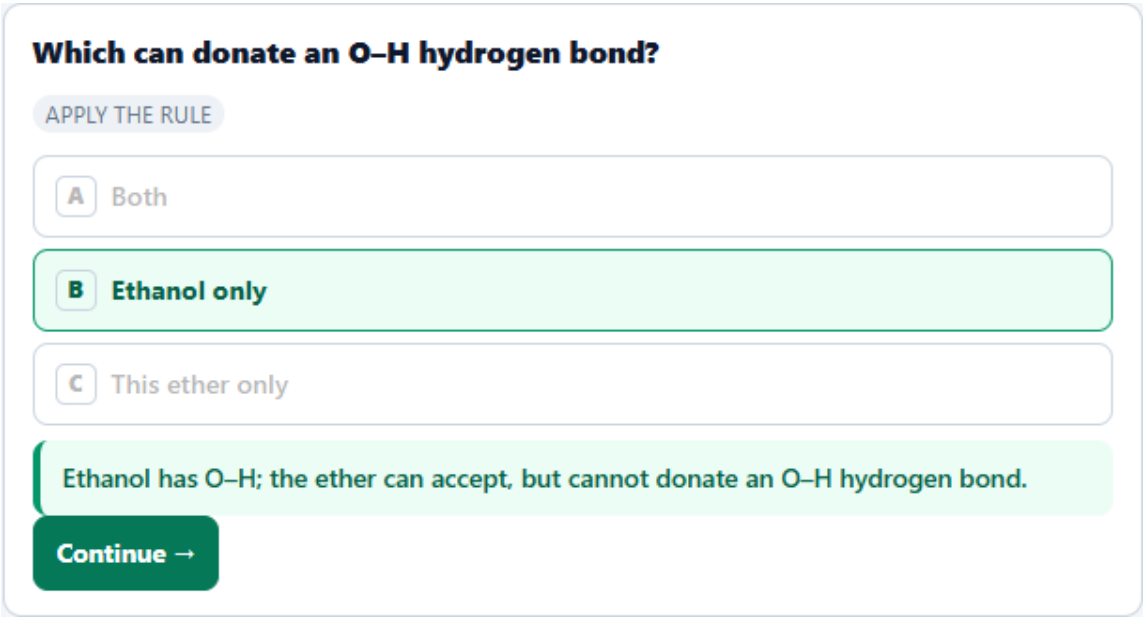
{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Donation requires H bonded to an electronegative atom.

Hint 2: Look for an O–H bond.

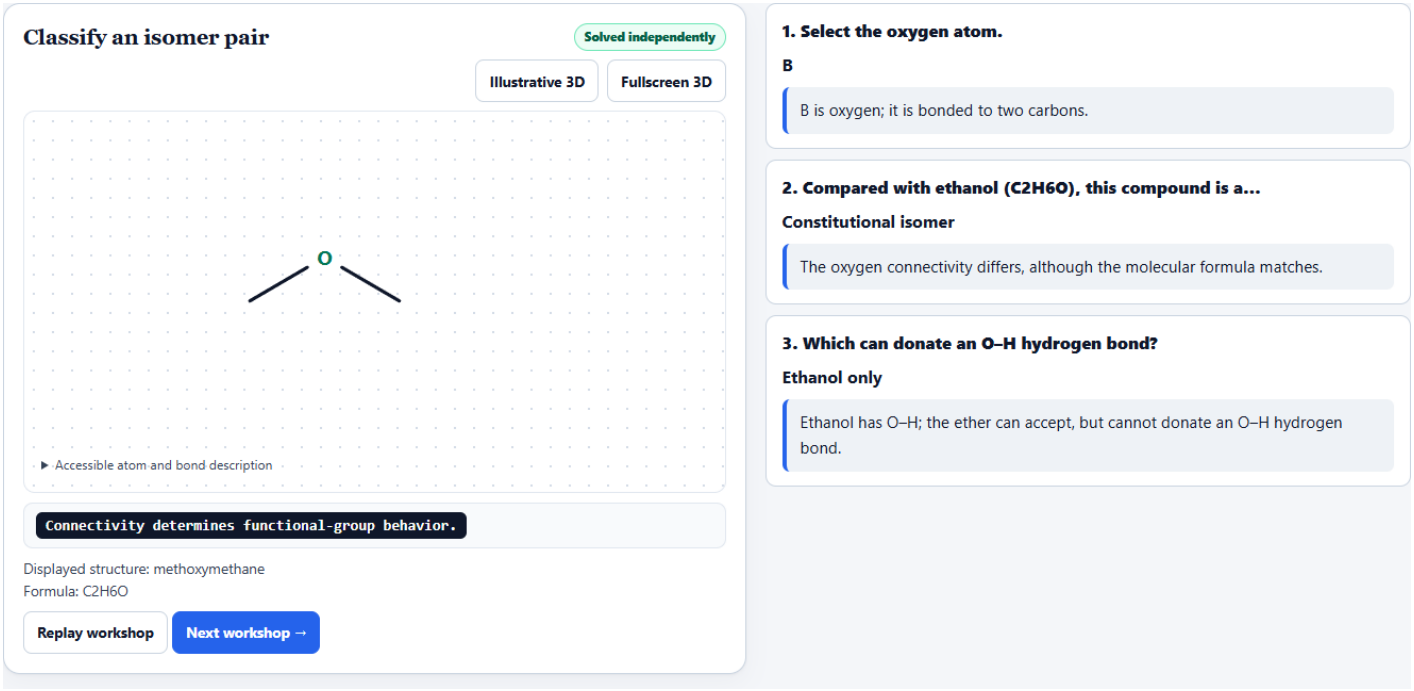
Explanation: Ethanol has O–H; the ether can accept, but cannot donate an O–H hydrogen bond.

Screenshot: isomer-pair-stage-3-feedback



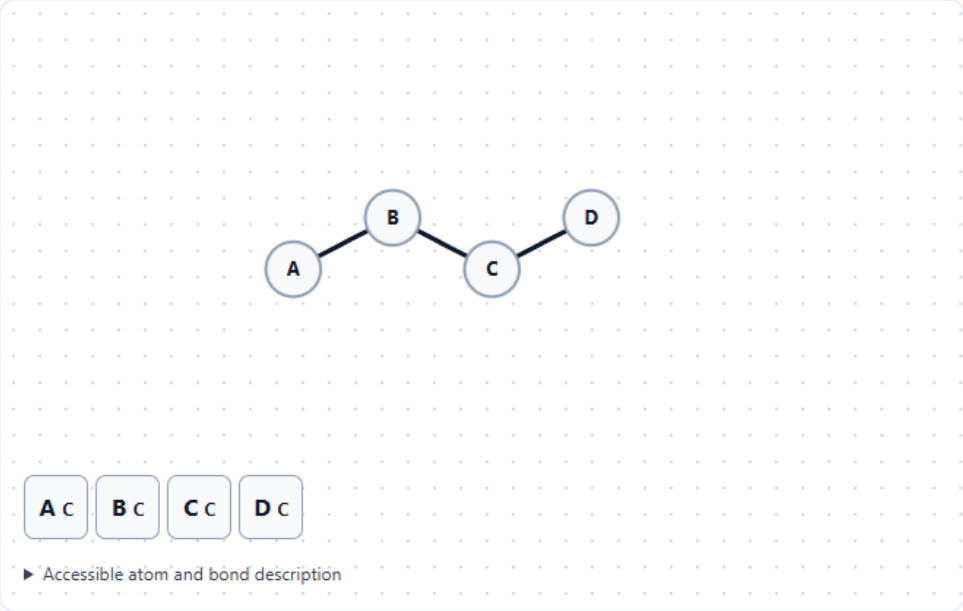
Completion summary: Connectivity determines functional-group behavior.

Screenshot: isomer-pair-complete

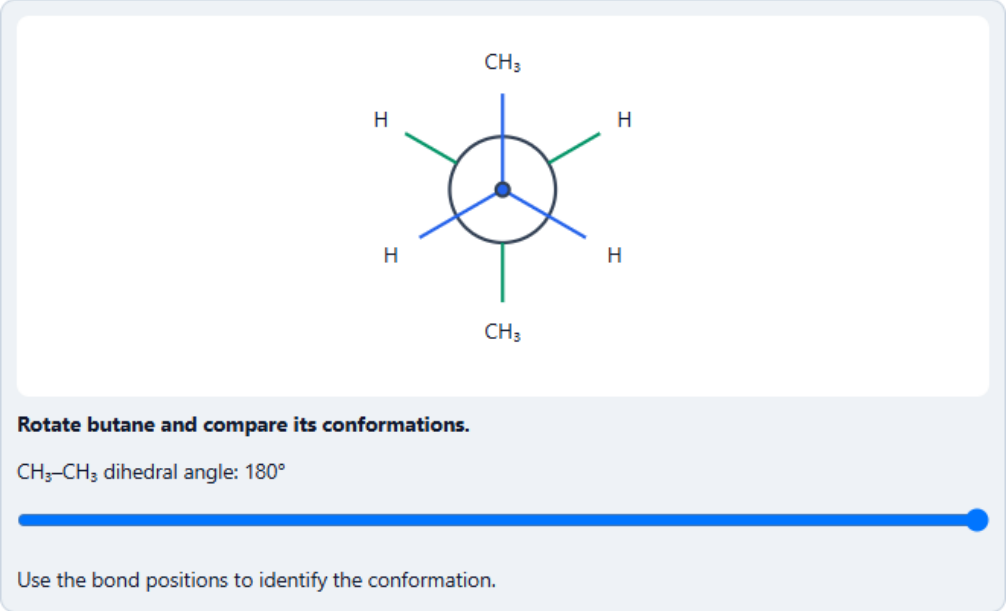


butane-rotation | Reason about conformational energy

Screenshot: butane-rotation-stage-1-molecule



Screenshot: butane-rotation-diagram

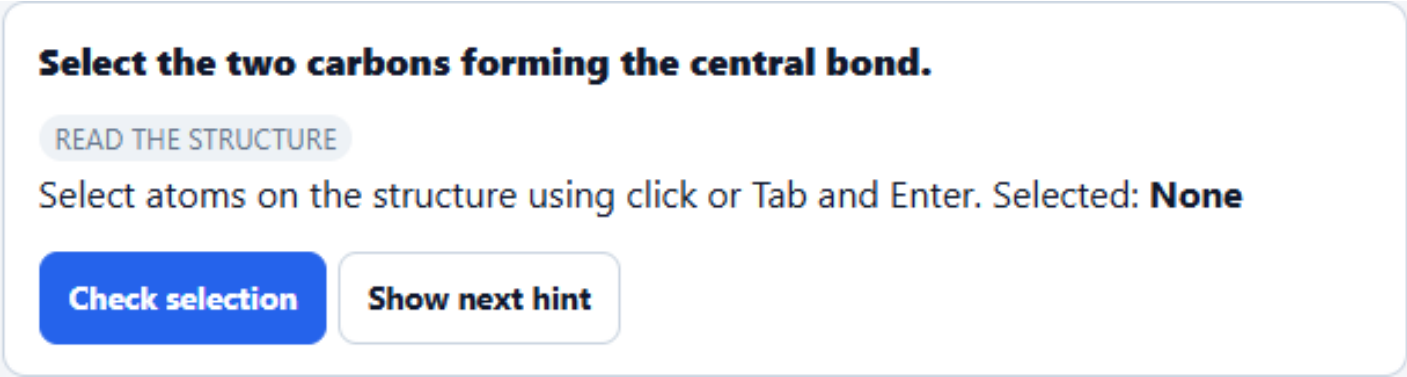


{"diagramSpecification": {"type": "newman", "caption": "Rotate butane and compare its conformations."}}

butane-rotation:stage-1 | Select the two carbons forming the central bond.

Answer type: atoms

Screenshot: butane-rotation-stage-1-question



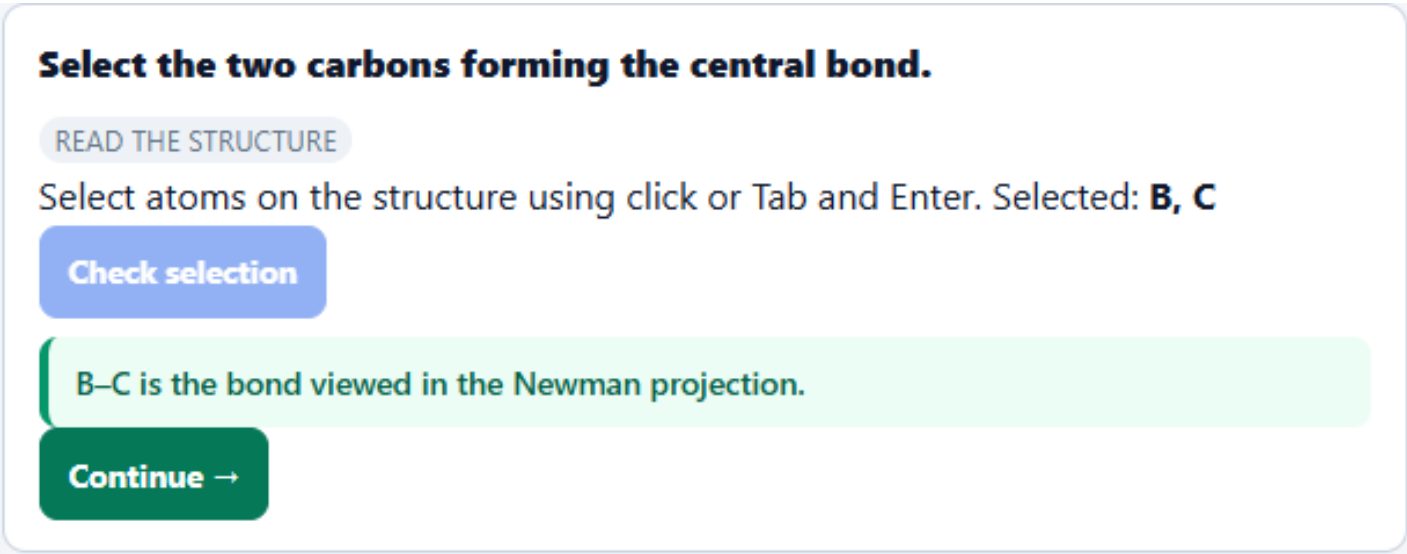
{"authorAnswer": [1, 2], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 2]}}

Hint 1: Find the four-carbon chain.

Hint 2: Choose the two internal atoms.

Explanation: B–C is the bond viewed in the Newman projection.

Screenshot: butane-rotation-stage-1-feedback



butane-rotation:stage-2 | At a 180° methyl–methyl dihedral, the conformation is…

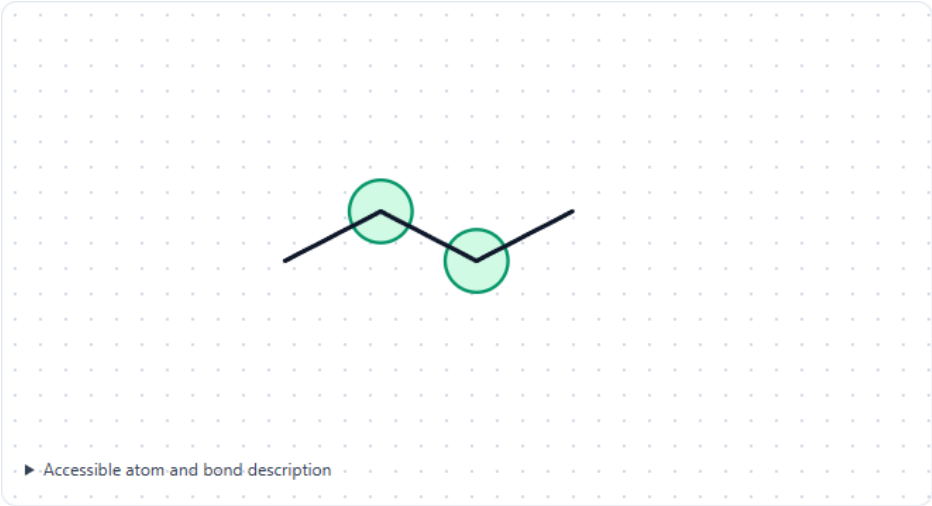
Answer type: choice

Option 0 (A): Anti

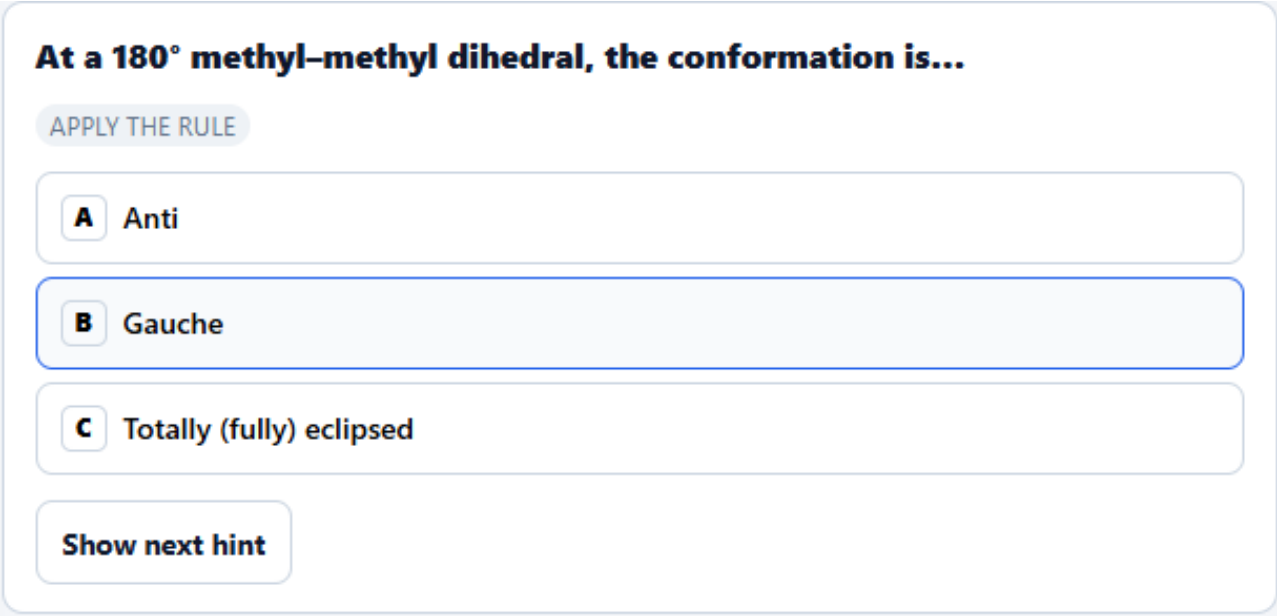
Option 1 (B): Gauche

Option 2 (C): Totally (fully) eclipsed

Screenshot: butane-rotation-stage-2-molecule



Screenshot: butane-rotation-stage-2-question



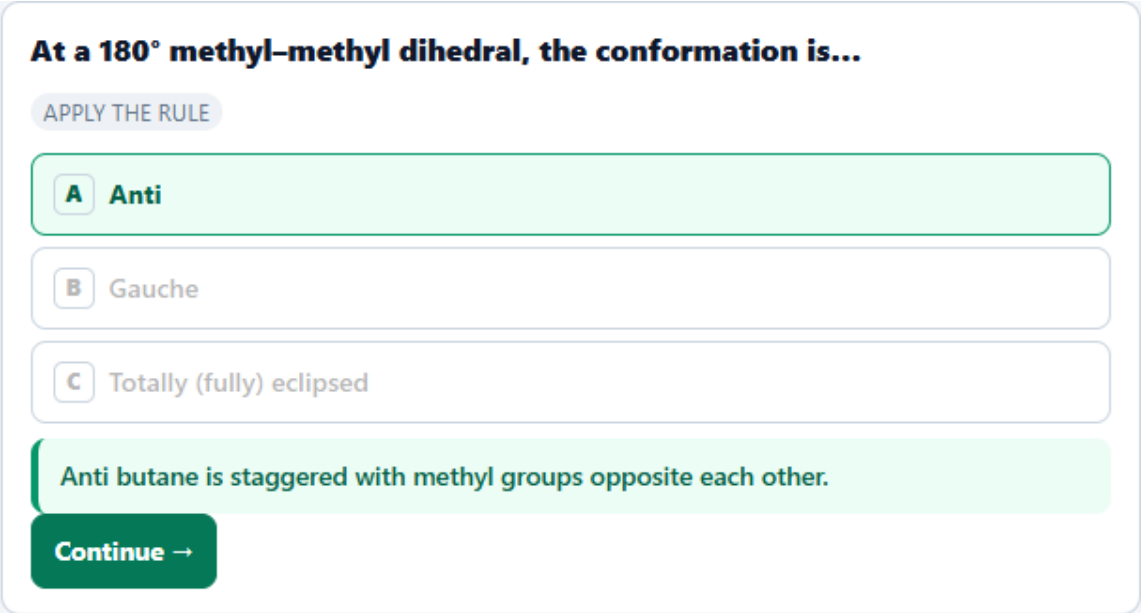
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Distinguish staggered from eclipsed.

Hint 2: Opposite groups are separated by half a turn.

Explanation: Anti butane is staggered with methyl groups opposite each other.

Screenshot: butane-rotation-stage-2-feedback



This stage uses a different molecule (connectivity shown).

butane-rotation:stage-3 | For cyclohexane (connectivity shown), a chair flip changes...

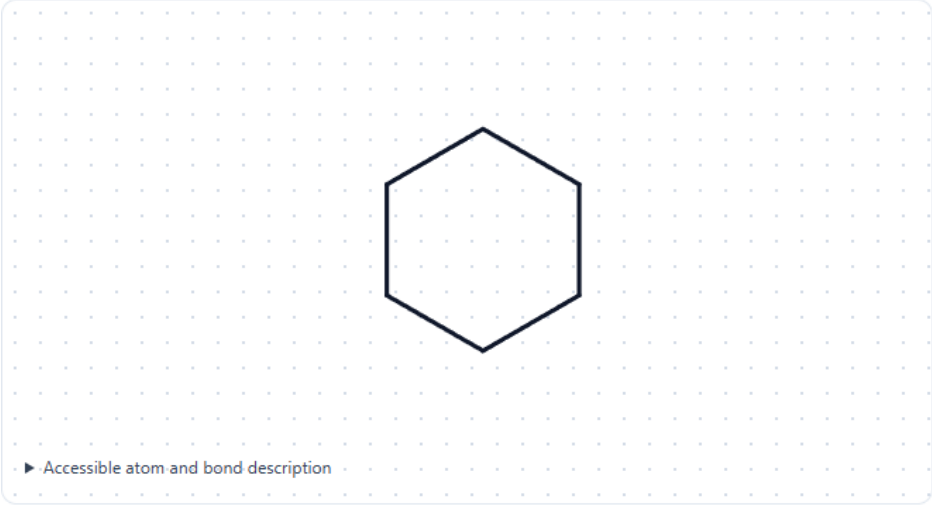
Answer type: choice

Option 0 (A): Up into down

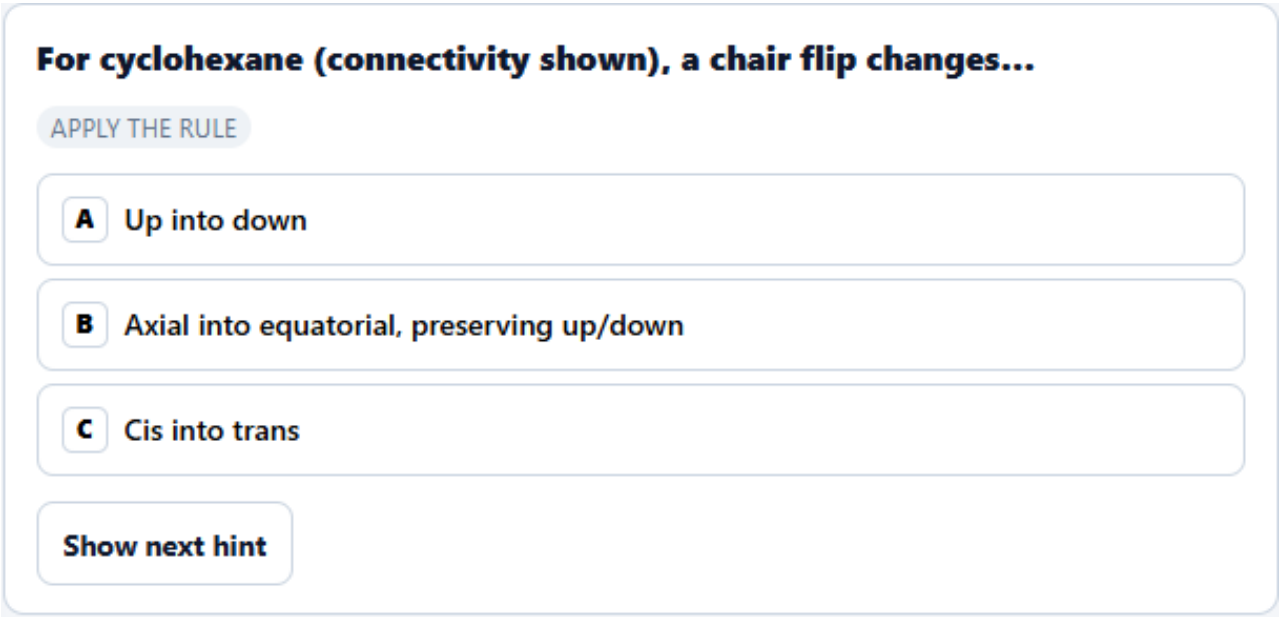
Option 1 (B): Axial into equatorial, preserving up/down

Option 2 (C): Cis into trans

Screenshot: butane-rotation-stage-3-molecule



Screenshot: butane-rotation-stage-3-question



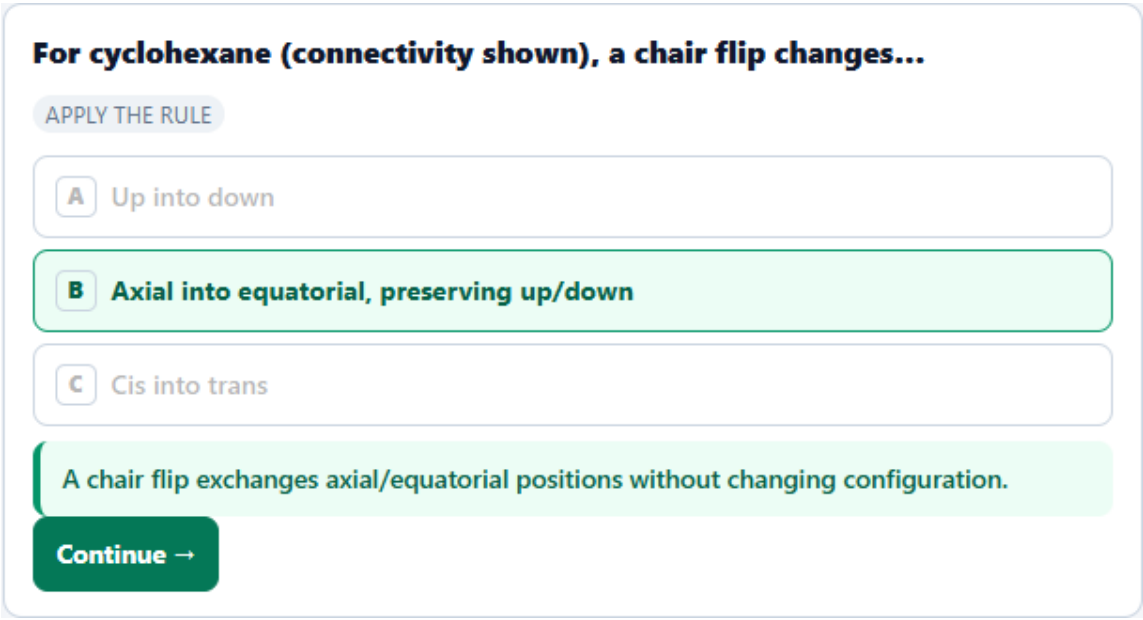
{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Track the bond orientation relative to the ring.

Hint 2: Up/down and axial/equatorial are different labels.

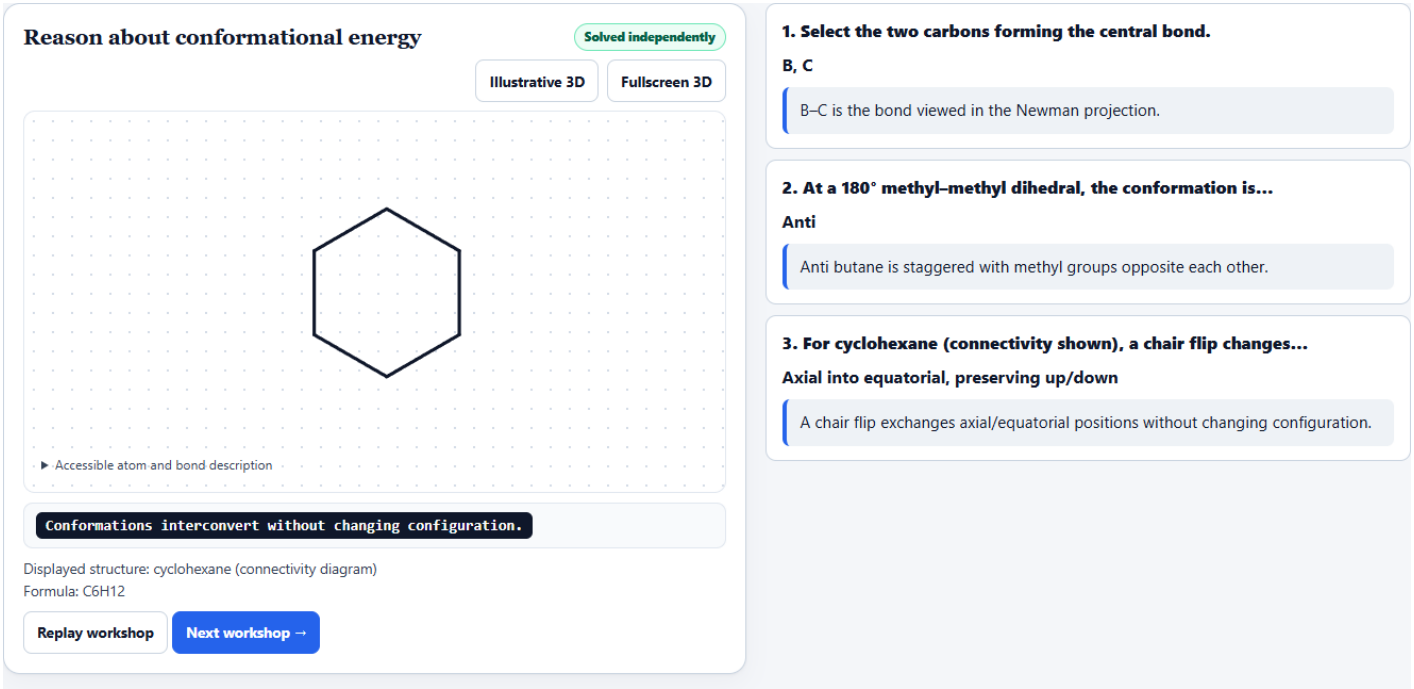
Explanation: A chair flip exchanges axial/equatorial positions without changing configuration.

Screenshot: butane-rotation-stage-3-feedback



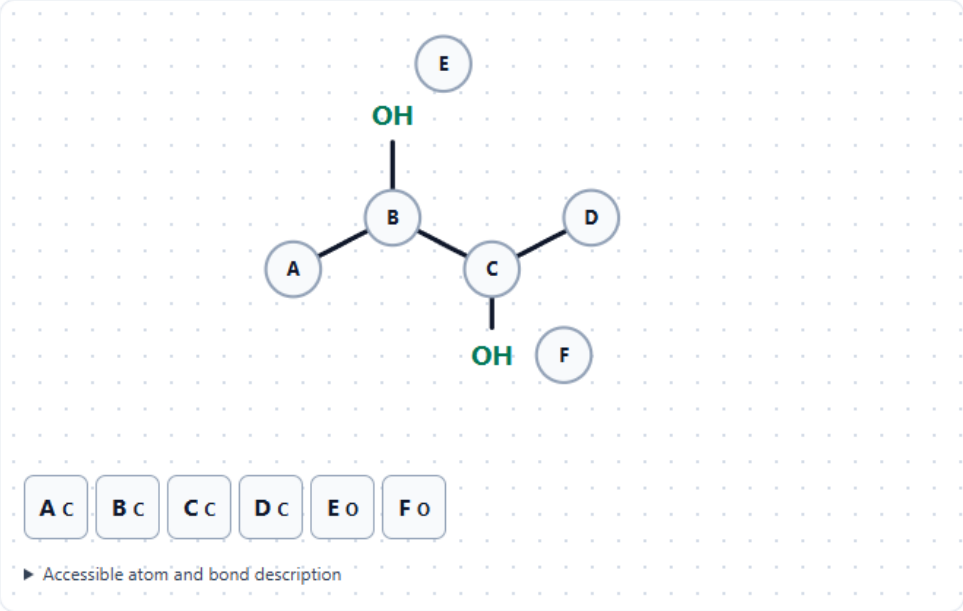
Completion summary: Conformations interconvert without changing configuration.

Screenshot: butane-rotation-complete



stereocenters | Count stereoisomers with symmetry

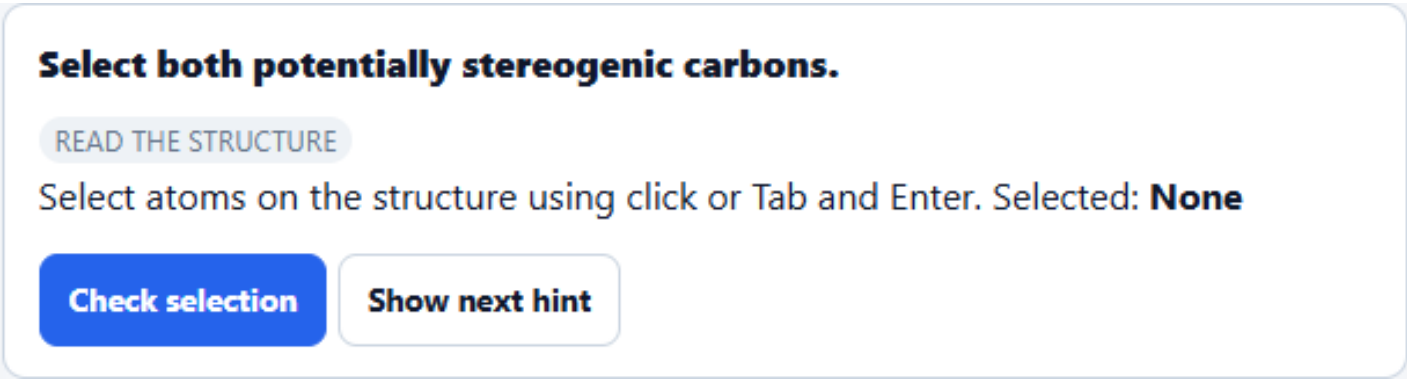
Screenshot: stereocenters-stage-1-molecule



stereocenters:stage-1 | Select both potentially stereogenic carbons.

Answer type: atoms

Screenshot: stereocenters-stage-1-question



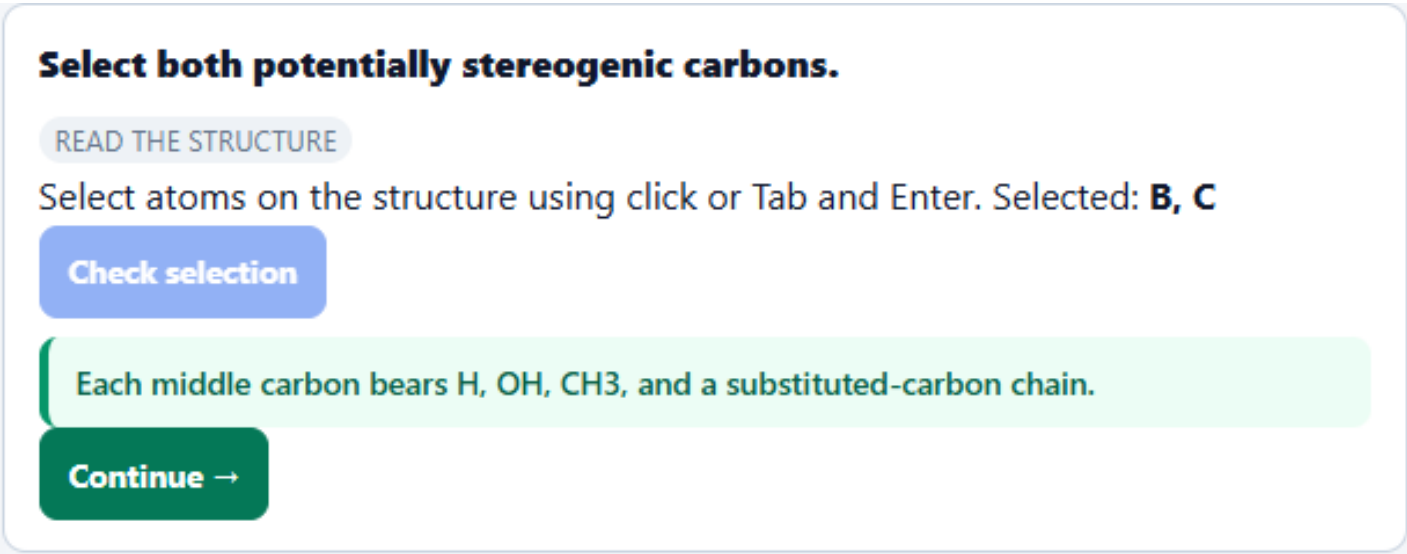
{"authorAnswer": [1, 2], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 2]}}

Hint 1: Look for tetrahedral atoms with four different groups.

Hint 2: The OH-bearing carbons are the candidates.

Explanation: Each middle carbon bears H, OH, CH3, and a substituted-carbon chain.

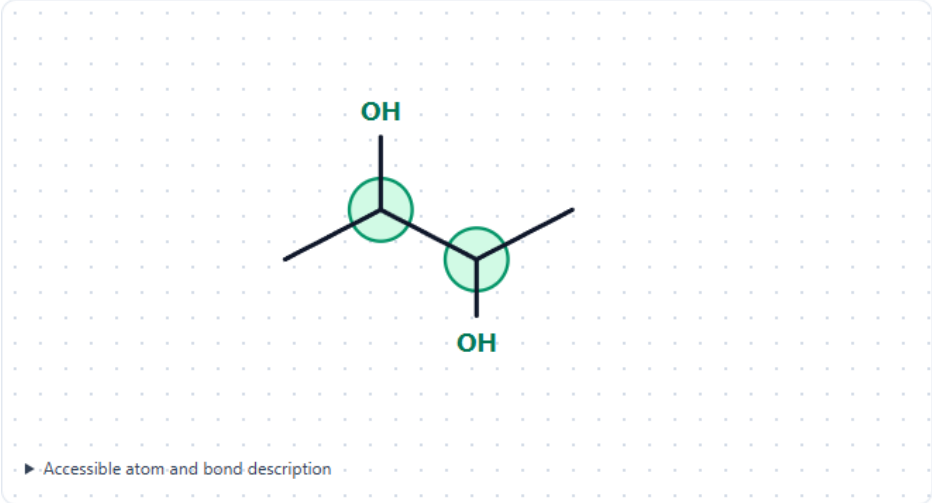
Screenshot: stereocenters-stage-1-feedback



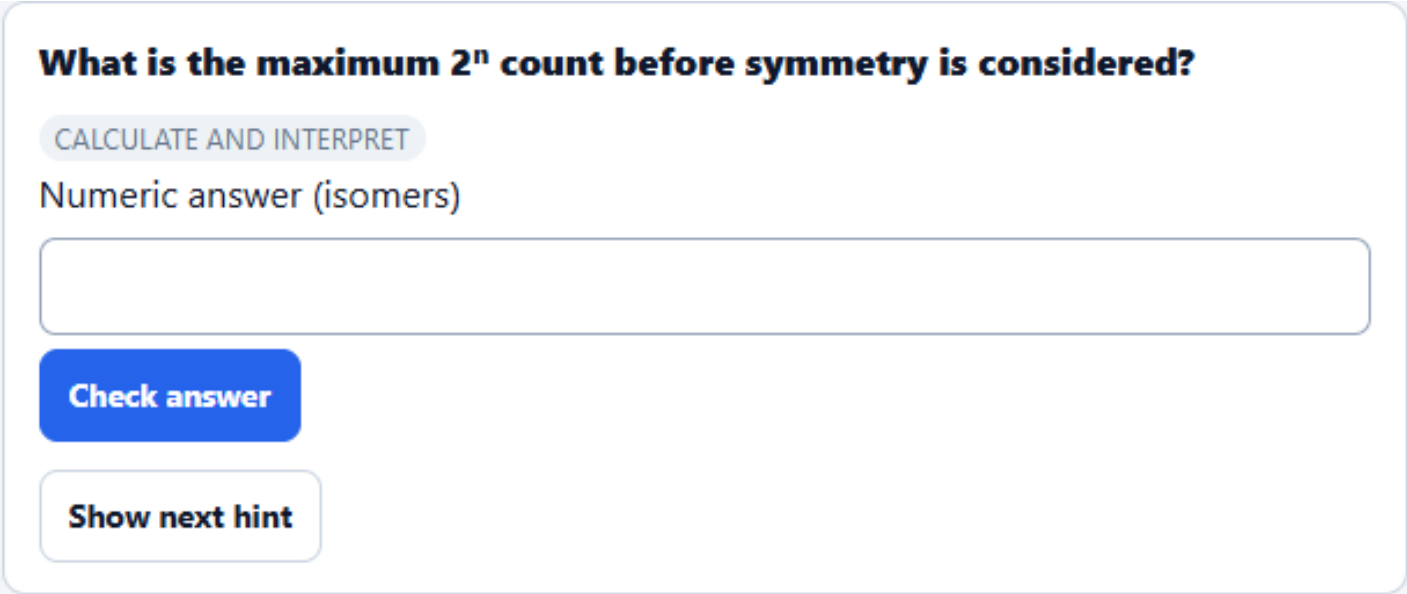
stereocenters:stage-2 | What is the maximum 2^n count before symmetry is considered?

Answer type: number | unit: isomers

Screenshot: stereocenters-stage-2-molecule



Screenshot: stereocenters-stage-2-question



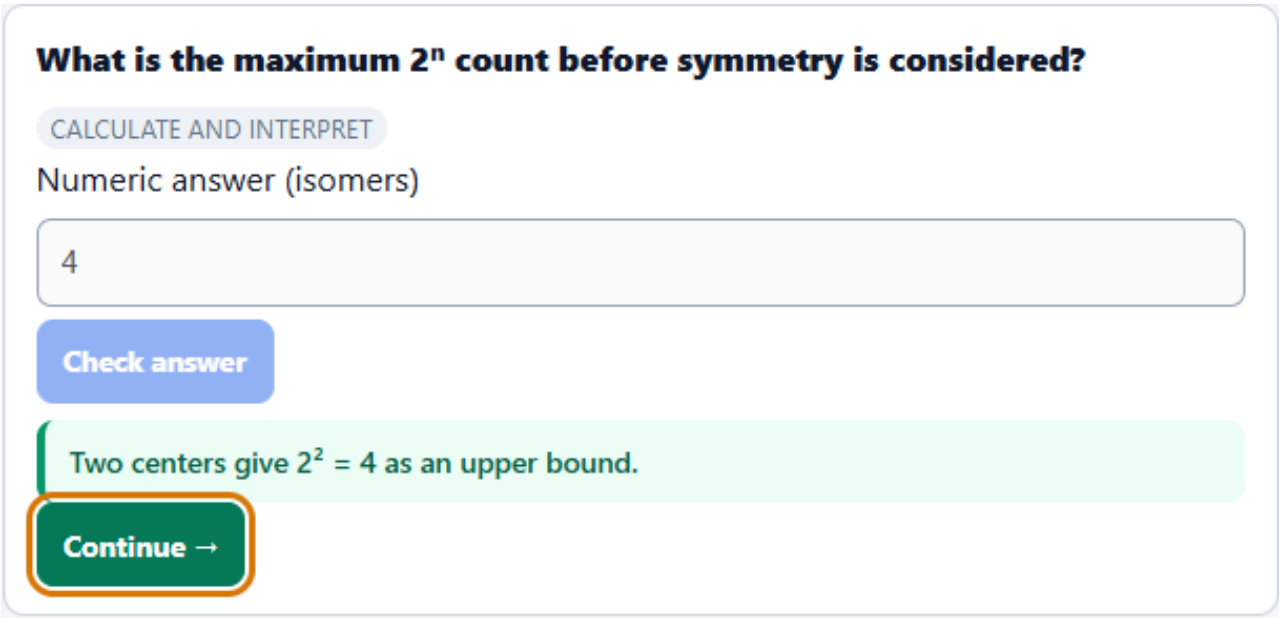
{"authorAnswer": 4, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: n is the number of independent stereocenters.

Hint 2: Raise two to the power of two.

Explanation: Two centers give $2^2 = 4$ as an upper bound.

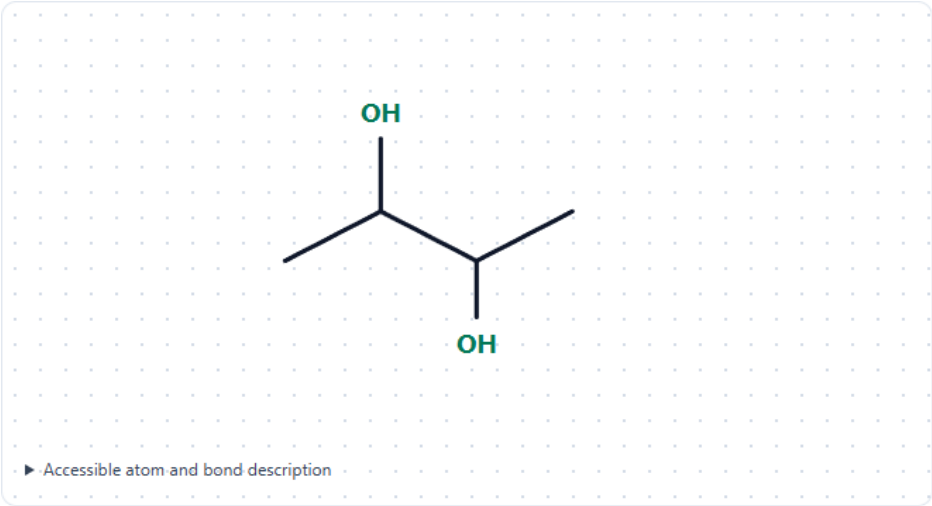
Screenshot: stereocenters-stage-2-feedback



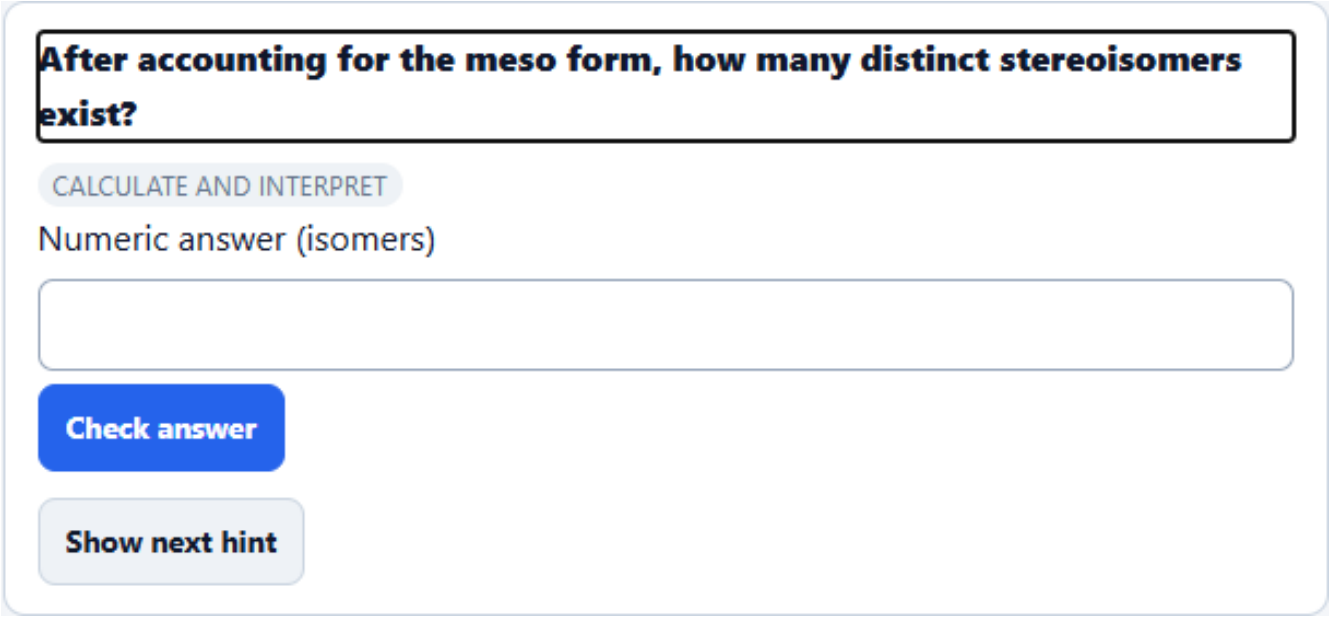
stereocenters:stage-3 | After accounting for the meso form, how many distinct stereoisomers exist?

Answer type: number | unit: isomers

Screenshot: stereocenters-stage-3-molecule



Screenshot: stereocenters-stage-3-question



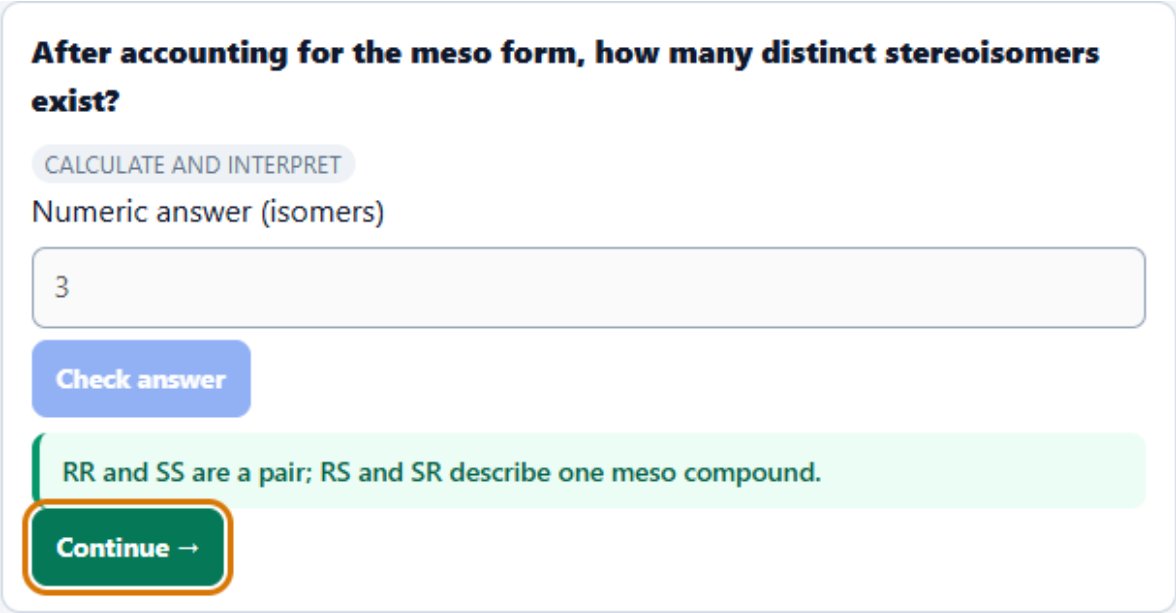
{"authorAnswer": 3, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Check whether exchanging the equivalent ends duplicates a structure.

Hint 2: There is one achiral meso form plus one mirror-image pair.

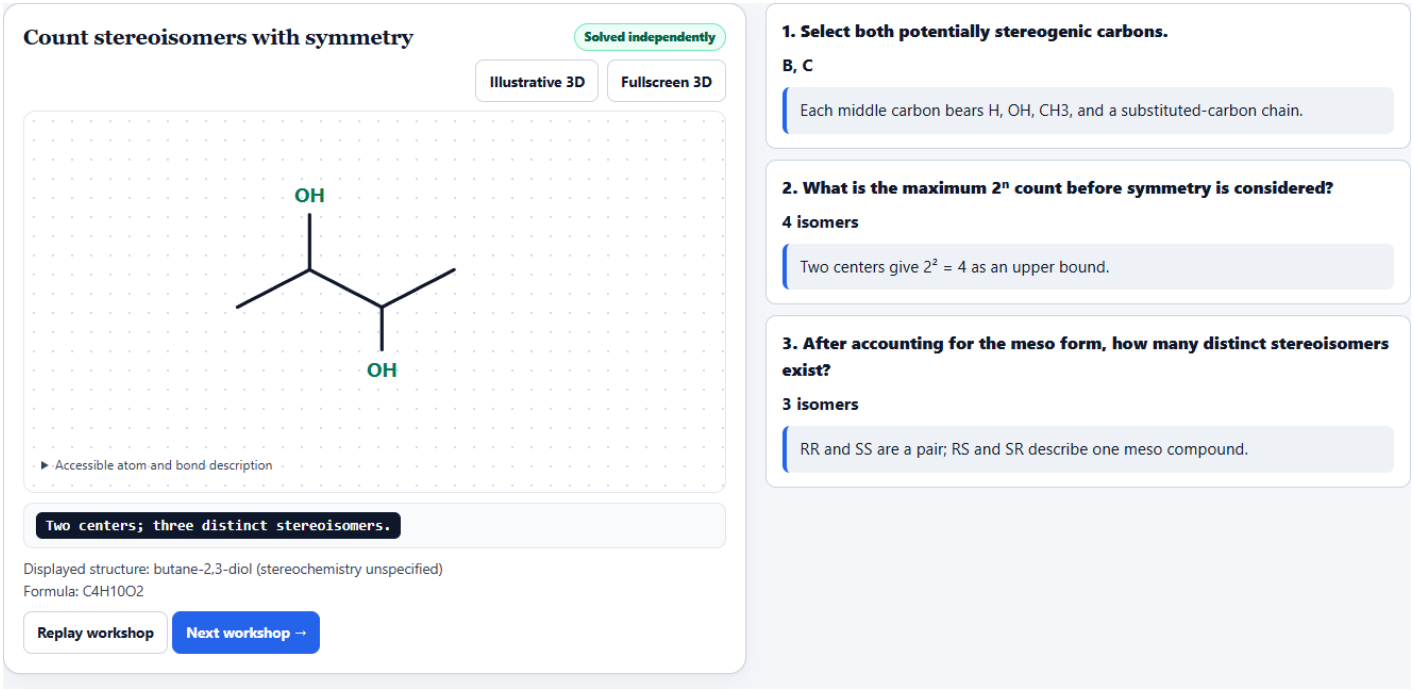
Explanation: RR and SS are a pair; RS and SR describe one meso compound.

Screenshot: stereocenters-stage-3-feedback



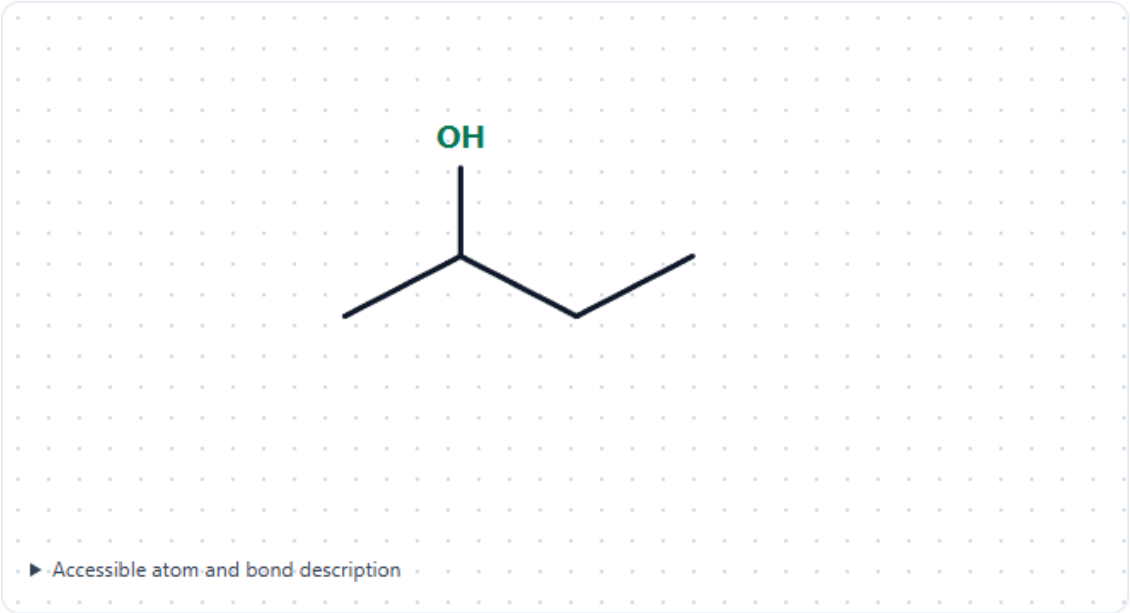
Completion summary: Two centers; three distinct stereoisomers.

Screenshot: stereocenters-complete

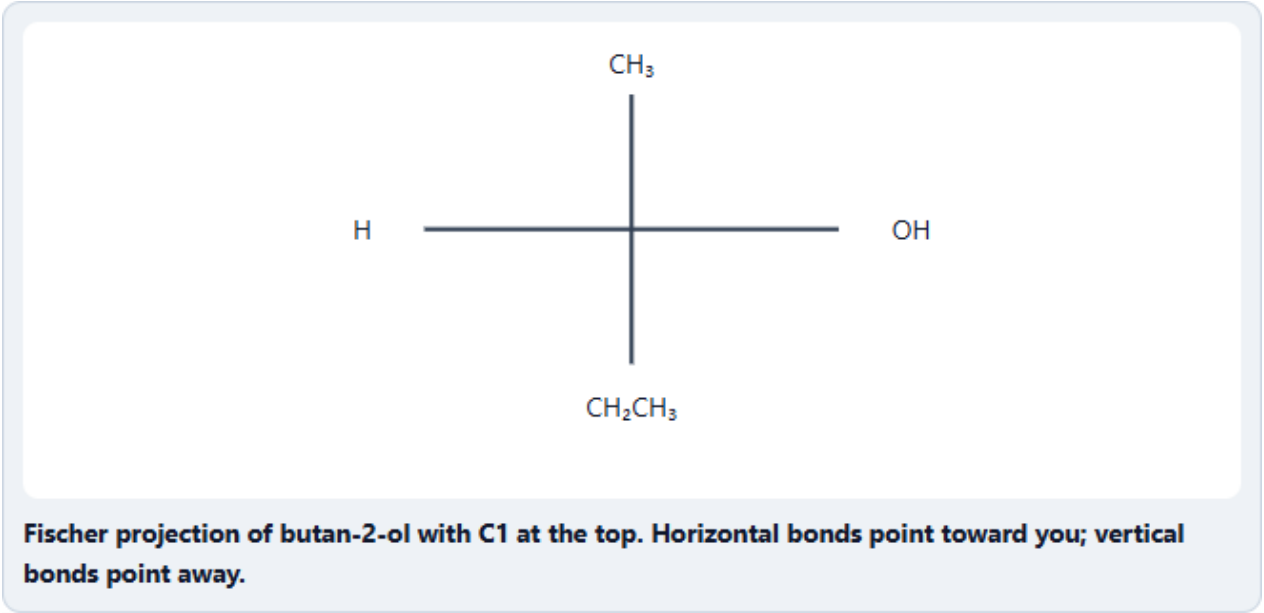


fischer-cip | Assign a Fischer configuration

Screenshot: fischer-cip-stage-1-molecule



Screenshot: fischer-cip-diagram



{"diagramSpecification": {"type": "fischer", "top": "CH₃", "bottom": "CH₂CH₃", "left": "H", "right": "OH", "caption": "Fischer projection of butan-2-ol with C1 at the top. Horizontal bonds point toward you; vertical bonds point away."}}

fischer-cip:stage-1 | Which substituent has highest CIP priority?

Answer type: choice

Option 0 (A): OH

Option 1 (B): Ethyl

Option 2 (C): Methyl

Option 3 (D): H

Screenshot: fischer-cip-stage-1-question



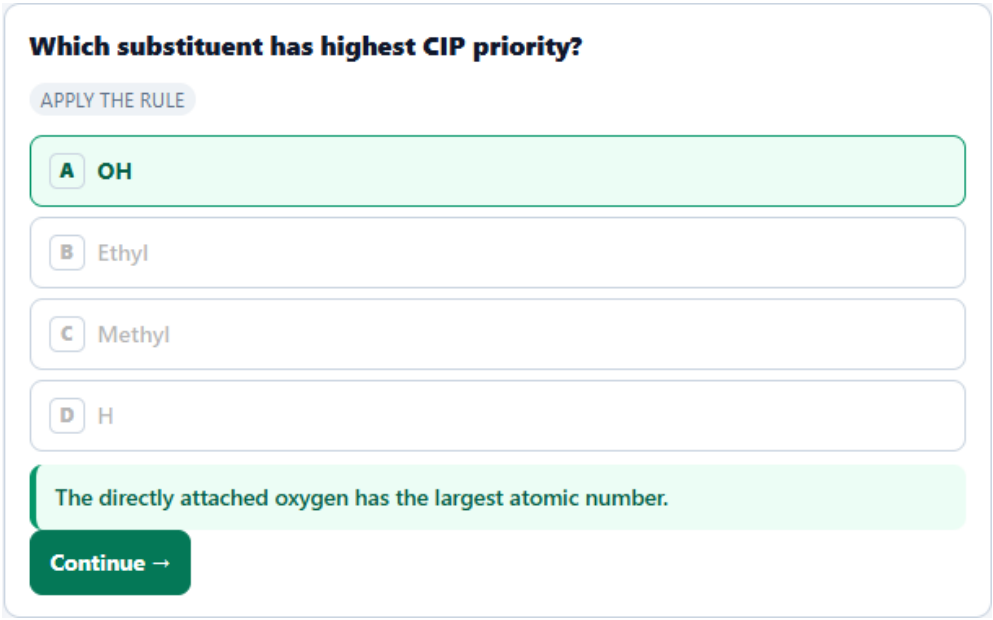
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Start at directly attached atoms.

Hint 2: O has atomic number 8; C is 6.

Explanation: The directly attached oxygen has the largest atomic number.

Screenshot: fischer-cip-stage-1-feedback



fischer-cip:stage-2 | In this projection, the lowest-priority H points…

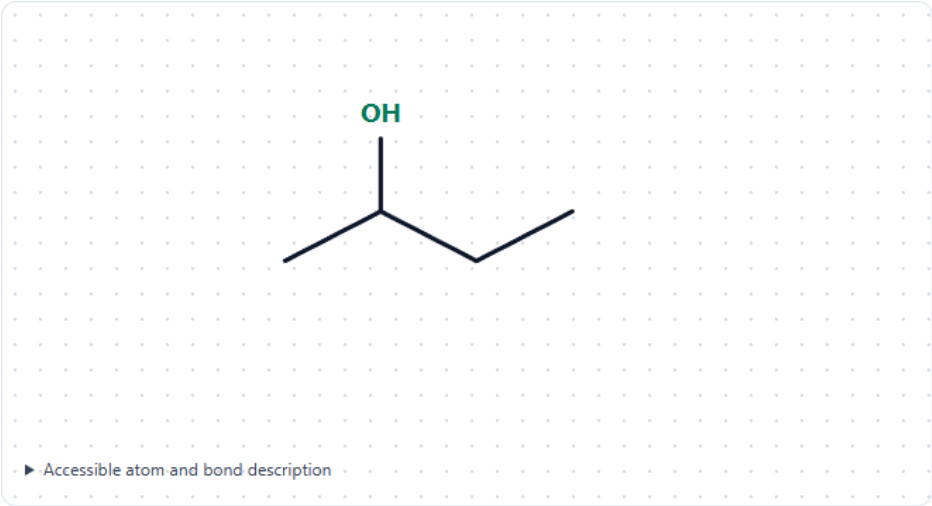
Answer type: choice

Option 0 (A): Away

Option 1 (B): Toward you

Option 2 (C): In the paper plane

Screenshot: fischer-cip-stage-2-molecule



Screenshot: fischer-cip-stage-2-question



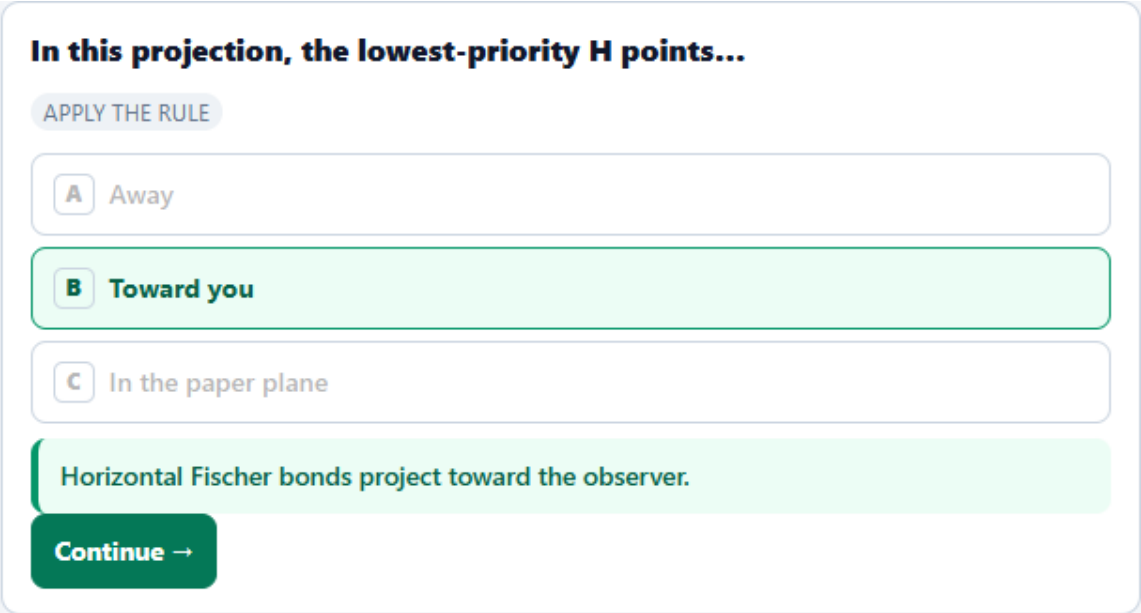
{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use the Fischer convention.

Hint 2: Vertical and horizontal bonds have opposite depth directions.

Explanation: Horizontal Fischer bonds project toward the observer.

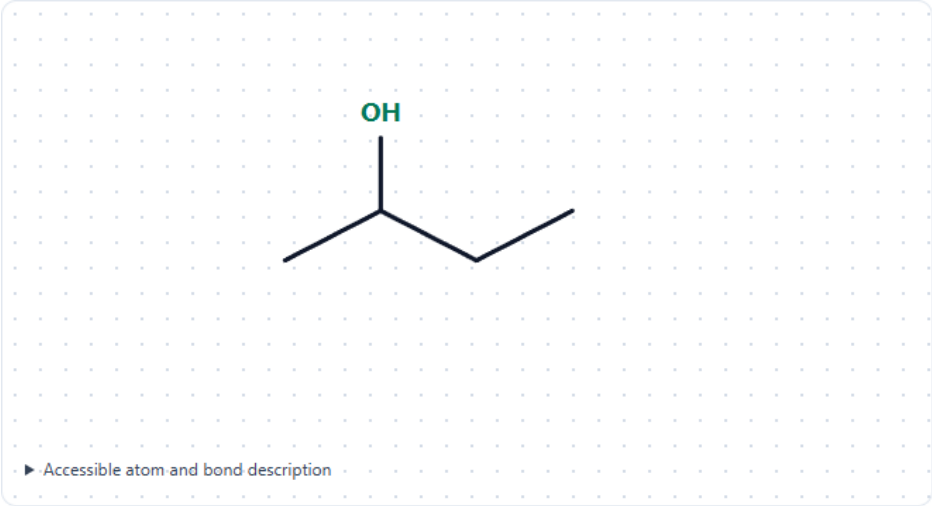
Screenshot: fischer-cip-stage-2-feedback



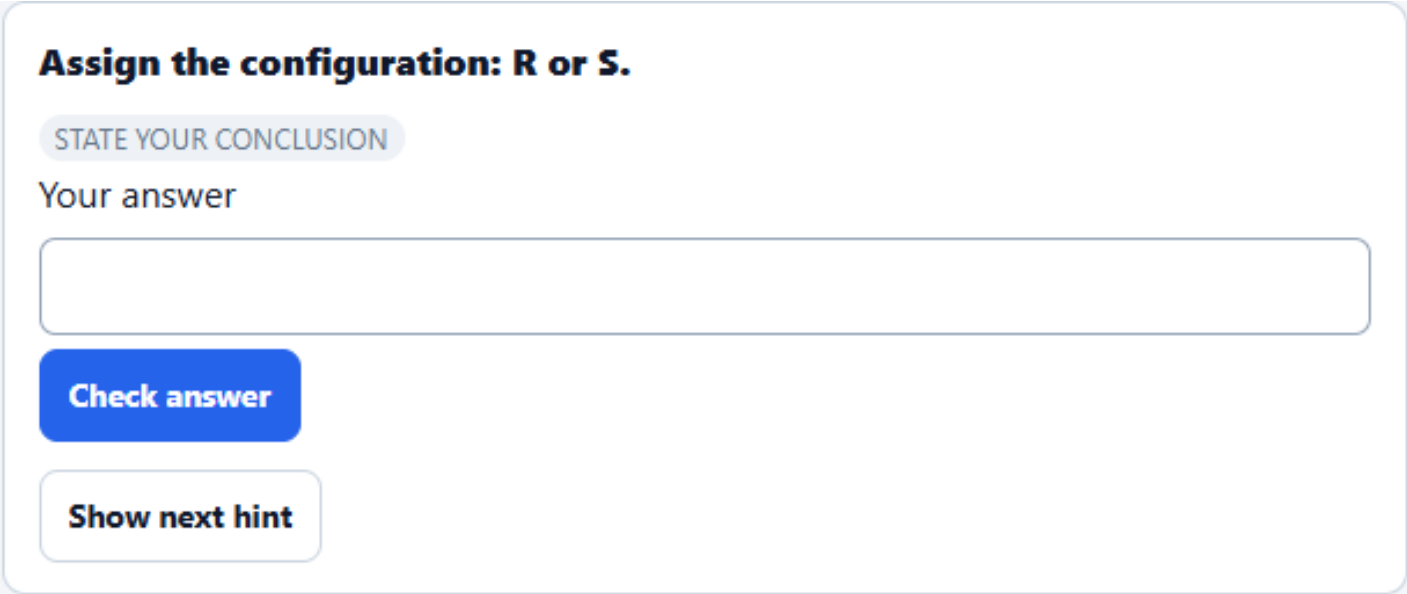
fischer-cip:stage-3 | Assign the configuration: R or S.

Answer type: text

Screenshot: fischer-cip-stage-3-molecule



Screenshot: fischer-cip-stage-3-question



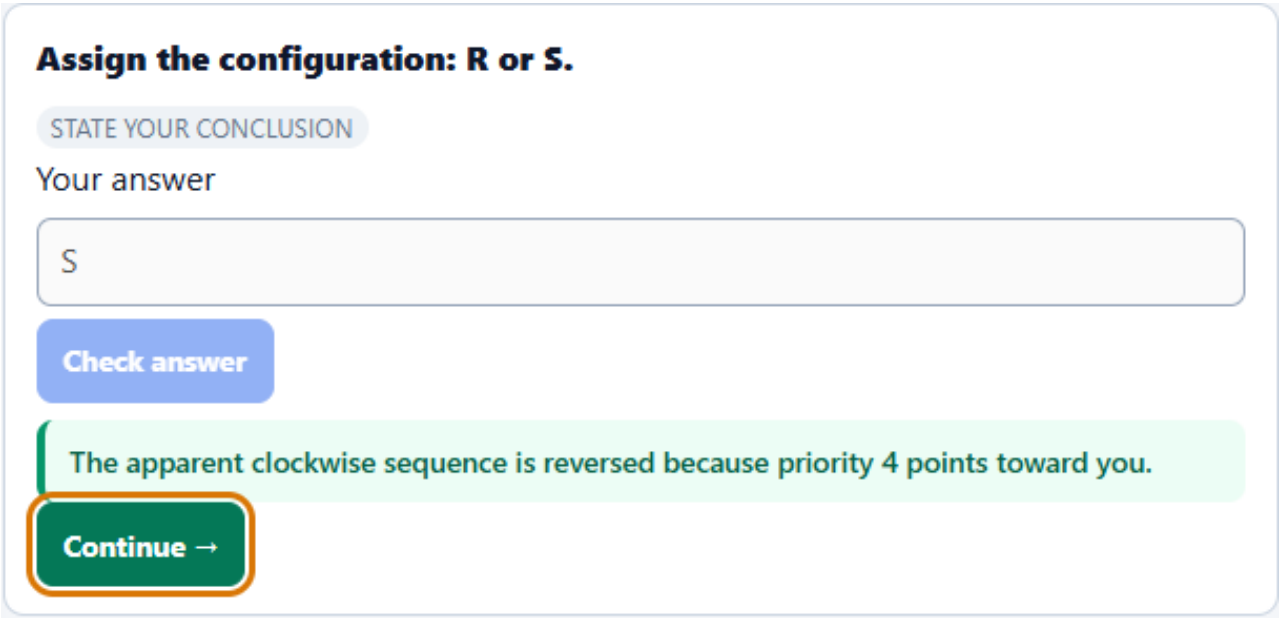
{"authorAnswer": ["S", "(S)", "2S", "(2S)", "(S)-butan-2-ol", "(2S)-butan-2-ol", "(S)-2-butanol", "(2S)-2-butanol"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Rank OH > ethyl > methyl > H.

Hint 2: Trace right → bottom → top, then reverse because H is horizontal.

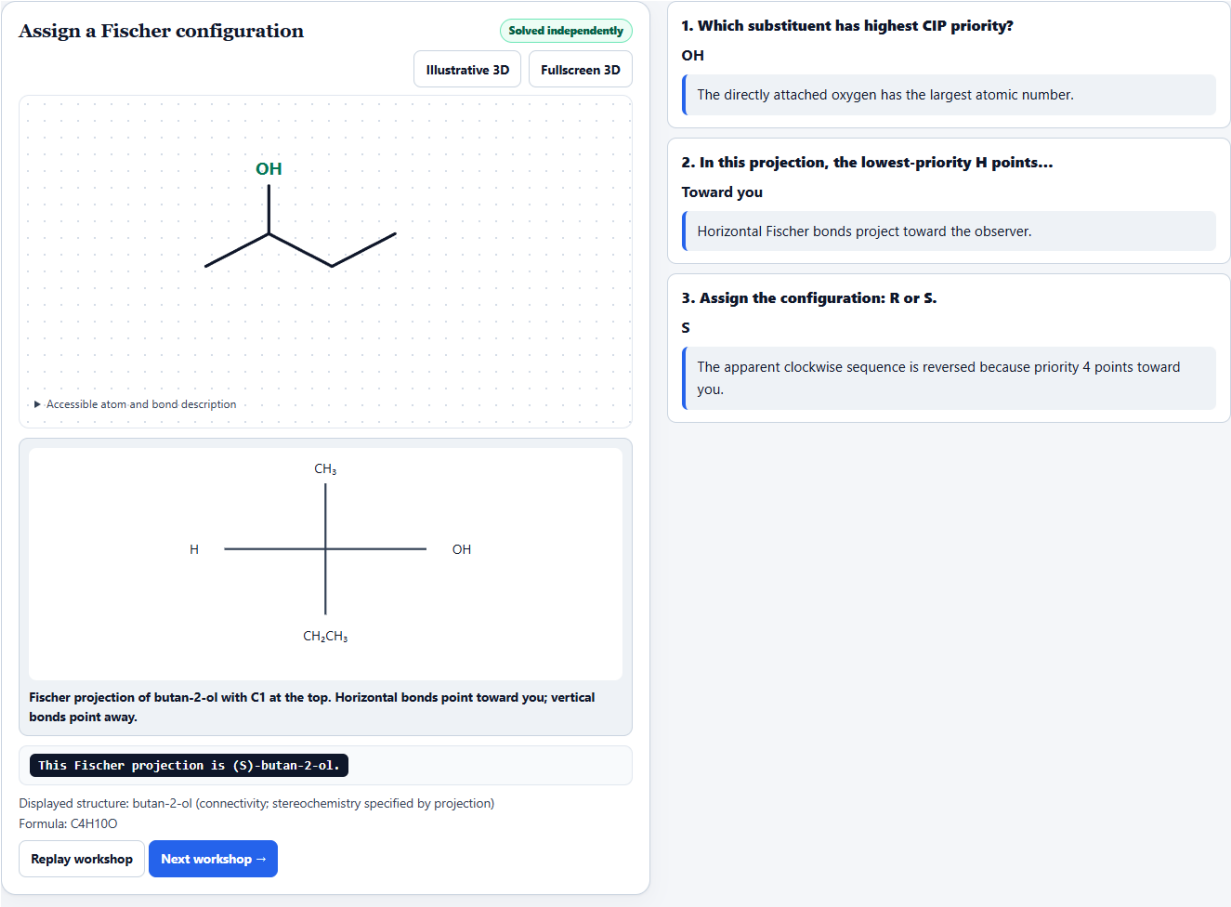
Explanation: The apparent clockwise sequence is reversed because priority 4 points toward you.

Screenshot: fischer-cip-stage-3-feedback



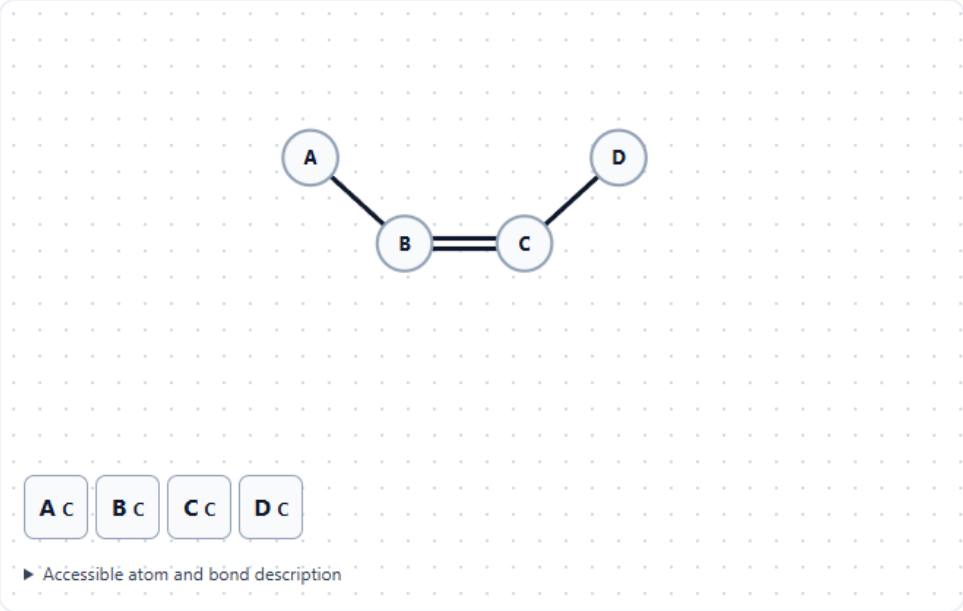
Completion summary: This Fischer projection is (S)-butan-2-ol.

Screenshot: fischer-cip-complete



ez-chirality | Separate E/Z geometry from chirality

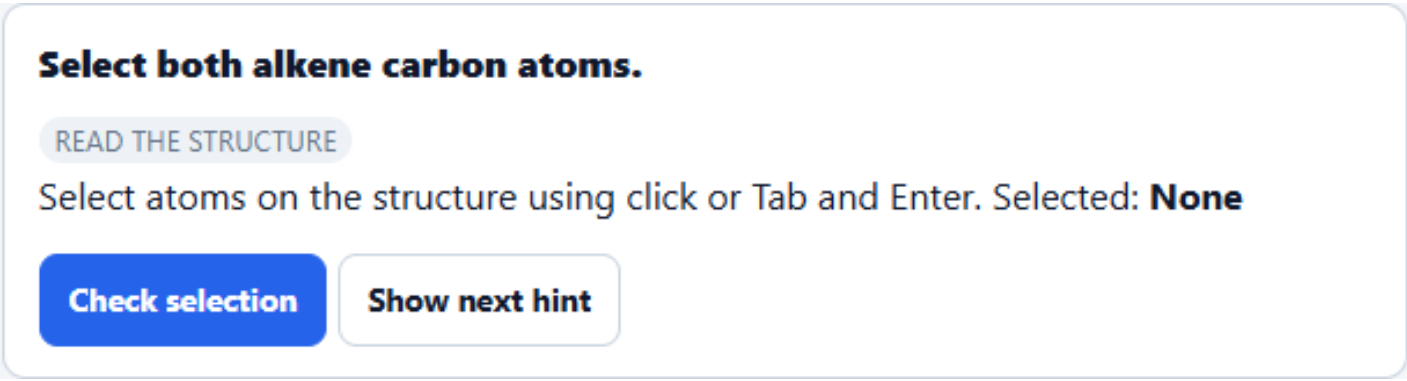
Screenshot: ez-chirality-stage-1-molecule



ez-chirality:stage-1 | Select both alkene carbon atoms.

Answer type: atoms

Screenshot: ez-chirality-stage-1-question



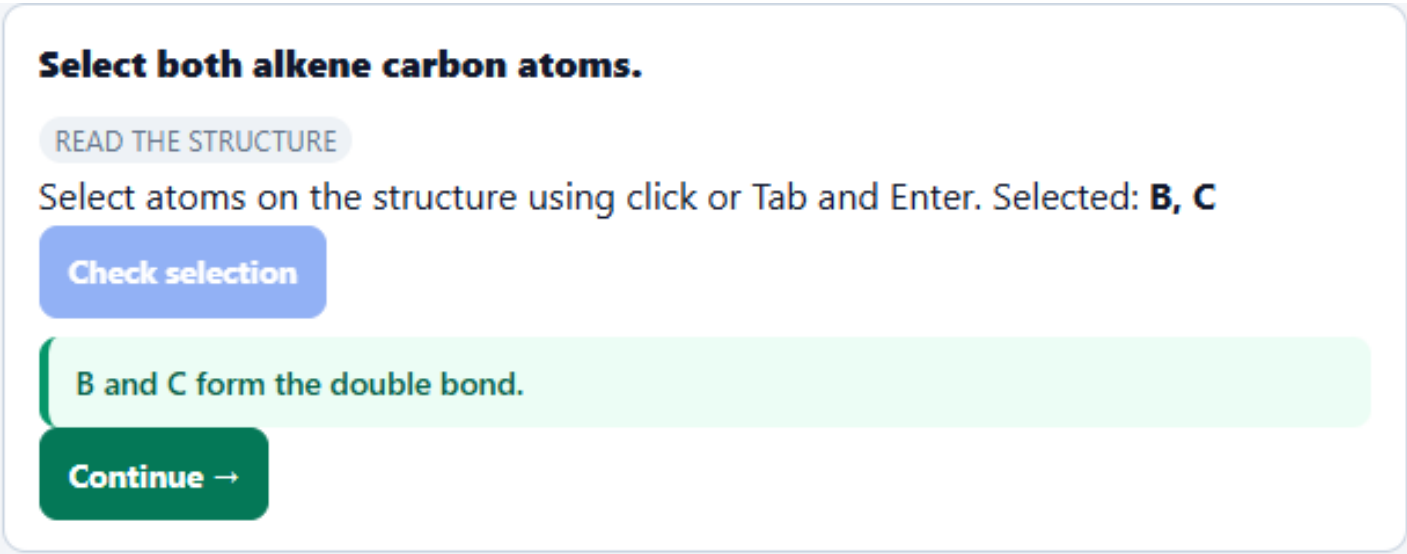
{"authorAnswer": [1, 2], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 2]}}

Hint 1: Look for two parallel bond lines.

Hint 2: Choose both endpoints of that bond.

Explanation: B and C form the double bond.

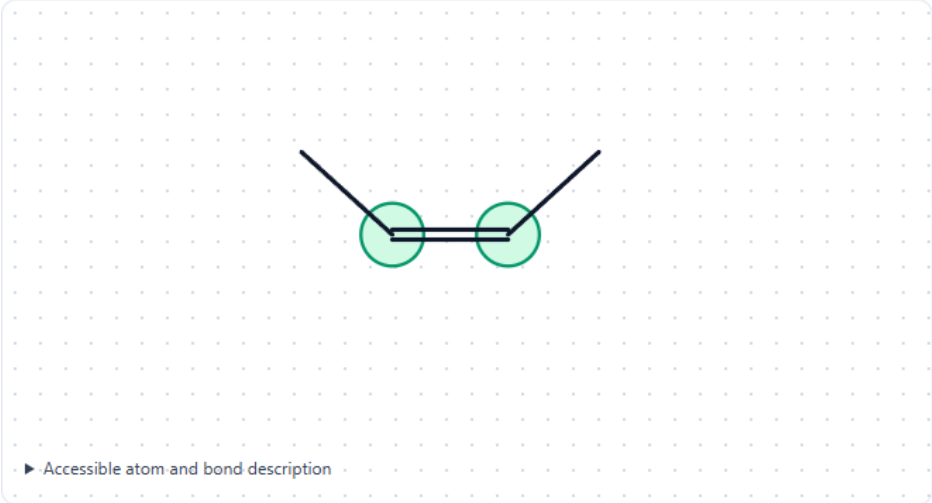
Screenshot: ez-chirality-stage-1-feedback



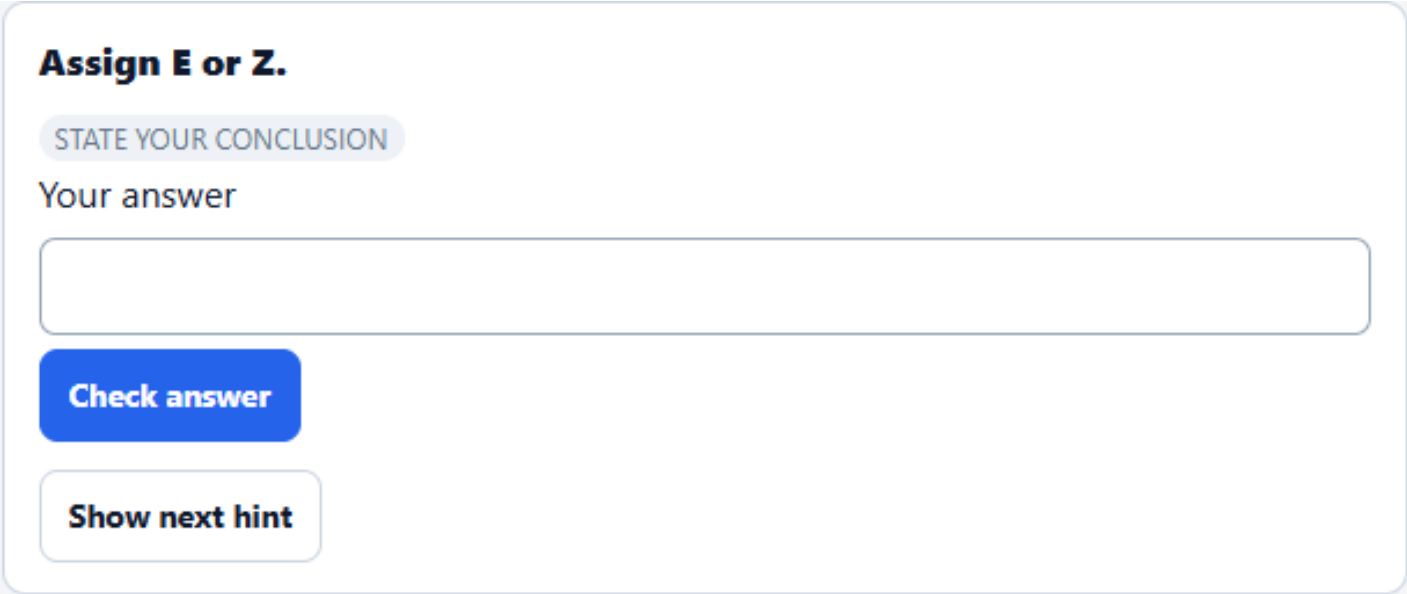
ez-chirality:stage-2 | Assign E or Z.

Answer type: text

Screenshot: ez-chirality-stage-2-molecule



Screenshot: ez-chirality-stage-2-question



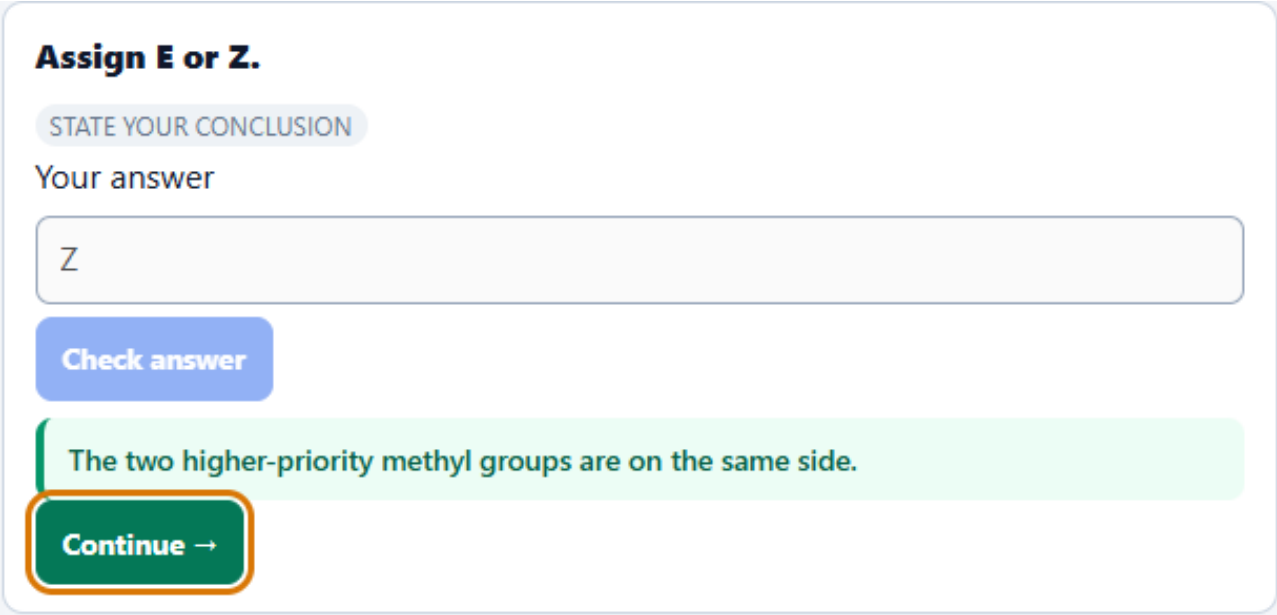
{"authorAnswer": ["Z", "(Z)", "2Z", "(2Z)", "(Z)-but-2-ene", "(2Z)-but-2-ene", "(Z)-2-butene", "(2Z)-2-butene"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Rank methyl versus H on each end.

Hint 2: Same-side higher-priority groups determine the descriptor.

Explanation: The two higher-priority methyl groups are on the same side.

Screenshot: ez-chirality-stage-2-feedback



ez-chirality:stage-3 | Why does a pure sample of this alkene isomer show no ordinary optical rotation?

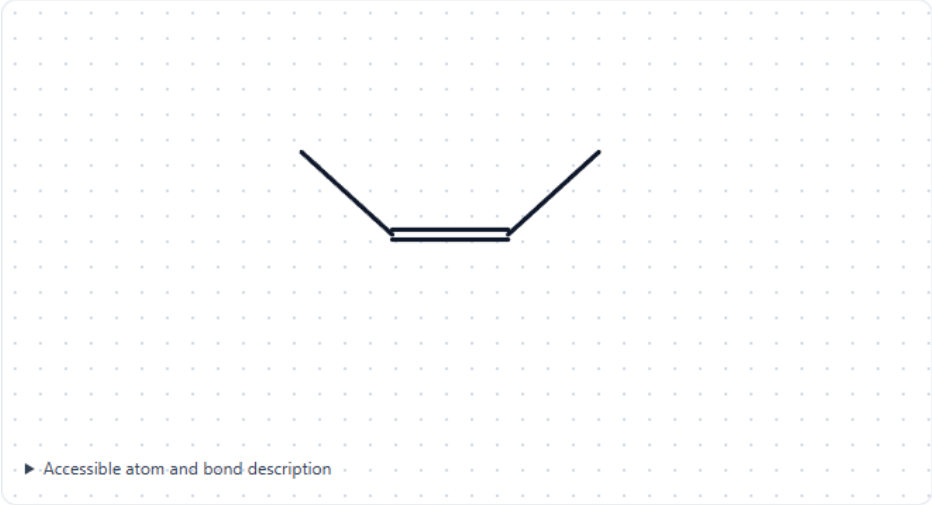
Answer type: choice

Option 0 (A): It is achiral

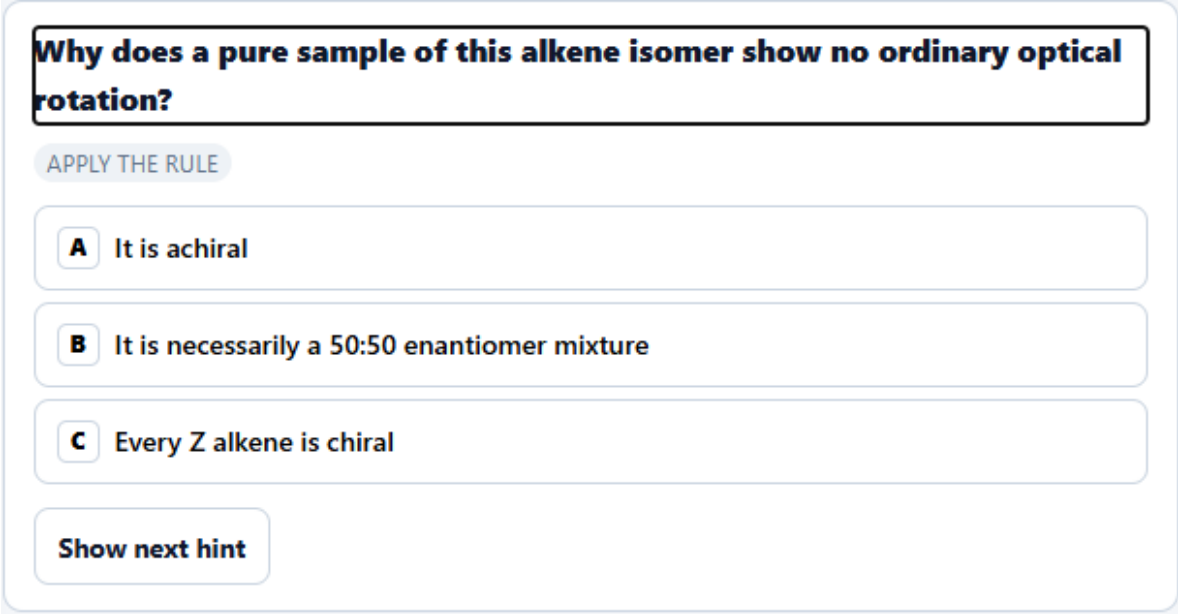
Option 1 (B): It is necessarily a 50:50 enantiomer mixture

Option 2 (C): Every Z alkene is chiral

Screenshot: ez-chirality-stage-3-molecule



Screenshot: ez-chirality-stage-3-question



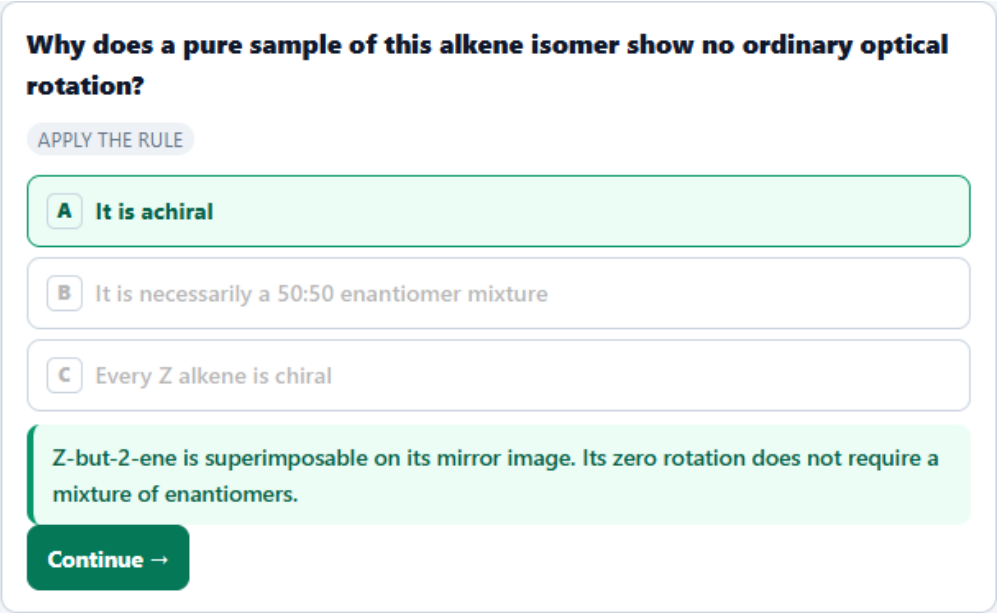
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Distinguish a property of one molecule from cancellation in a mixture.

Hint 2: Inspect the symmetry of the entire planar but-2-ene structure.

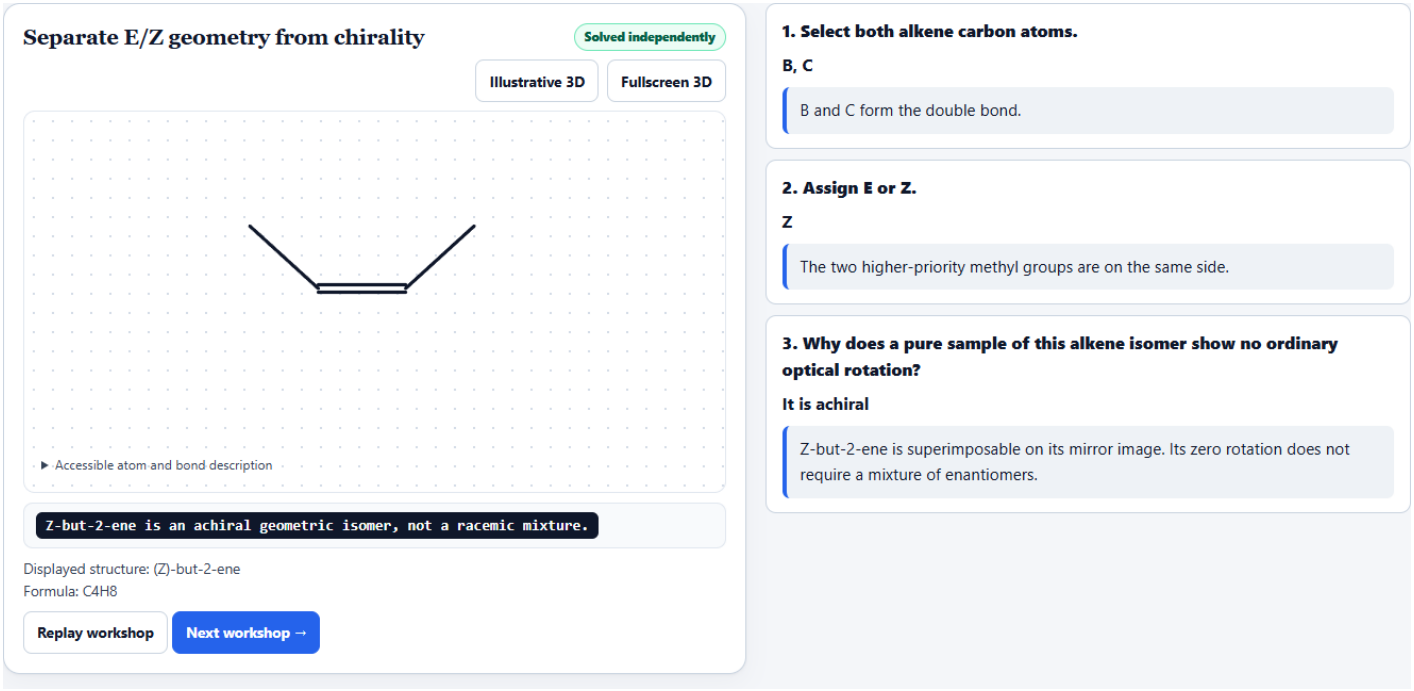
Explanation: Z-but-2-ene is superimposable on its mirror image. Its zero rotation does not require a mixture of enantiomers.

Screenshot: ez-chirality-stage-3-feedback



Completion summary: Z-but-2-ene is an achiral geometric isomer, not a racemic mixture.

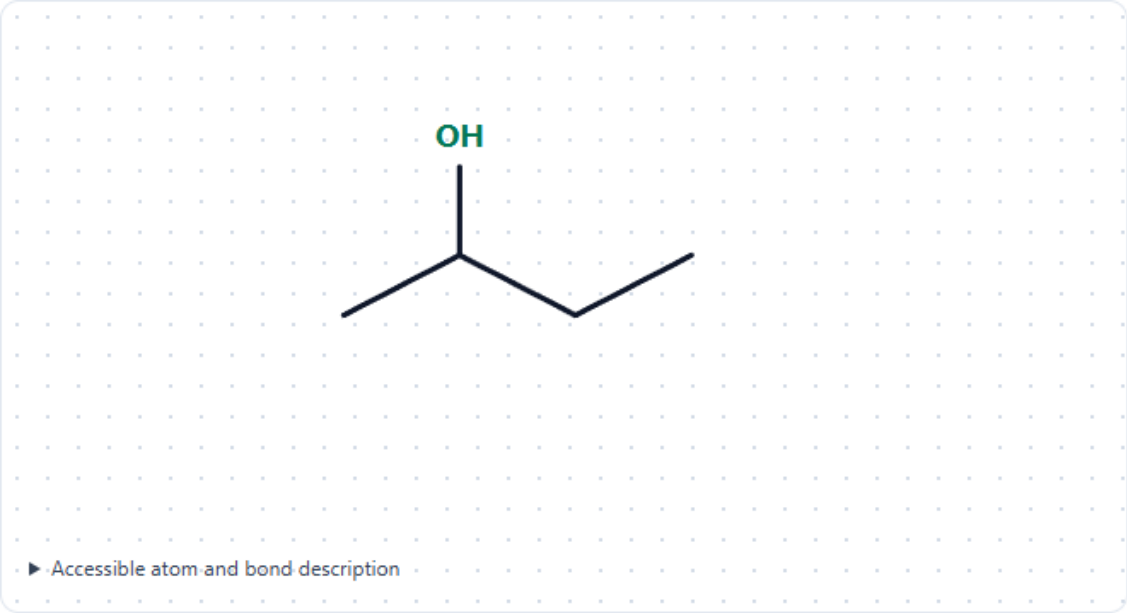
Screenshot: ez-chirality-complete



specific-rotation | Normalize a polarimeter measurement

Hypothetical polarimetry data for an enantiomerically pure chiral sample with the illustrated connectivity: αobs = +2.4°, c = 0.080 g/mL, path length = 5.0 cm. These are practice data, not literature measurements for butan-2-ol. Do not infer the sign from the flat molecular drawing.

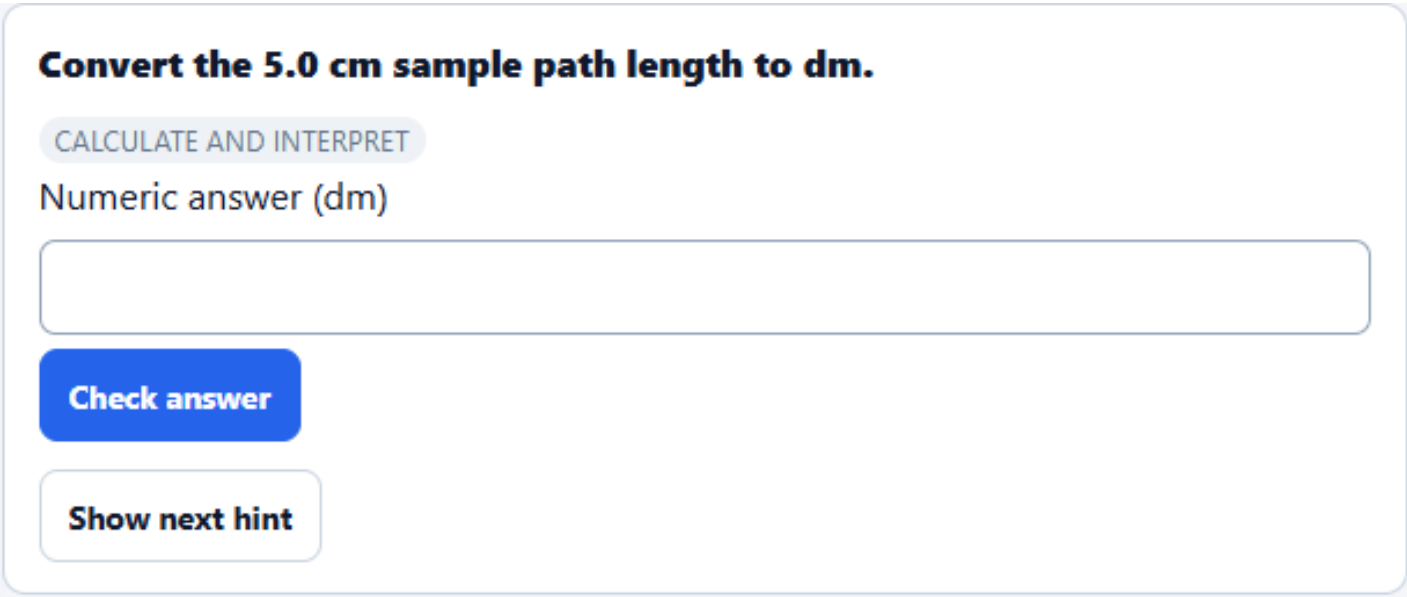
Screenshot: specific-rotation-stage-1-molecule



specific-rotation:stage-1 | Convert the 5.0 cm sample path length to dm.

Answer type: number | unit: dm

Screenshot: specific-rotation-stage-1-question



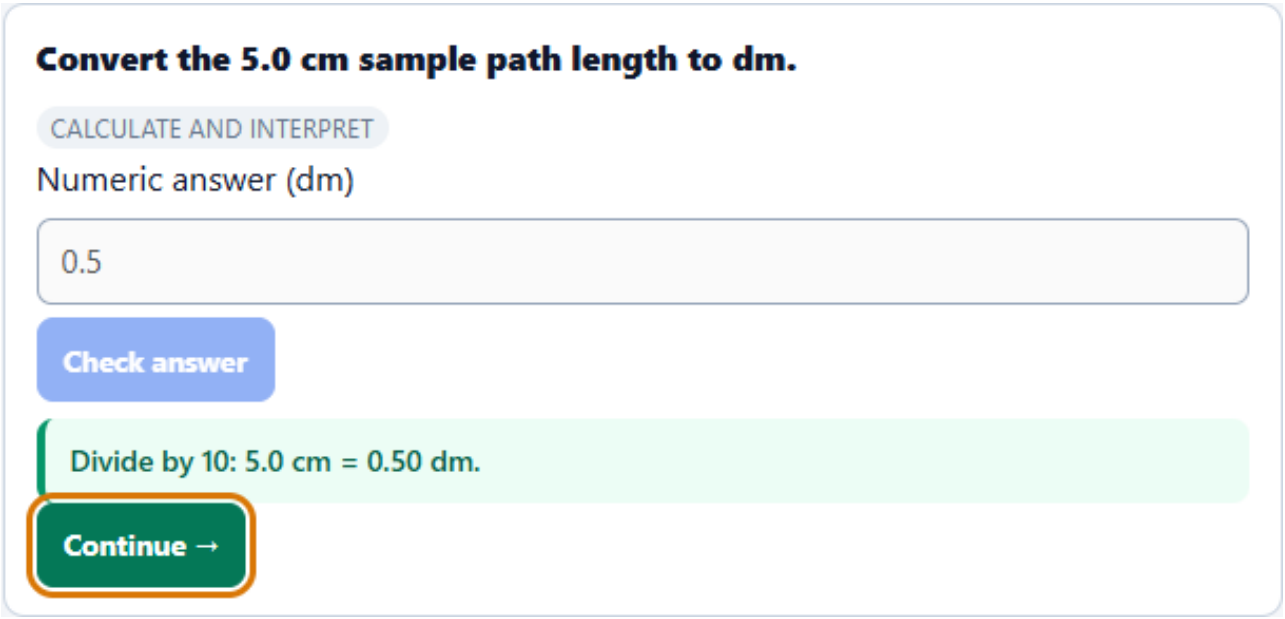
{"authorAnswer": 0.5, "tolerance": 0.0001, "highlightsAfterCorrectAnswer": {}}

Hint 1: The specific-rotation formula here requires dm.

Hint 2: One decimeter is ten centimeters.

Explanation: Divide by 10: 5.0 cm = 0.50 dm.

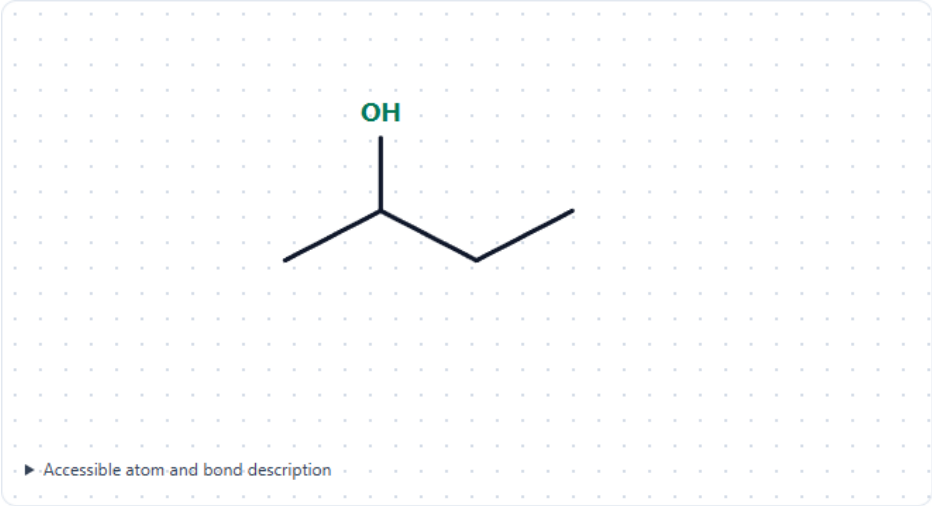
Screenshot: specific-rotation-stage-1-feedback



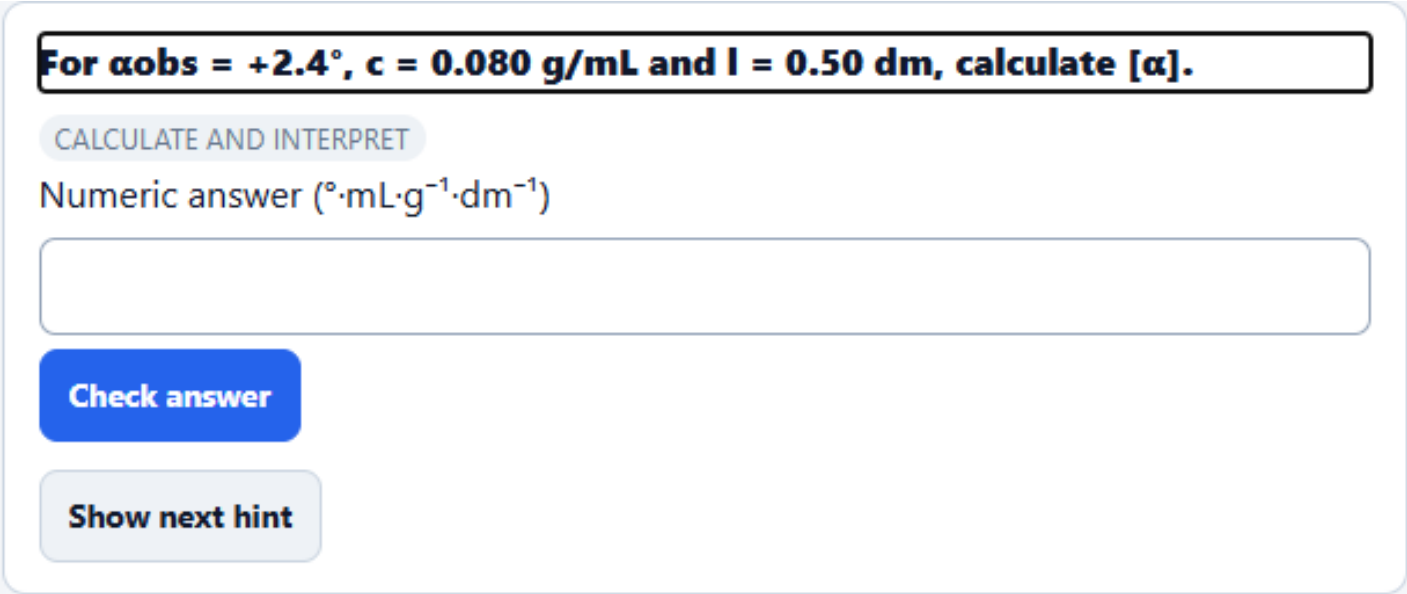
specific-rotation:stage-2 | For αobs = +2.4°, c = 0.080 g/mL and l = 0.50 dm, calculate [α].

Answer type: number | unit: °·mL·g^{-1}·dm^{-1}

Screenshot: specific-rotation-stage-2-molecule



Screenshot: specific-rotation-stage-2-question



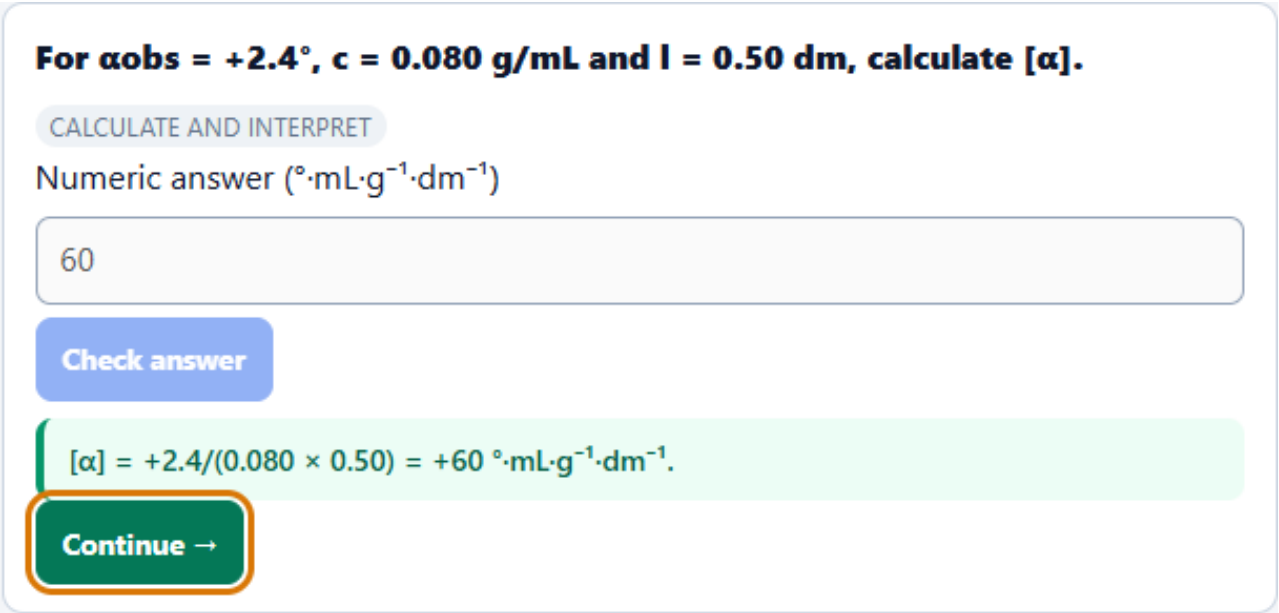
{"authorAnswer": 60, "tolerance": 0.01, "highlightsAfterCorrectAnswer": {}}

Hint 1: Divide observed rotation by concentration times length.

Hint 2: The denominator is 0.080 × 0.50 = 0.040.

Explanation: [α] = +2.4/(0.080 × 0.50) = +60 °·mL·g^{-1}·dm^{-1}.

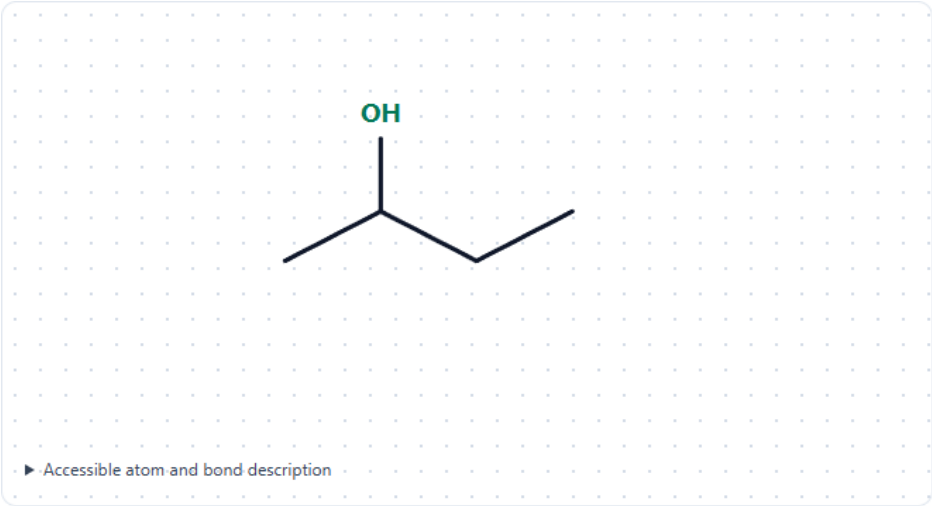
Screenshot: specific-rotation-stage-2-feedback



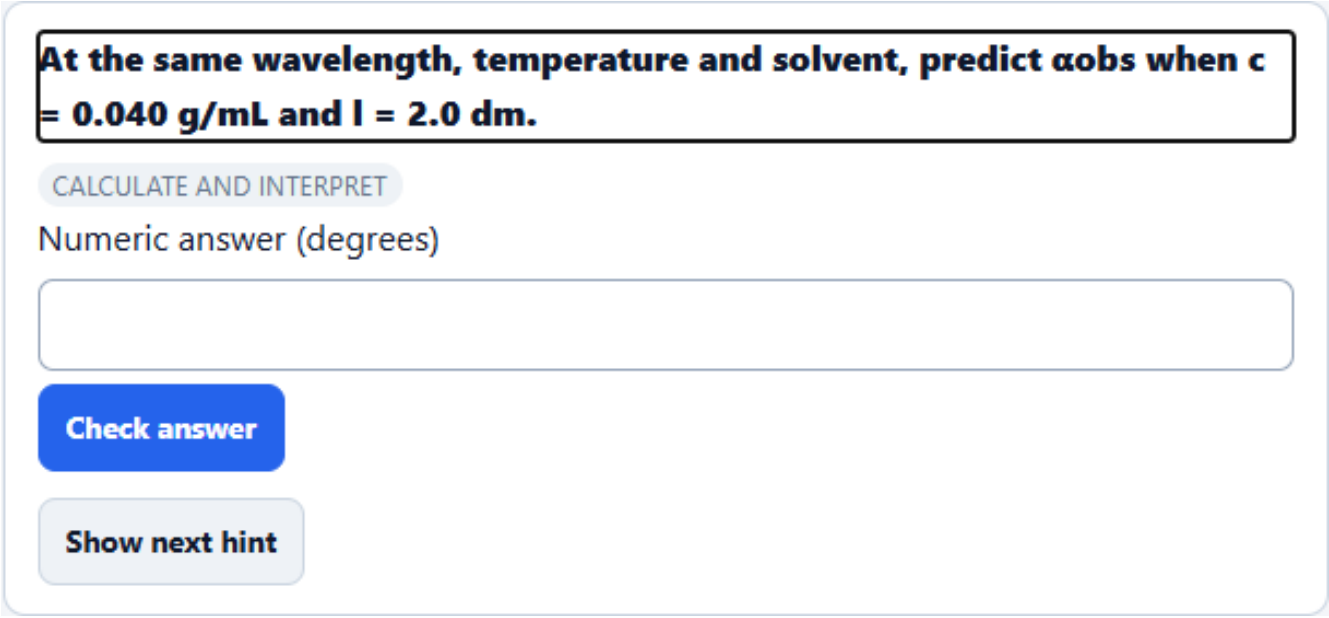
specific-rotation:stage-3 | At the same wavelength, temperature and solvent, predict αobs when c = 0.040 g/mL and l = 2.0 dm.

Answer type: number | unit: degrees

Screenshot: specific-rotation-stage-3-molecule



Screenshot: specific-rotation-stage-3-question



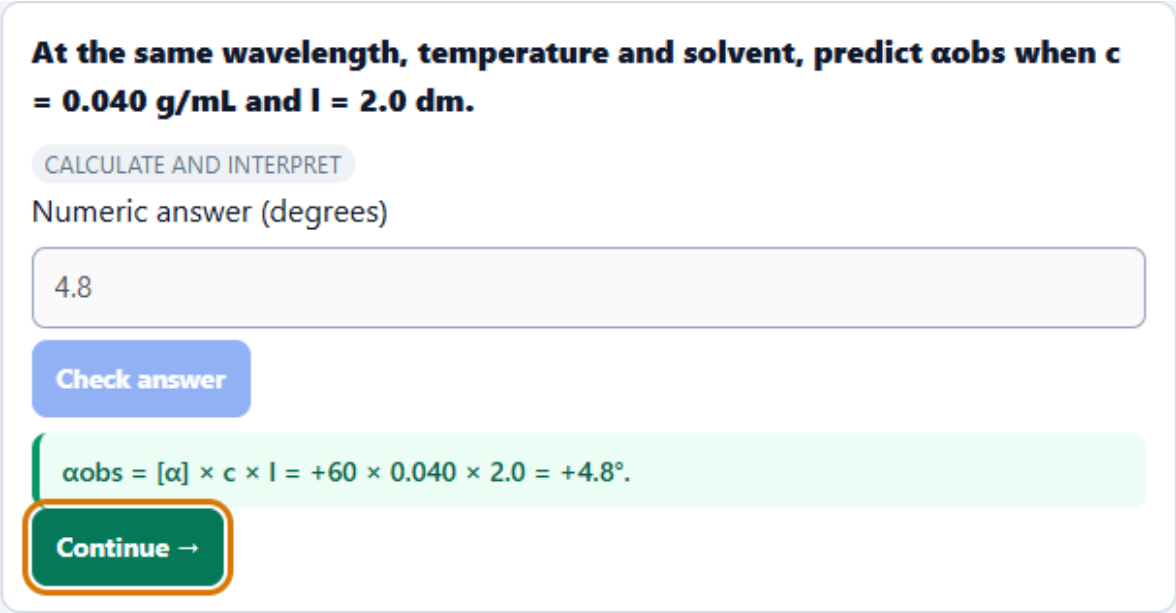
{"authorAnswer": 4.8, "tolerance": 0.001, "highlightsAfterCorrectAnswer": {}}

Hint 1: Rearrange the same normalization equation.

Hint 2: Use the previously calculated specific rotation with the new concentration and length.

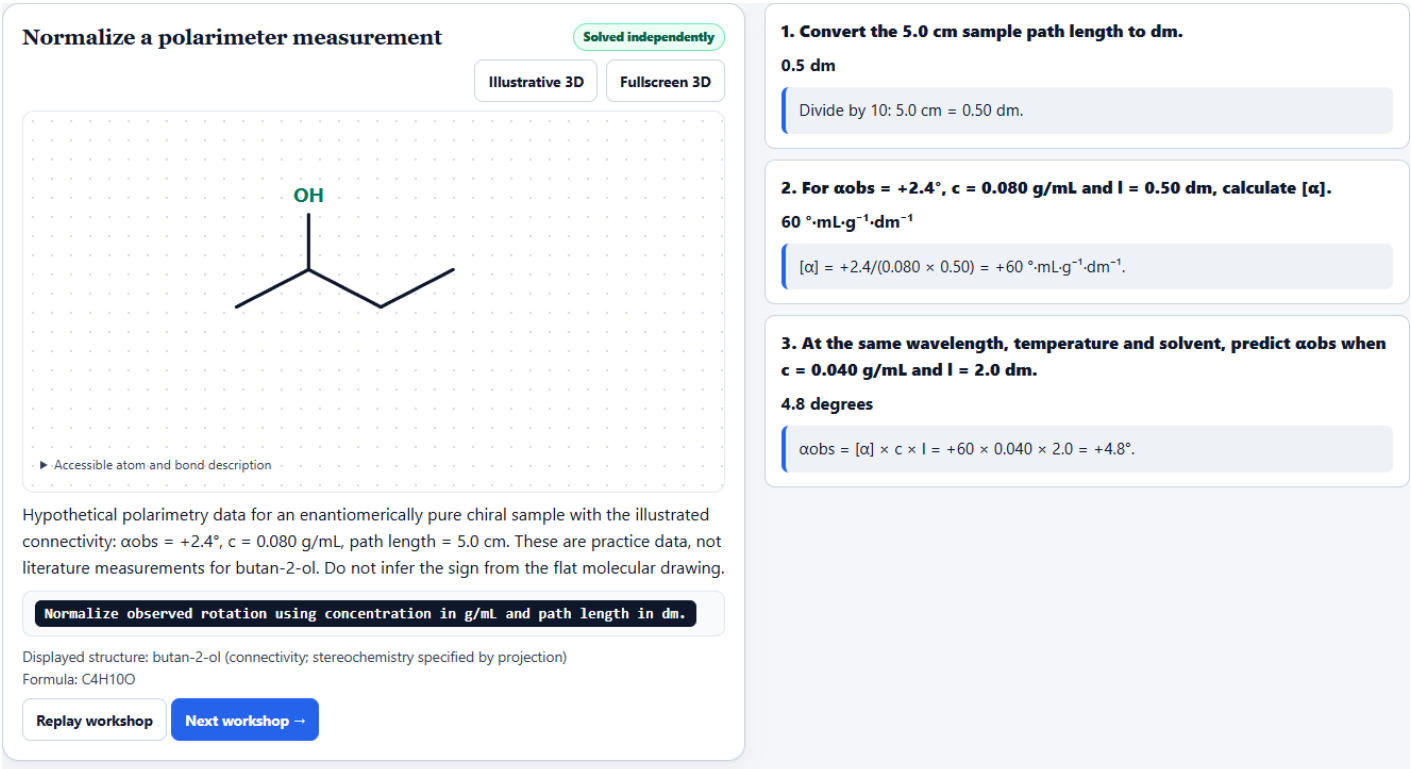
Explanation: αobs = [α] × c × l = +60 × 0.040 × 2.0 = +4.8°.

Screenshot: specific-rotation-stage-3-feedback



Completion summary: Normalize observed rotation using concentration in g/mL and path length in dm.

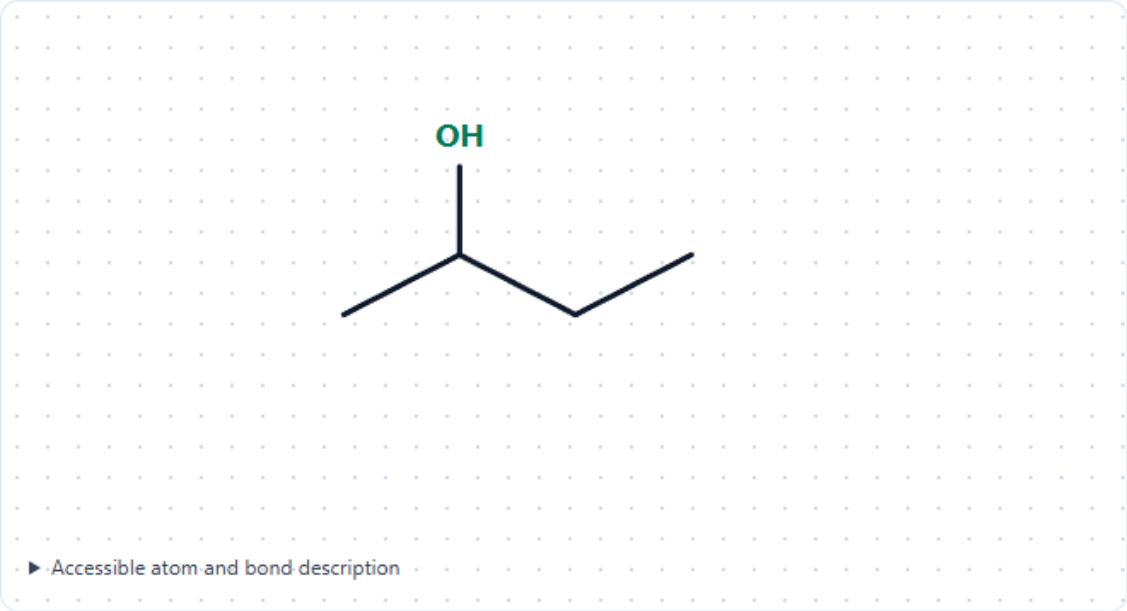
Screenshot: specific-rotation-complete



optical-mixture | Interpret an enantiomer mixture

The illustrated chiral connectivity represents a pair of enantiomers. The 85:15 composition and +40° reference rotation are hypothetical practice data; no (+)/(−) assignment is made from the drawing.

Screenshot: optical-mixture-stage-1-molecule



optical-mixture:stage-1 | The major enantiomer is dextrorotatory. Which sign describes its observed rotation?

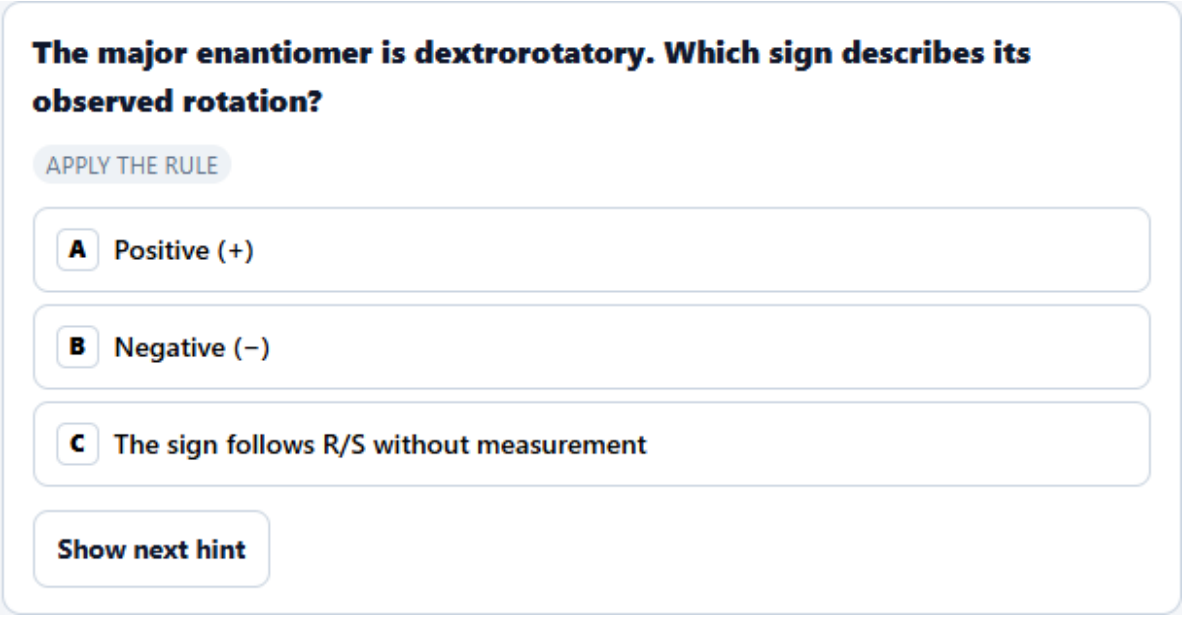
Answer type: choice

Option 0 (A): Positive (+)

Option 1 (B): Negative (−)

Option 2 (C): The sign follows R/S without measurement

Screenshot: optical-mixture-stage-1-question



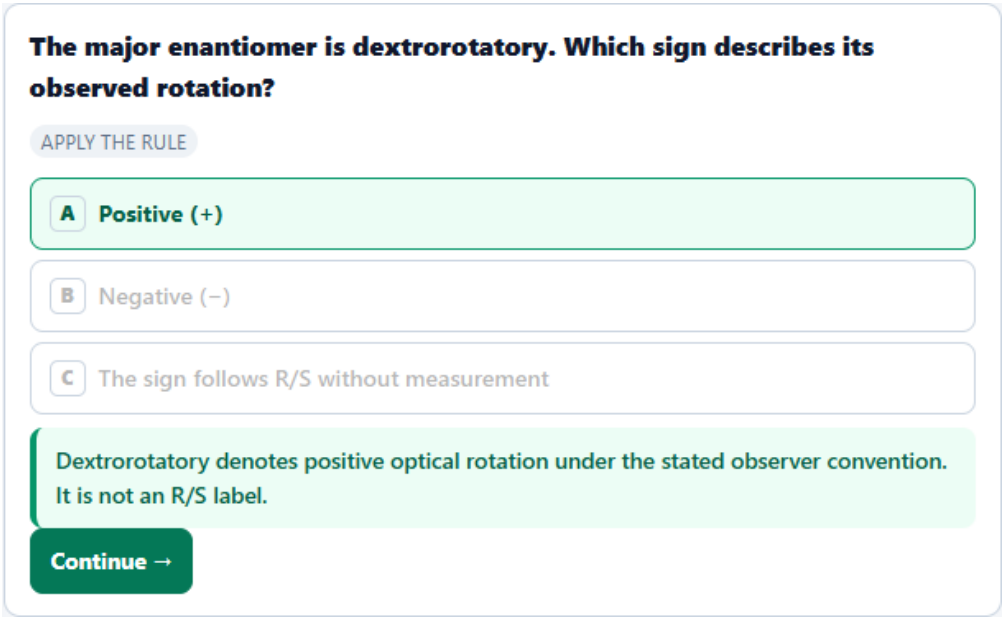
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use the optical-rotation vocabulary.

Hint 2: Configuration descriptors do not determine this sign.

Explanation: Dextrorotatory denotes positive optical rotation under the stated observer convention. It is not an R/S label.

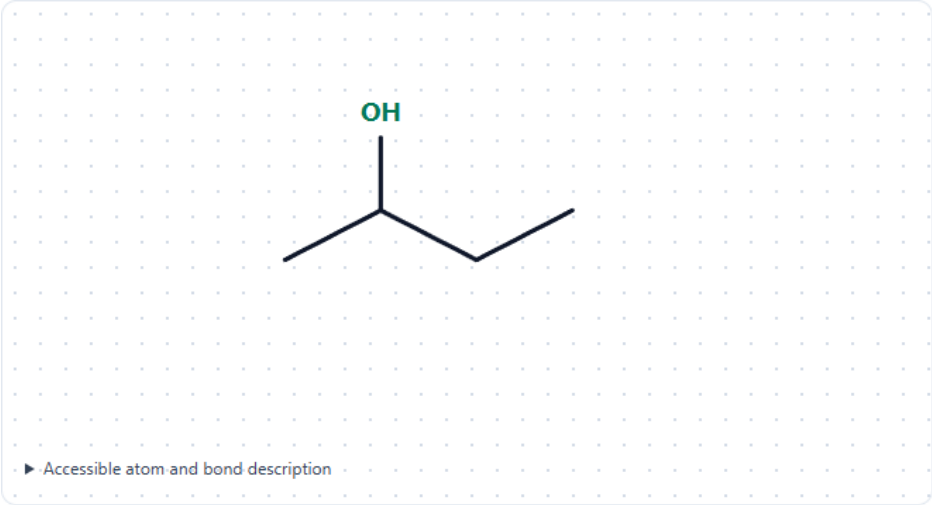
Screenshot: optical-mixture-stage-1-feedback



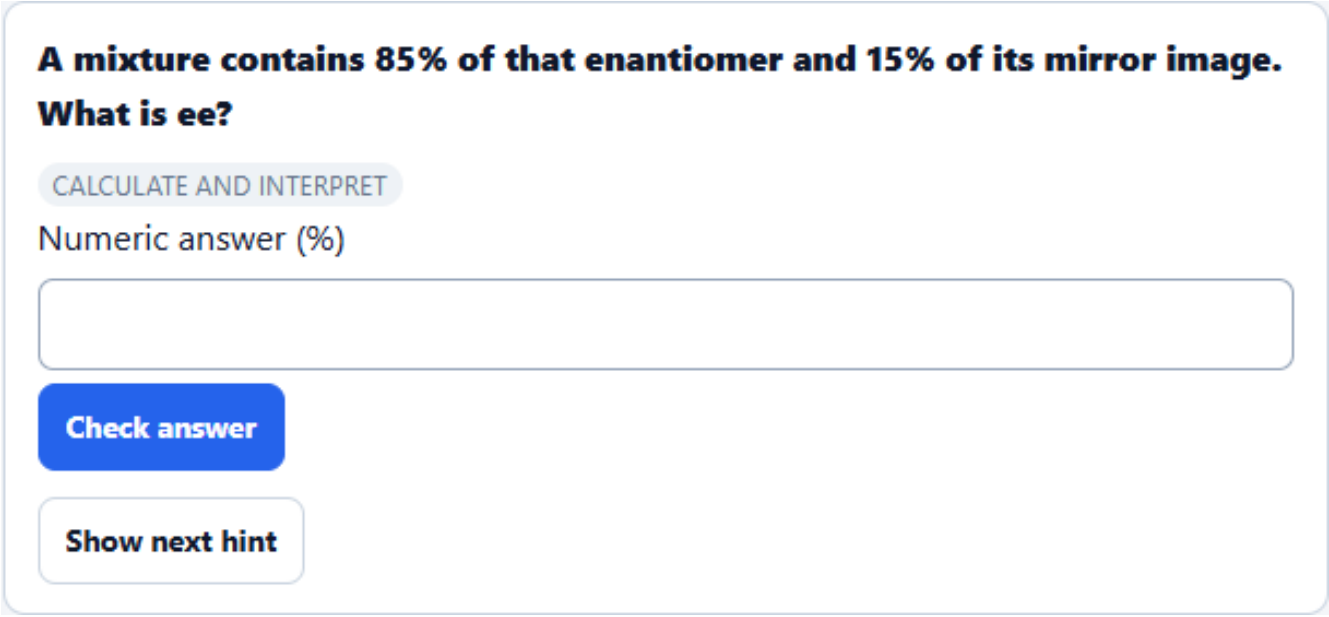
optical-mixture:stage-2 | A mixture contains 85% of that enantiomer and 15% of its mirror image. What is ee?

Answer type: number | unit: %

Screenshot: optical-mixture-stage-2-molecule



Screenshot: optical-mixture-stage-2-question



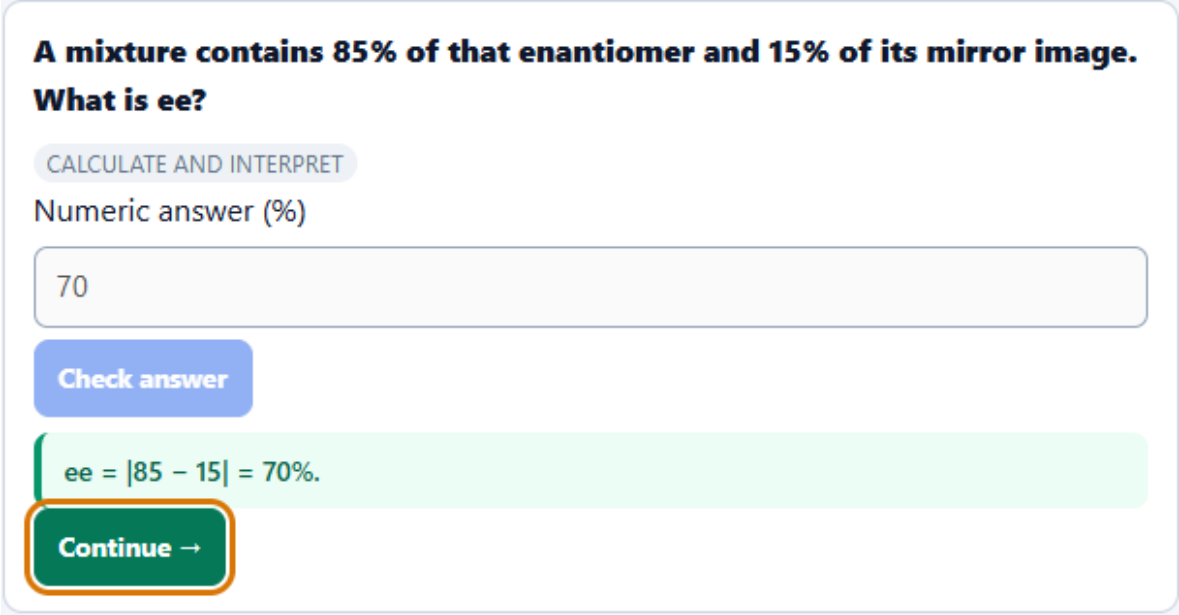
{"authorAnswer": 70, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Subtract the smaller enantiomer percentage from the larger.

Hint 2: Do not use the 75:25 worked example without recalculating.

Explanation: ee = |85 − 15| = 70%.

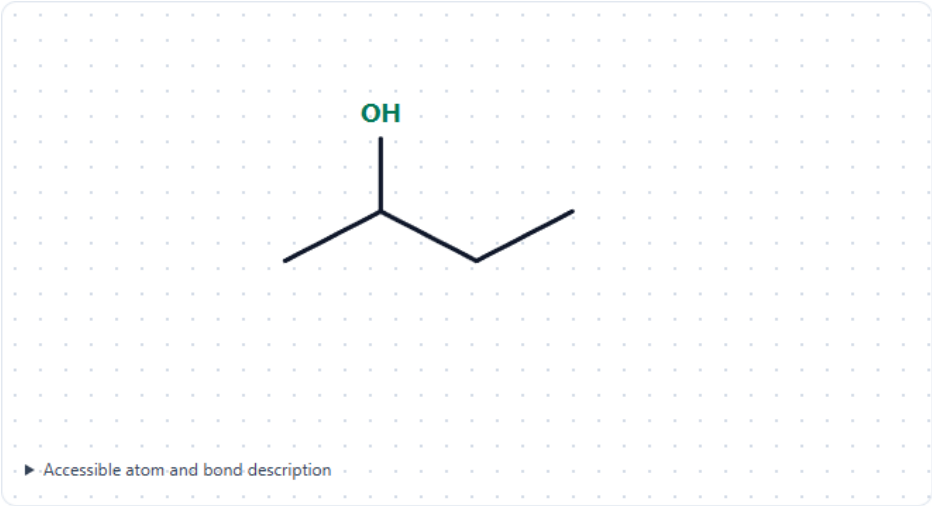
Screenshot: optical-mixture-stage-2-feedback



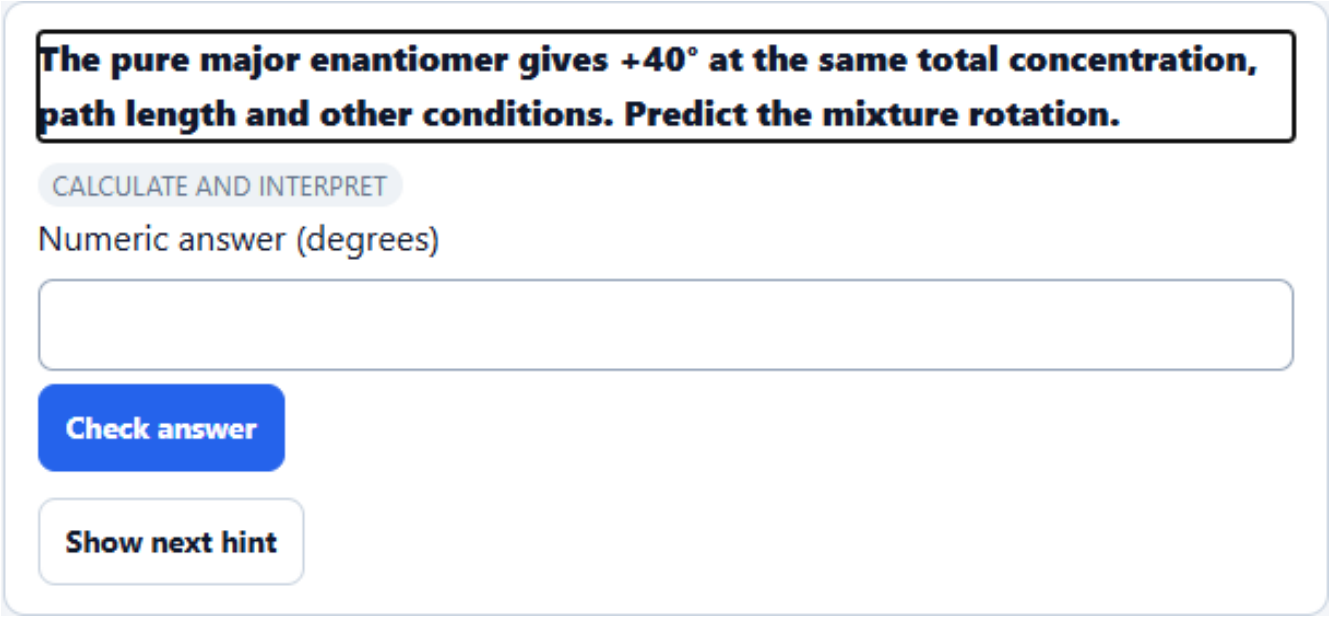
optical-mixture:stage-3 | The pure major enantiomer gives +40° at the same total concentration, path length and other conditions. Predict the mixture rotation.

Answer type: number | unit: degrees

Screenshot: optical-mixture-stage-3-molecule



Screenshot: optical-mixture-stage-3-question



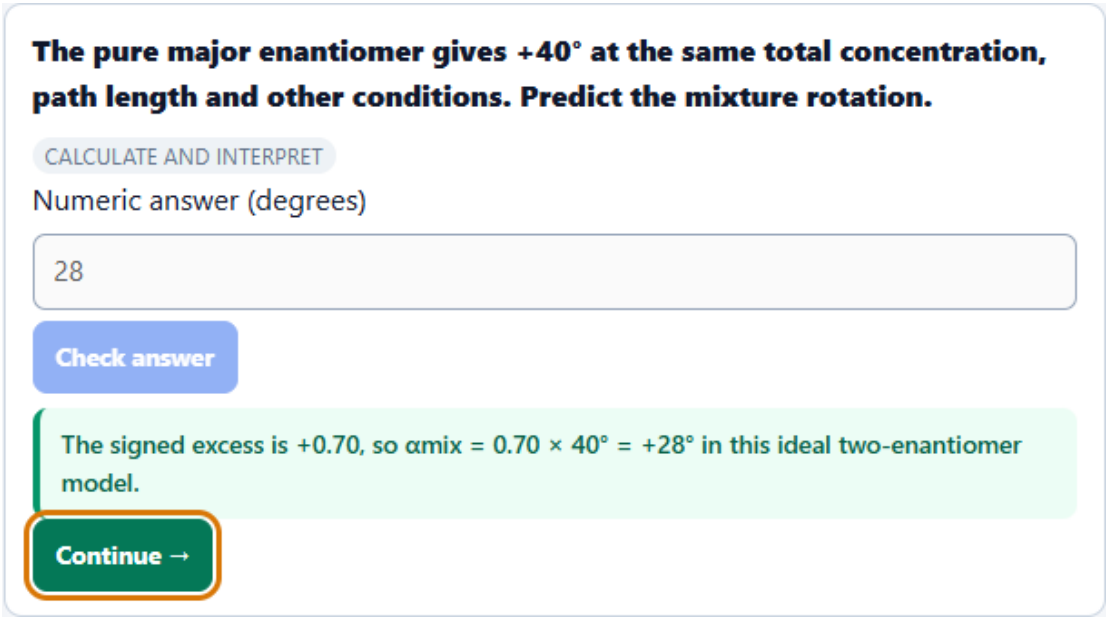
{"authorAnswer": 28, "tolerance": 0.001, "highlightsAfterCorrectAnswer": {}}

Hint 1: The two enantiomers rotate equally and oppositely.

Hint 2: Multiply the pure-enantiomer angle by the fractional ee.

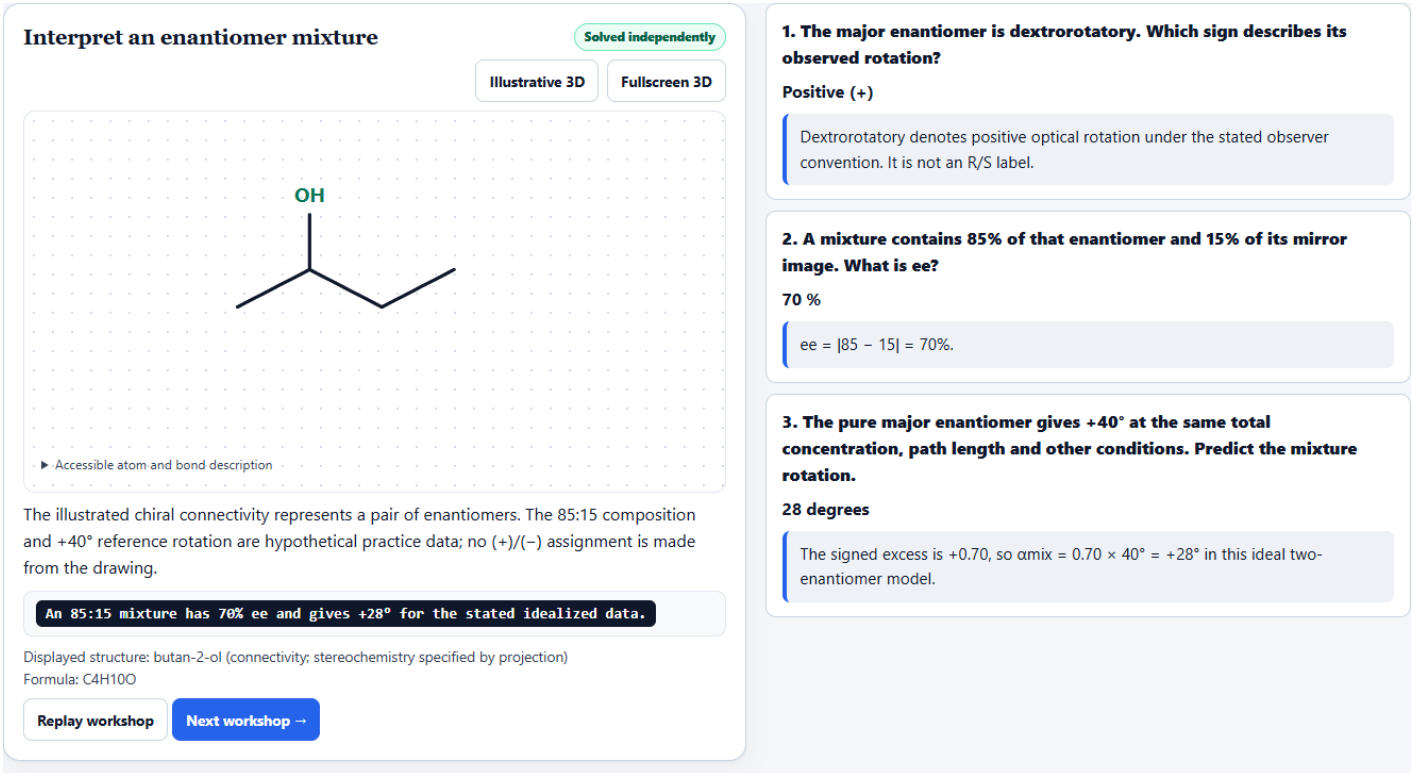
Explanation: The signed excess is +0.70, so αmix = 0.70 × 40° = +28° in this ideal two-enantiomer model.

Screenshot: optical-mixture-stage-3-feedback



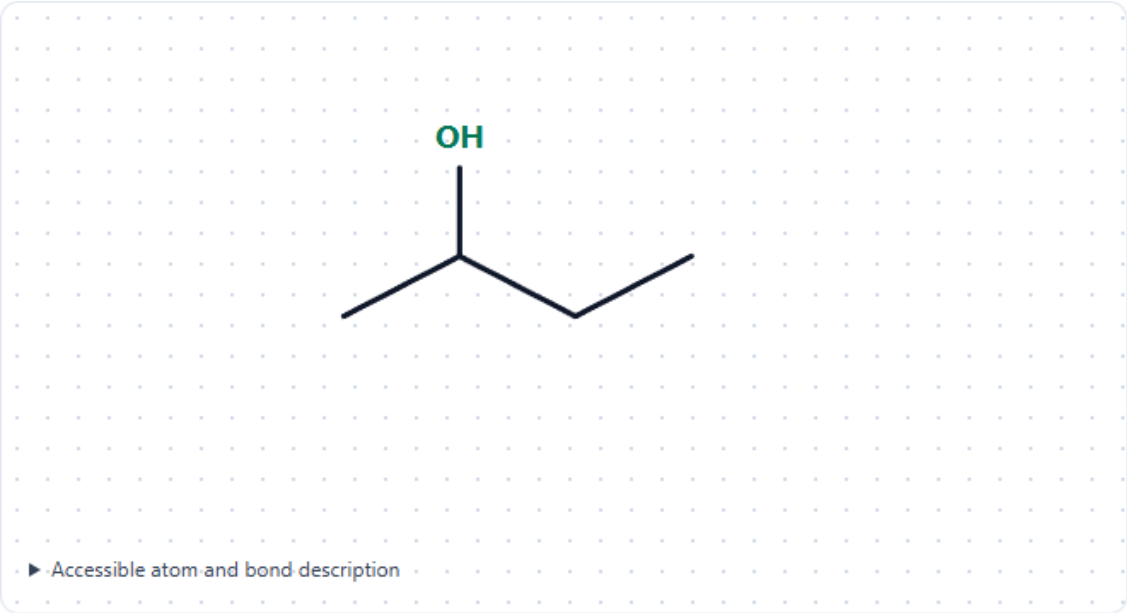
Completion summary: An 85:15 mixture has 70% ee and gives +28° for the stated idealized data.

Screenshot: optical-mixture-complete



chiral-resolution | Choose a chiral resolution strategy

Screenshot: chiral-resolution-stage-1-molecule



chiral-resolution:stage-1 | Why does ordinary achiral distillation generally fail to resolve two enantiomers?

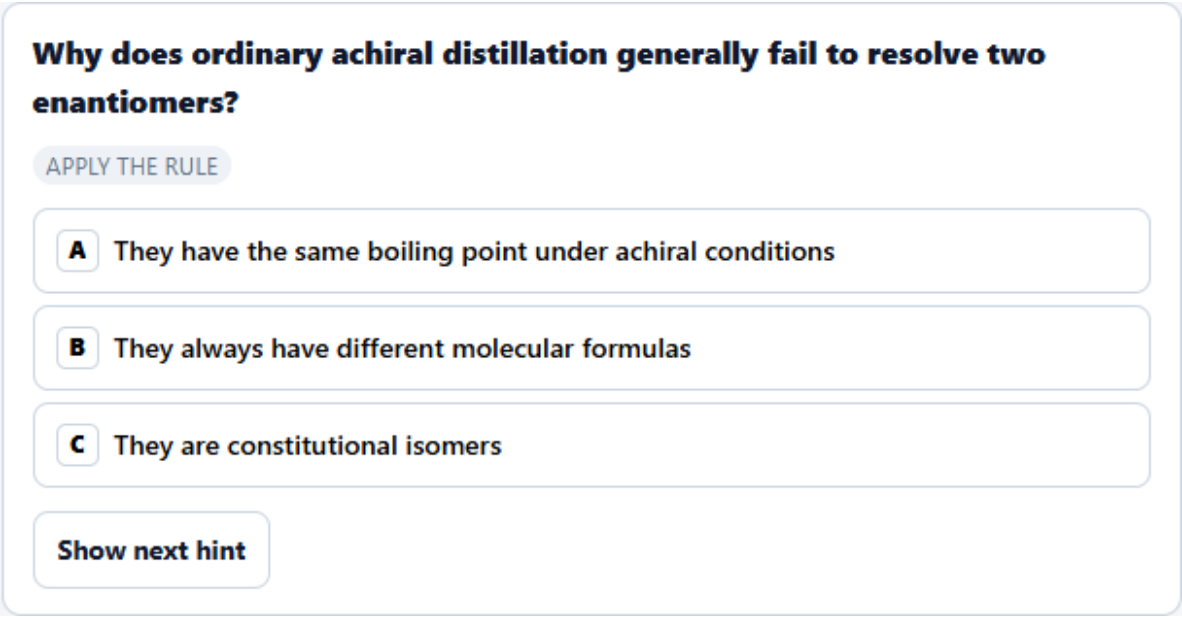
Answer type: choice

Option 0 (A): They have the same boiling point under achiral conditions

Option 1 (B): They always have different molecular formulas

Option 2 (C): They are constitutional isomers

Screenshot: chiral-resolution-stage-1-question



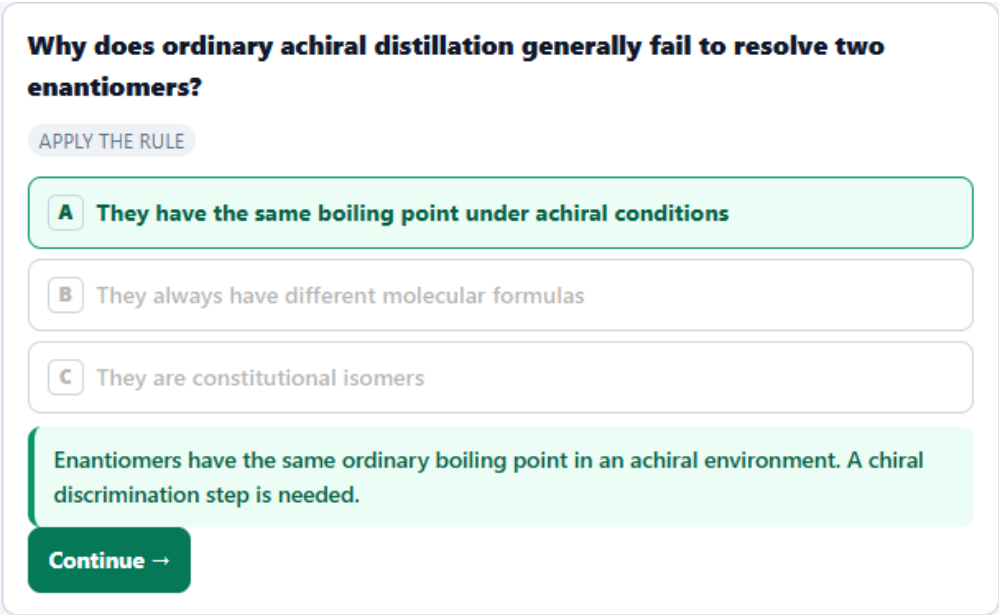
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Compare their properties in an achiral environment.

Hint 2: Opposite optical rotation does not imply different boiling points.

Explanation: Enantiomers have the same ordinary boiling point in an achiral environment. A chiral discrimination step is needed.

Screenshot: chiral-resolution-stage-1-feedback



chiral-resolution:stage-2 | For a suitable substrate, reaction with one enantiomer of a resolving agent can create…

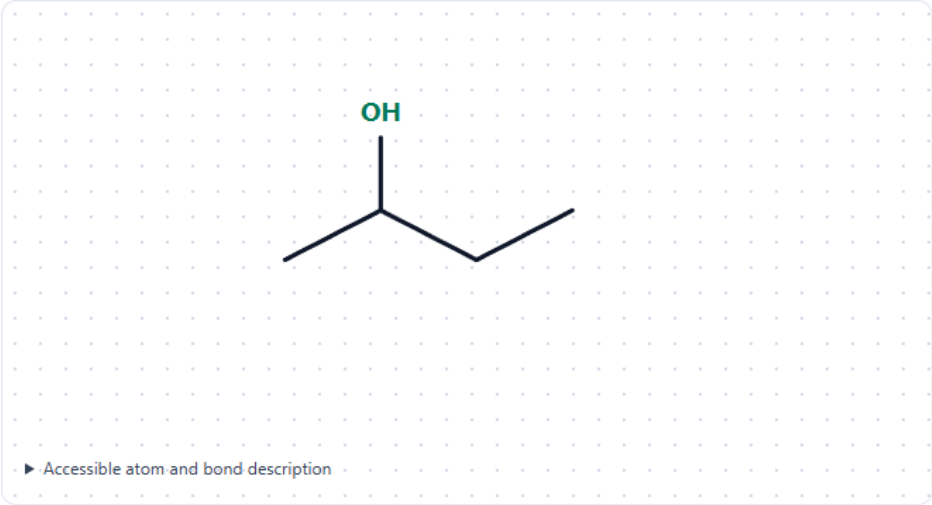
Answer type: choice

Option 0 (A): Separable diastereomeric derivatives or salts

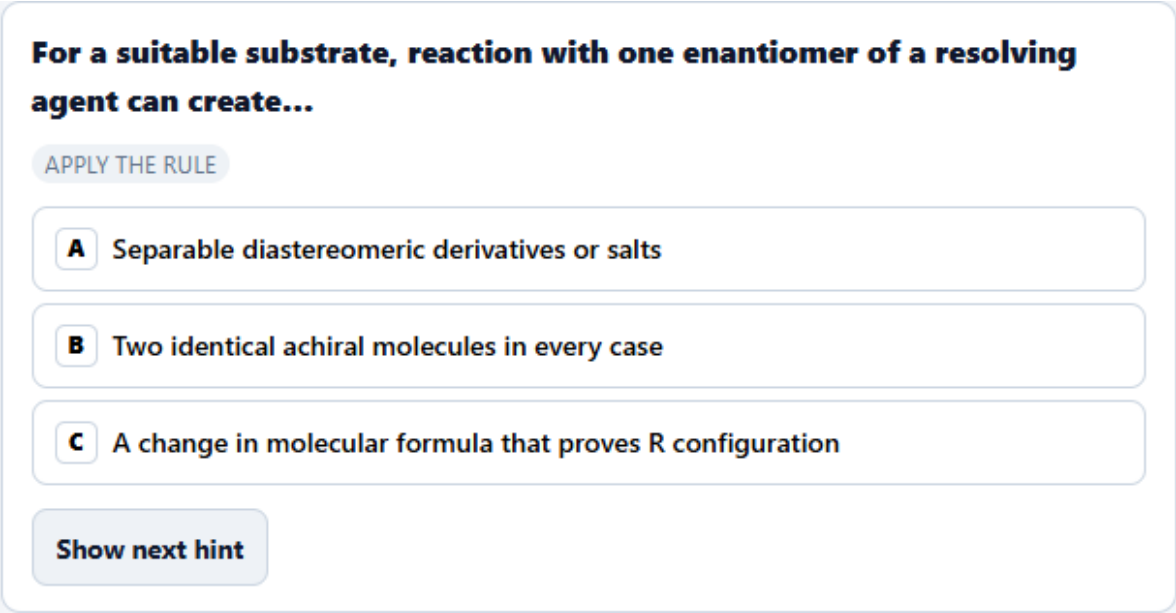
Option 1 (B): Two identical achiral molecules in every case

Option 2 (C): A change in molecular formula that proves R configuration

Screenshot: chiral-resolution-stage-2-molecule



Screenshot: chiral-resolution-stage-2-question



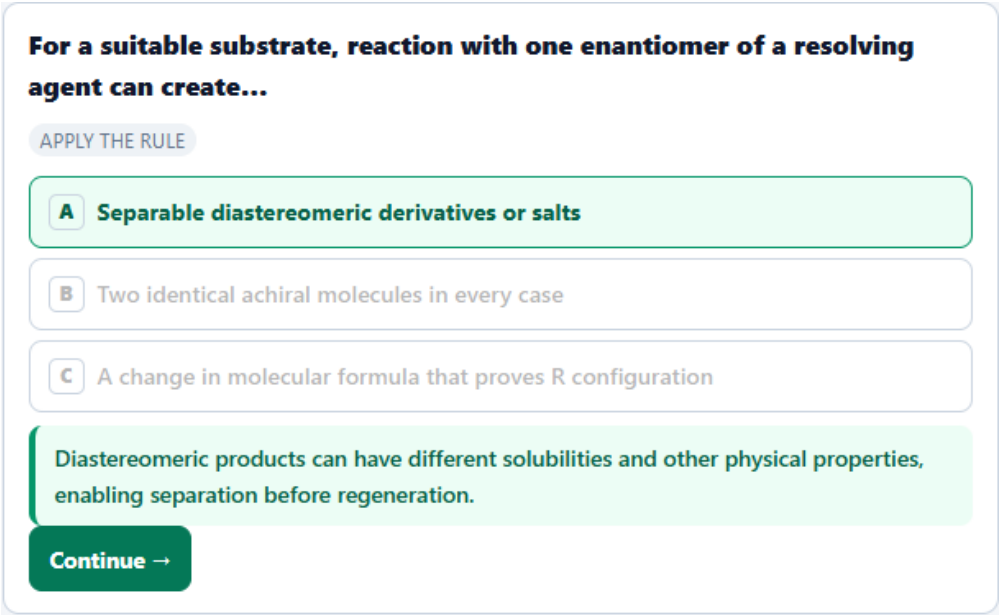
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Add one fixed chiral influence to both enantiomers.

Hint 2: The two resulting products are no longer mirror-image counterparts.

Explanation: Diastereomeric products can have different solubilities and other physical properties, enabling separation before regeneration.

Screenshot: chiral-resolution-stage-2-feedback



chiral-resolution:stage-3 | A remote substituent change retains the local arrangement at a tetrahedral stereocenter but changes its CIP priorities. Its R/S label…

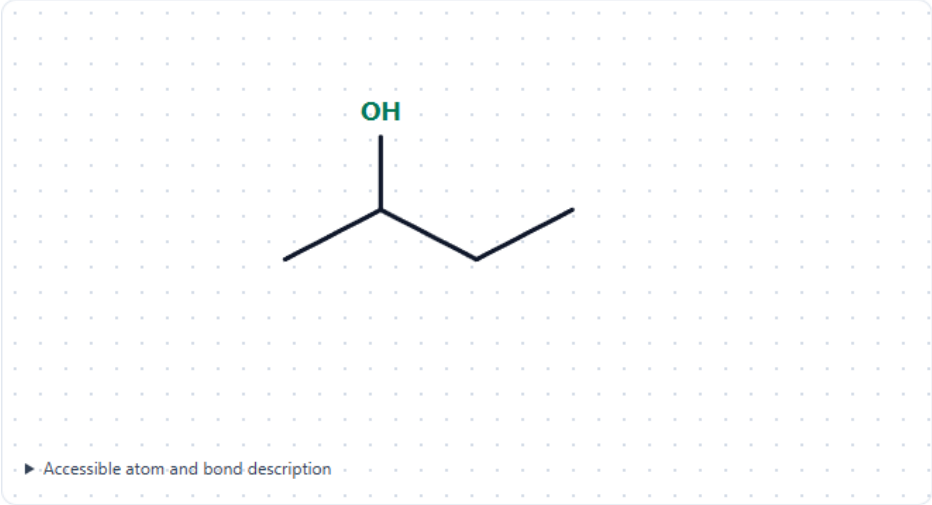
Answer type: choice

Option 0 (A): May change without physical inversion

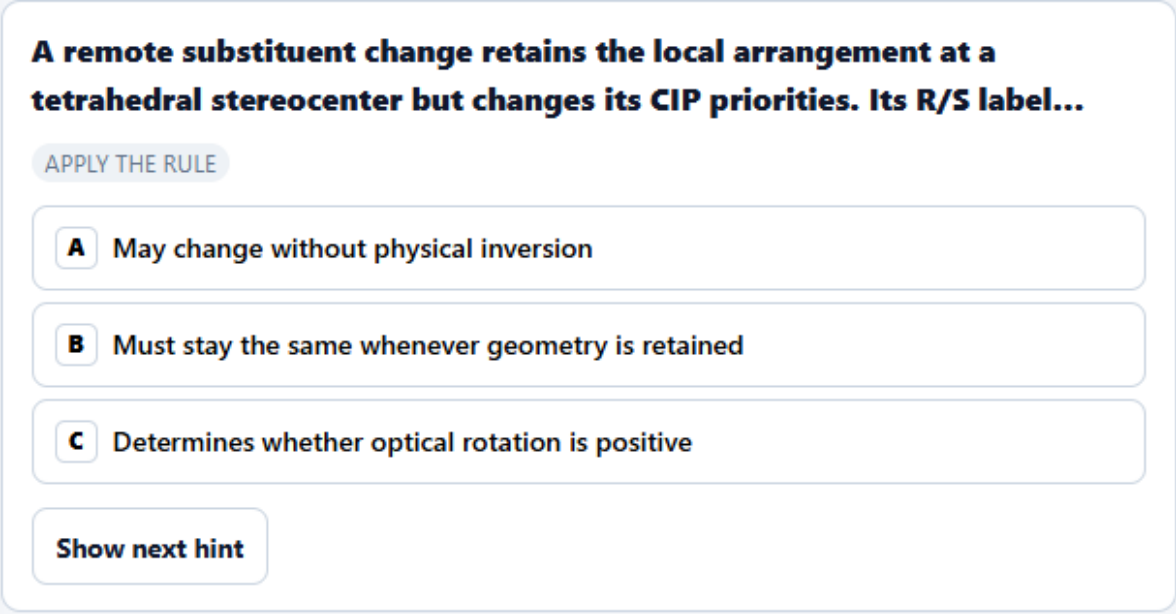
Option 1 (B): Must stay the same whenever geometry is retained

Option 2 (C): Determines whether optical rotation is positive

Screenshot: chiral-resolution-stage-3-molecule



Screenshot: chiral-resolution-stage-3-question



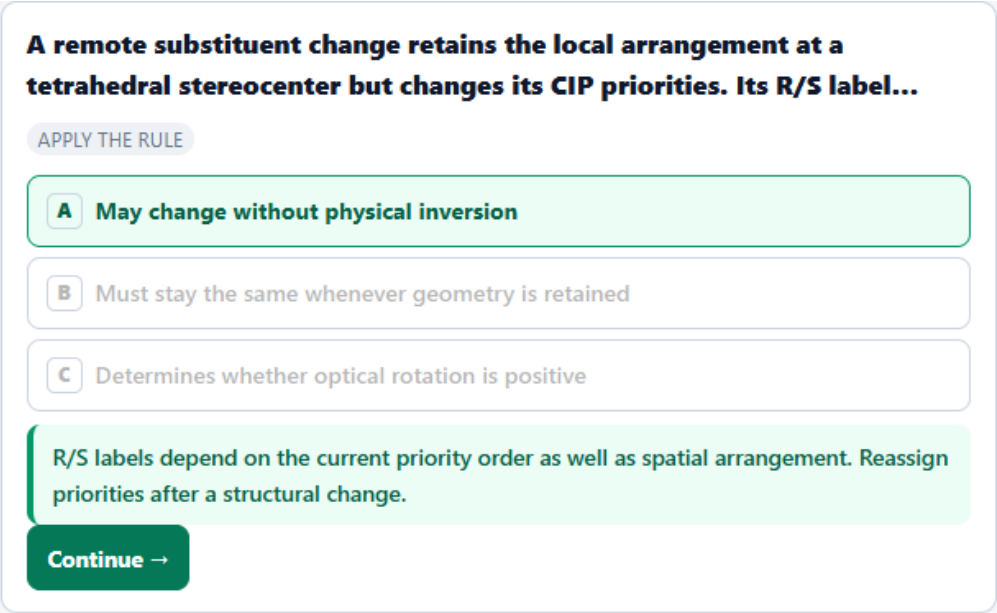
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Keep the geometry and priority ranking as separate inputs.

Hint 2: A descriptor change does not by itself prove inversion.

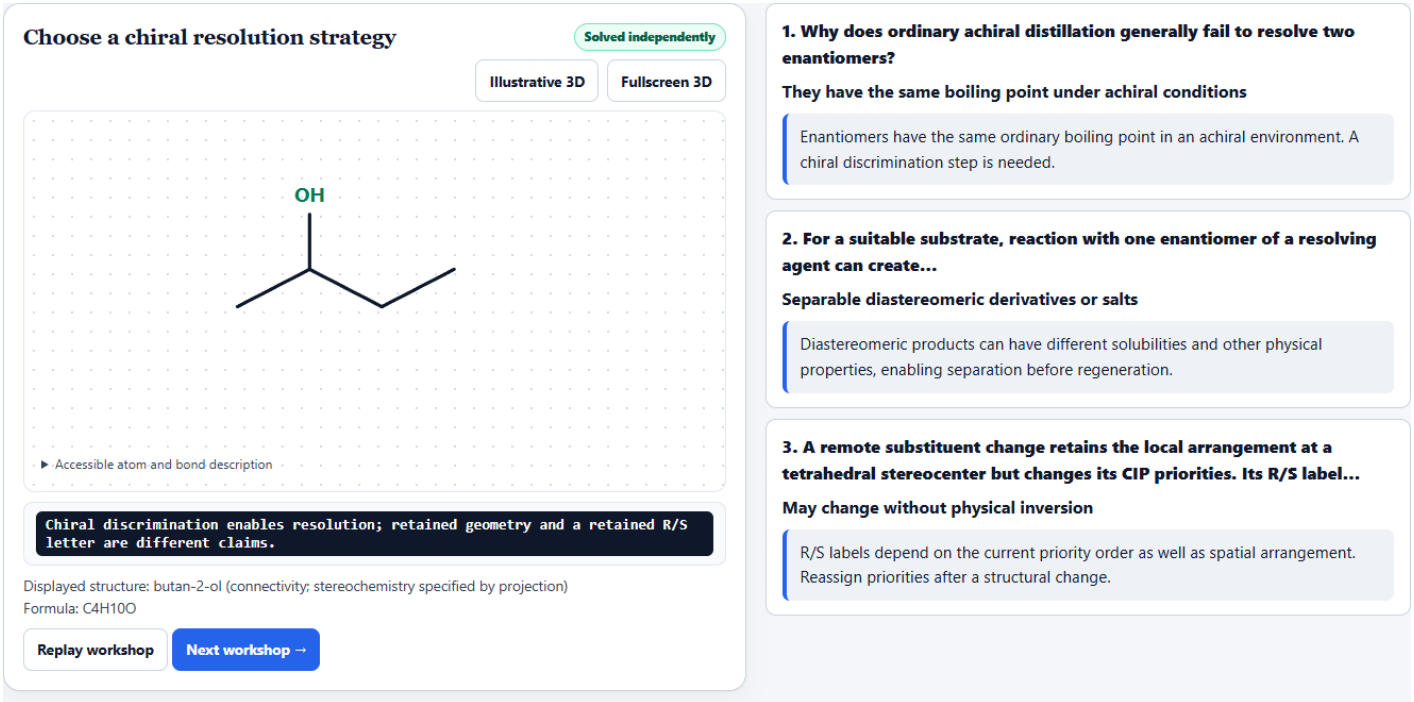
Explanation: R/S labels depend on the current priority order as well as spatial arrangement. Reassign priorities after a structural change.

Screenshot: chiral-resolution-stage-3-feedback



Completion summary: Chiral discrimination enables resolution; retained geometry and a retained R/S letter are different claims.

Screenshot: chiral-resolution-complete



Molecular repository: author claims to verify

ethanol

Author claims: ethanol | formula C2H6O | canonical composition C2H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1]]

ether

Author claims: methoxymethane | formula C2H6O | canonical composition C2H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=60.00, y=110.00; charge=0; explicit H override=none

1 (B): O; x=120.00, y=75.00; charge=0; explicit H override=none

2 (C): C; x=180.00, y=110.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1]]

isobutane

Author claims: 2-methylpropane | formula C4H10 | canonical composition C4H10 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=108.00, y=30.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1]]

butane

Author claims: butane | formula C4H10 | canonical composition C4H10 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1]]

cyclohexane

Author claims: cyclohexane (connectivity diagram) | formula C6H12 | canonical composition C6H12 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=170.00, y=37.00; charge=0; explicit H override=none

1 (B): C; x=228.89, y=71.00; charge=0; explicit H override=none

2 (C): C; x=228.89, y=139.00; charge=0; explicit H override=none

3 (D): C; x=170.00, y=173.00; charge=0; explicit H override=none

4 (E): C; x=111.11, y=139.00; charge=0; explicit H override=none

5 (F): C; x=111.11, y=71.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [4, 5, 1], [5, 0, 1]]

butanol2

Author claims: butan-2-ol (connectivity; stereochemistry specified by projection) | formula C4H10O | canonical composition C4H10O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [1, 4, 1]]

butanediol

Author claims: butane-2,3-diol (stereochemistry unspecified) | formula C4H10O2 | canonical composition C4H10O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=224.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=108.00, y=25.00; charge=0; explicit H override=none

5 (F): O; x=166.00, y=165.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [1, 4, 1], [2, 5, 1]]

e-butene

Author claims: (E)-but-2-ene | formula C4H8 | canonical composition C4H8 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=60.00, y=50.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=100.00; charge=0; explicit H override=none

2 (C): C; x=185.00, y=100.00; charge=0; explicit H override=none

3 (D): C; x=240.00, y=150.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [2, 3, 1]]

z-butene

Author claims: (Z)-but-2-ene | formula C4H8 | canonical composition C4H8 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=60.00, y=50.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=100.00; charge=0; explicit H override=none

2 (C): C; x=185.00, y=100.00; charge=0; explicit H override=none

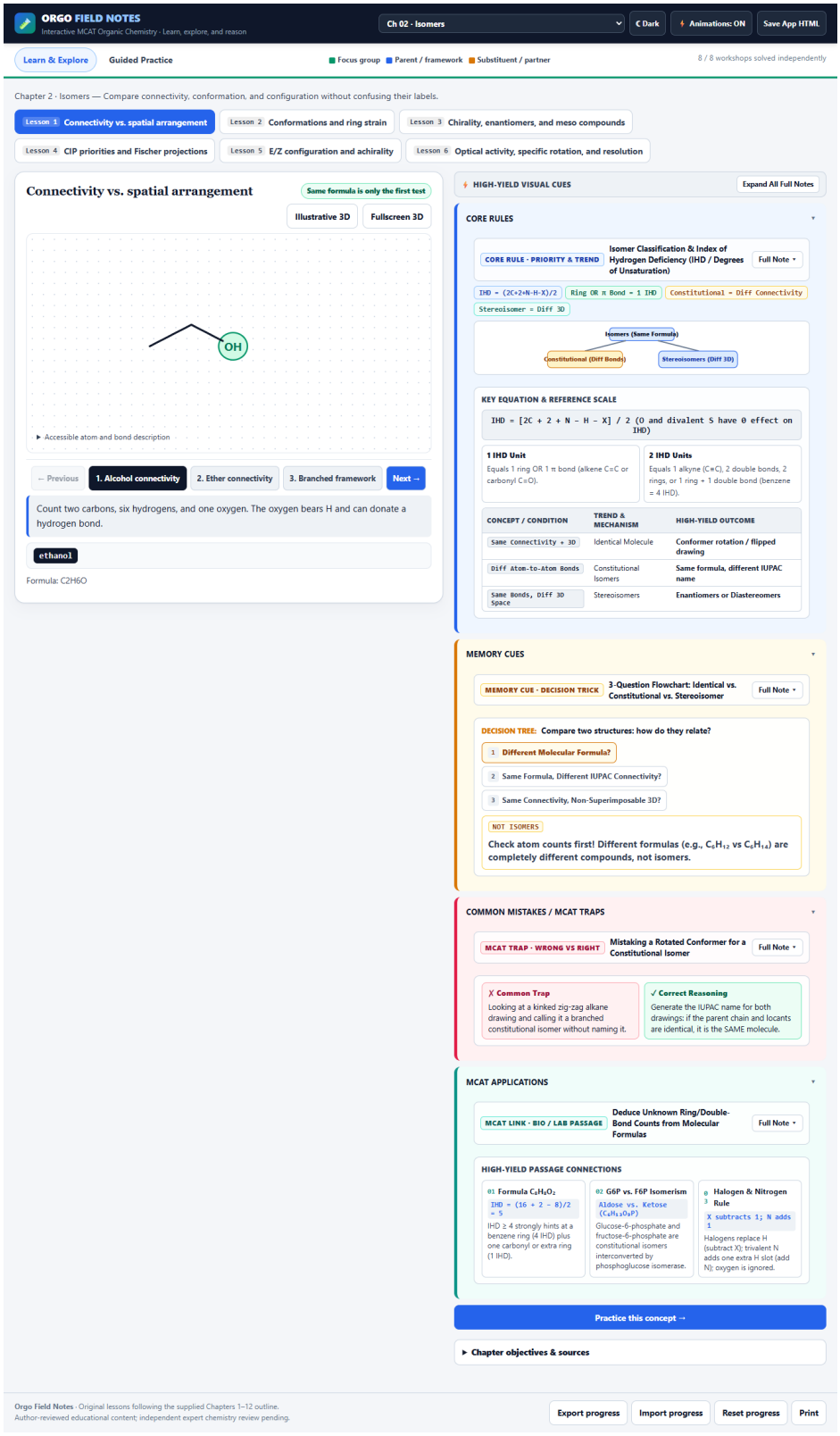
3 (D): C; x=240.00, y=50.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [2, 3, 1]]

Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

Screenshot: layout-desktop-light



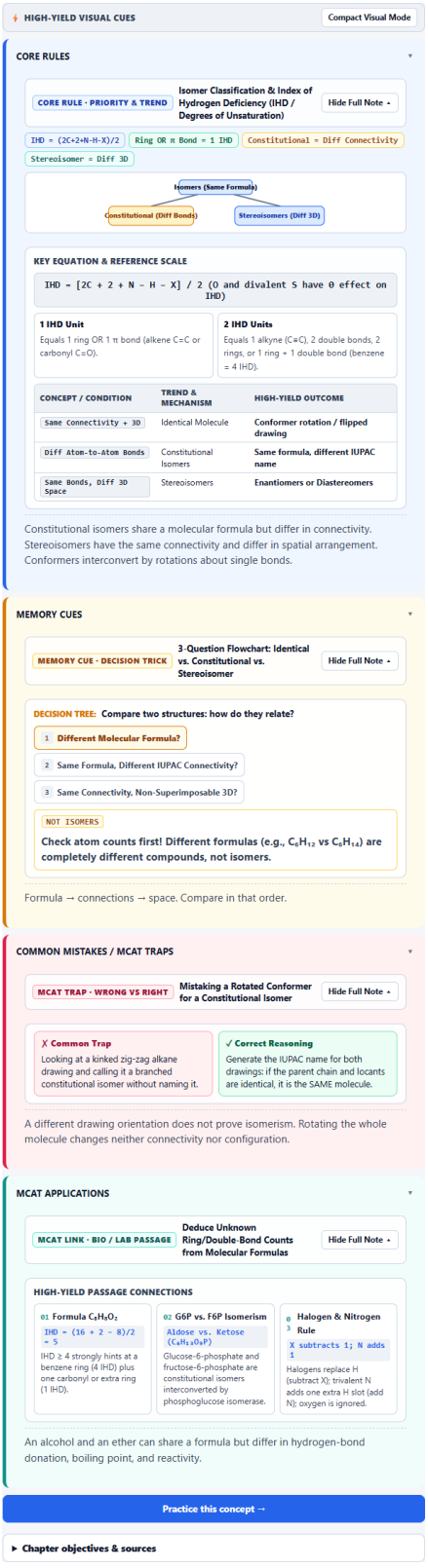
Screenshot: layout-mobile-light



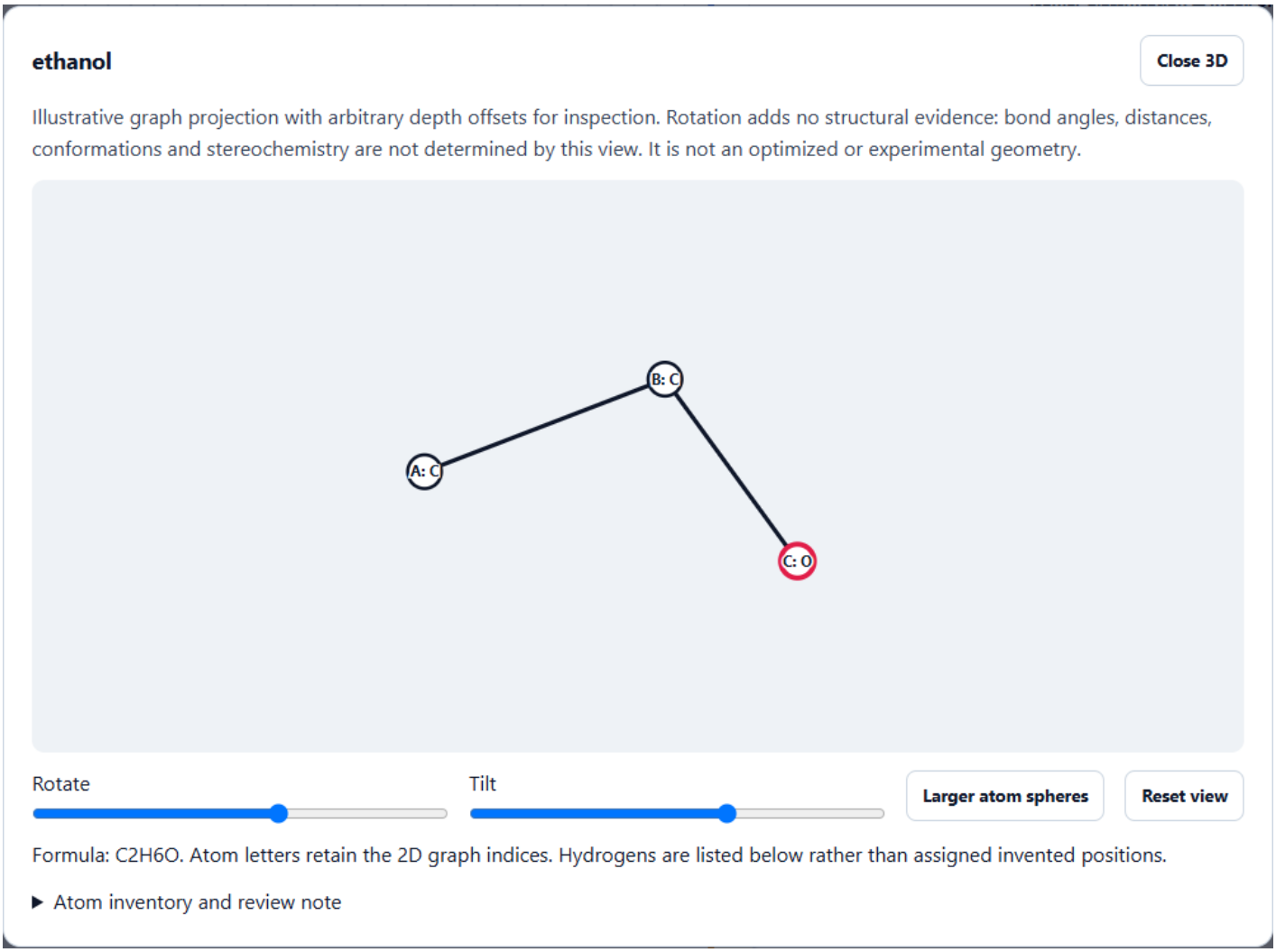
Screenshot: layout-mobile-dark



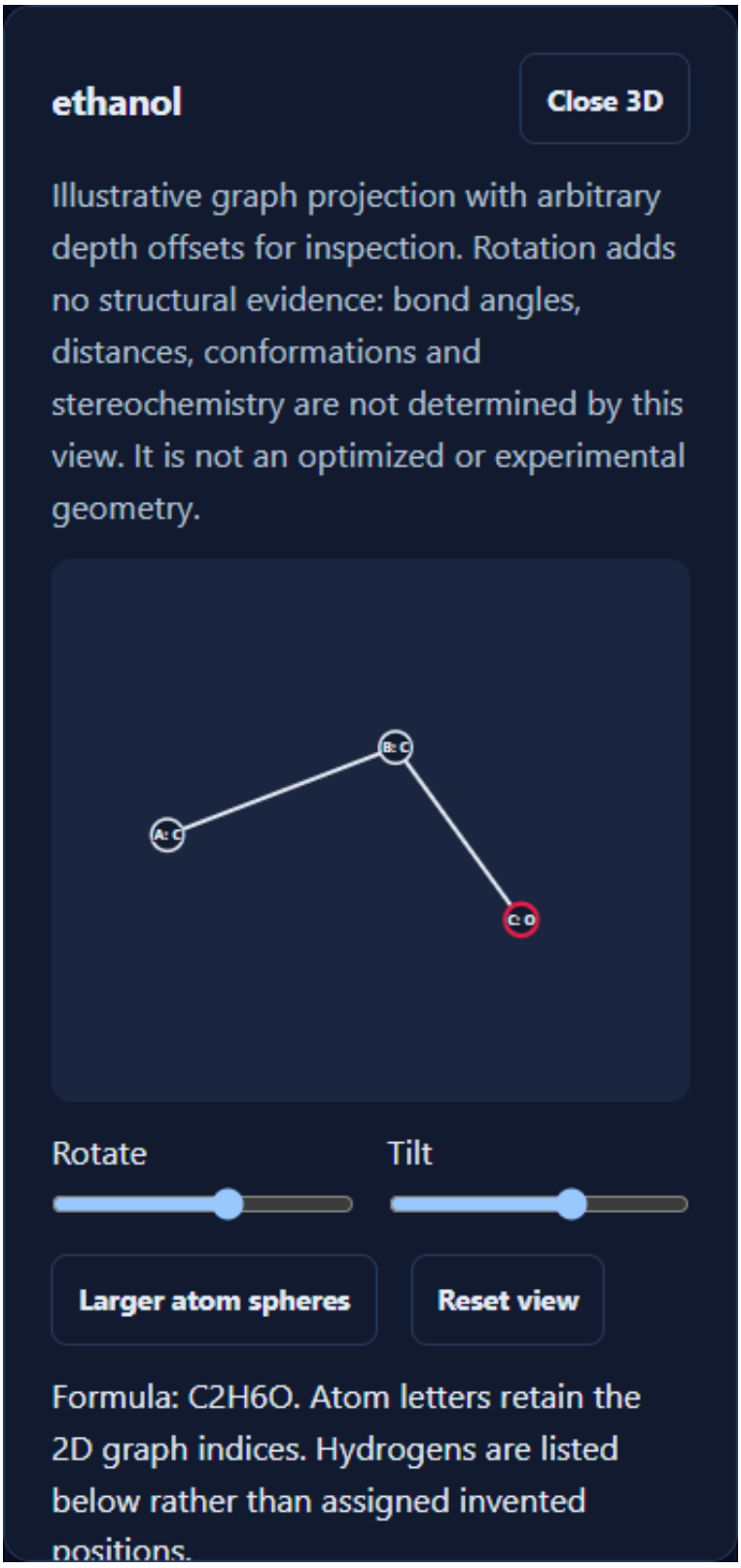
Screenshot: visual-notes-expanded



Screenshot: projection-desktop-light



Screenshot: projection-mobile-dark



Execution evidence

115 screenshots and 25 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.